Counting Complexity of PureCircuit -1

CS907 Dissertation Project

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Abstract

When this thing gets up to 88 mph, you’re gonna see some serious s· · ·.

Keywords: Flux Capacitor, 1.21 Gigawatts, Calvin Klein

Acknowledgements

Always good to acknowledge people.

Abbreviations

Turing Machine	TM
Non-Deterministic Turing Machine	NTM

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List of Symbols

$[n]$ List of numbers $1, ..., n$.

$[n]_0$ Defined as $[n] \cup \{0\}$.

$\mathbb{N}$ Set of natural numbers excluding 0.

$\mathbb{N}_0$ Natural numbers including 0.

$n^{O(1)}$ Set of all functions $f : \mathbb{N} \to \mathbb{N}$, where $f \in O(n^c)$ for some $c \in \mathbb{N}$.

x_{-i} Given $x \in A^d$, we use x_{-i} to denote all members excluding index i.

$x_{-i}(e) \equiv x(x_{-i}, e)$ Given $x \in A^d$, we denote $x_{-i}(e)$ or $x(x_{-i}, e)$ to indicate n dimensional vector, where

$$\forall j \in [n] : [x_{-i}(e)]_j = [x(x_{-i}, e)]_j = \begin{cases} e & \text{if } j = i \\ x_i & \text{otherwise} \end{cases}$$

Will be used to indicate element replacement.

$[x]_j$ Given $x \in A^d$ for some $j \in [d]$, we have $[x]_j = x_j$.

$\mathbb{B}$ The binary set $\{0, 1\}$.

$\mathbb{T}$ Defines the set $\{0, 1, \perp\}$ which depicts the kleene algebra.

Chapter 1

Introduction

1.1 Introduction

Combinatorics is a field of study in mathematics that primarily focuses on the notion of counting objects with certain properties. Over time, this notion has shifted, especially in the subfield of algebraic combinatorics, where there is no clear notion of the object that we are counting, and the numbers express something more abstract [43]. Researchers have started asking whether it is possible to ask the inverse question, or more informally, can we associate our sequence of numbers to sets of objects? [33, 43, 34]

To understand this notion, we provide an example from the papers by Ikenmeyer and Pak [43, 33]: Assume that $f(G)$ counts the number of 3-colourings of a graph G. Now one can, ask: does the function $h(G) \triangleq f(G)/6$ also count something? The answer is yes as argued in [43, 33] and the idea is, as for every colouring χ, we can compute 3! additional colourings since we can permute the colours we assigned. Therefore, we only need to count one of the permutations, or the smallest colouring. We can therefore argue that $h(G)$ counts the smallest possible 3-colourings of a graph G. We say that this is a "combinatorial interpretation of $h(G)$".

But not all non-negative functions have such a clear interpretation. We use an example by Ikenmeyer et al. [33]. We recall a combinatorial variant of the *Pigeonhole principle*, where given a circuit $C : \{0,1\}^n \to \{0,1\}$, find x such that $C(x) = 0^n$ or distinct $x, y \in \{0,1\}^n$ such that $C(x) = C(y)$ and $C(C(x)) \neq 0$. C essentially maps $\{0,1\}^n \to \{0,1\}^n \setminus \{0^n\}$. Every point that is mapped to 0^n is considered a violation. The problem counts every pair of points that is mapped to the same bin or to a point that is a violation. We add the restriction of $C(C(x)) \neq 0$ to avoid double counting. We can clearly observe that, if $e(C)$ counts the number of solutions, under the pigeonhole principle, it must be the case $e(C) \geqslant 1$ since we either have a violation or there exists at least one pair of points that is mapped to the same bin by the pigeonhole principle. So now we can ask: does the function $e'(C) \doteq e(C) - 1$ also have a set of object which we can enumerate over? Ikenmeyer et al. [33] demonstrated that this problem does not have a combinatorial interpretation. This leads to the questions of how can we formally represent the meaning of combinatorial interpretations, and what can we do to verify whether a function has one?

In recent years, there has been a revolutionary initiative to answer the aforementioned questions, with the help of counting complexity theory and the complexity class of **#P**. In fact, being able to find such definitions or interpretations is crucial, as it allows us to utilise tools from enumerative combinatorics to understand and reveal hidden structures and properties of such functions [43, 33]. Moreover, there are several problems or numbers, such as *Kronecker coefficients* [39], whose combinatorial interpretation could lead to groundbreaking breakthroughs, such as a step closer to the resolution of the $\mathbf{P} \neq \mathbf{NP}$ conjecture [33, 32].

In our current work, we focus on extending the work done by Ikenmeyer et al. [33], where they focused on the creation of frameworks that determine whether $f \stackrel{?}{\in}$ **#P**, by looking at the subclasses of **#TFNP** $- 1$ problems, a class of problems whose combinatorial nature guarantee the existence of a solution. More specifically, we look into the **PPAD** class which seems to have several problems where under decrementation, may or may not have a combinatorial interpretation [33]. This implies that **PPAD** acts as some sort of barrier, to **#P** which motivates the research as to what causes

To accomplish this task, we studied *PureCircuit* which was created by Deligkas et al. [21], as we hope its **PPAD**-completeness and circuit-like nature may give us useful insights to the counting complexity of **PPAD** $- 1$.

We will now give an extensive introduction to the literature and the related problems and then offer formal and precise objectives that capture our goal. We will then go over the main results of our theory research with the conjectures and possible directions that we uncovered. Moreover, we will give an overview of the visualisation software we developed to verify our theoretical results and give an overview of its capabilities and limitations. Lastly we will give brief analysis of our project management, the lessons we learnt and a conclusion that summarises our project as a whole.

1.2 Theoretical Preliminaries and Literature Overview

In the current section, we introduce all the necessary preliminaries with respect to search problems and counting complexity theory. Moreover we give an overview of the core problem of the research *PureCircuit*, and its connections with search problems and *Kleene logic*.

1.2.1 Search problems

In complexity theory, we are interested in different kinds of problems. A common example are problems founds in **P** and **NP** classes, known as decision problems, where given an input, the output is a boolean value of whether the specific instance is in the language or not. Function problems or search problems are extensions of decision problems, where we can represent functions or relations as explained in the definitions 1.2.1. In addition, we reference an additional subclass of these problems known as *Total Search problems*, in which for every input, there must exist at least one solution.

Definition 1.2.1: Search Problems

Search problems can be defined as relations $R \subseteq \mathbb{B}^* \times \mathbb{B}^*$, where given $x \in \mathbb{B}^*$, we want to find $y \in \mathbb{B}^*$ such that xRy.
Total Search problems are search problems such that for each input, there must exist at least one solution.

We mainly focus on the search variants of **NP** know an **FNP** and its subclass for *total problems* known as **TFNP** using the definitions in 1.2.1.

Definition 1.2.2: FNP and TFNP

FNP are *search problems* such that there exists poly-time *TM* $M : \mathbb{B}^* \to \mathbb{B}$ and a polynomial function $p : \mathbb{N} \to \mathbb{N}$ such that:

$$\forall x \in \mathbb{B}^*, y \in \mathbb{B}^{p(|x|)} : xRy \iff M(x, y) = 1$$

Lastly $\mathbf{TFNP} = \{L \in \mathbf{FNP} \mid L \text{ is total}\}$

Adding on, in the decision problem space, it is common to relate a problem to each other with the usage of reductions such as *Kerp reductions*. We introduce its search problem variant known as *Levin Reductions* using the definition in 1.2.1.

Definition 1.2.3: Levin Reductions

Let A and B be two search problems with relations $R_A \subseteq \{0,1\}^* \times \{0,1\}^*$ and $R_B \subseteq \{0,1\}^* \times \{0,1\}^*$, respectively.
We say that A is *Levin reducible* to B, denoted $A \leqslant_L B$, if there exist polynomial-time computable functions $f : \{0,1\}^* \to \{0,1\}^*$ and $g : \{0,1\}^* \times \{0,1\}^* \to \{0,1\}^*$ such that:

1. f applies a *Kerp reduction* from problems of A to problems of B such that:

$$\forall x \in \{0,1\}^* : x \in S_{R_A} \implies f(x) \in S_{R_B}$$

2. Using an element $R_b(f(x))$, we can construct a solution to R_A such that:

$$\forall x \in S_{R_A}, y' \in R_B(f(x)) : (x, g(x,y)) \in R_A$$

Where we denote the sets S_R and $R(\cdot)$ as the domain and co-domain of the relations R, or more formally

$$S_R \triangleq \{x \mid \exists y : xRy\}$$
$$R(x) \triangleq \{y \mid \exists x : xRy\}$$

Various **TFNP** were introduced by various researchers which form the hierarchy as seen in 1.2.1 [45, 35, 17, 24]. In the current project we are mainly interested in **PPAD**, known as “Polynomial Parity of Augmented DiGraphs”, created by Papadimitriou [45]. We will give a formal definition of the class as well as several cornerstone problems that we will refer to in the upcoming chapters. We will first introduce the *EndOfLine* problem 1.2.1 by Papadimtirou [45], which takes advantage of the combinatorial fact that if there exists an unbalanced node in a directed graph, there must always exist a different one.

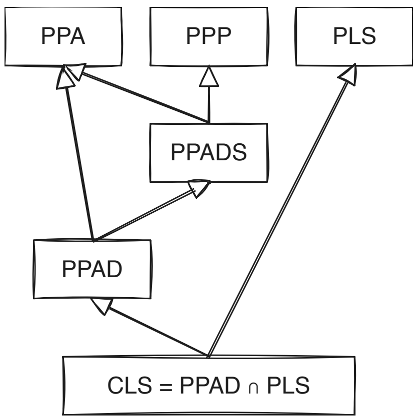


Figure 1.2.1: TFNP hierarchy. Figure re-created by [33].

Definition 1.2.4: *EndOfLine* problem [45]

Given poly-sized circuits $S, P \in \mathbb{B}^n \to \mathbb{B}^n$, we define a digraph $G = (V, E)$ such that $V = \mathbb{B}^n$ and E defined as:

$$E = \{(x,y) \in V^2 : S(x) = y \wedge P(y) = x\}$$

We define source nodes as $v \in V : deg(v) = (0,1)$ and sink nodes as $deg(v) = (1,0)$. We also syntactically ensure that the 0^n node is always a source, meaning $S(P(0^n)) \neq 0 \wedge P(S(0^n)) = 0^n$. A node $v \in V$ is a solution *iff* *in-deg*$(v) \neq$ *out-deg*(v).

An illustrative example of an *EndOfLine* instance can be seen in figure 1.2.2. The idea that of the

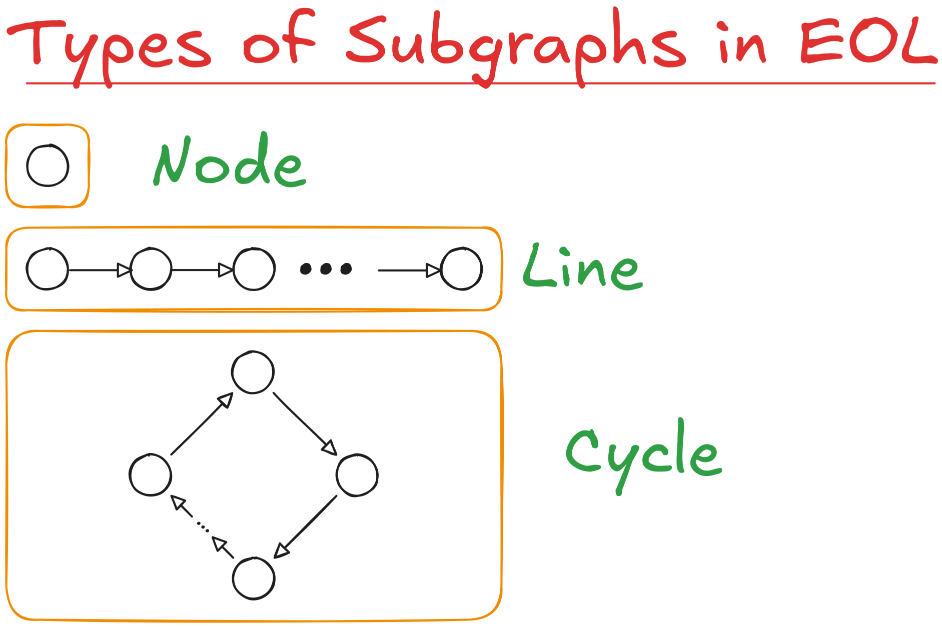


Figure 1.2.2: Types of subgraphs in *EndOfLine*

problem is that its solutions will always be at the beginning and the end of some line, hence its name. We also always guarantee a solution since we ensure that the 0^n is always a source and therefore there must exist a pairing sink. From this problem, we define the **PPAD** complexity class 1.2.5.

Definition 1.2.5: *PPAD* complexity class [45]

PPAD is defined as the set of search problems that are *levin reducible* 1.2.1 to the *EndOfLine* problem 1.2.1.

EndOfLine variants and extensions

We introduce several key variants of the *EndOfLine* that have been created in the past. Alexandros et al. in the paper [29] was able to demonstrate that for several modifications of the original problem such as adding k known sources, where k can be a polynomial function with respect to the number of sources. Moreover, people have experimented with changing the degrees of the graphs, as demonstrated by Ikenmeyer et al. [33], with the problems such as $\mathit{SourceOrExcess}(k, 1)$ where k it the maximum in-degree for each node as defined in 1.2.1. All these problems remain **PPAD** complete. More interestingly, works by Papadimitriou [44], Beame et al. [4] and Alexandros et al. [29] were able to demonstrate that in the more general case of superpolynomial graphs, if we bound the in-degree and out-degree of each node to some polynomial of the input size, then the problem remains **PPAD**.

Definition 1.2.6: *SourceOrExcess* problem [33]

We define as $\mathit{SourceOrExcess}(k, 1)$ for $k \in \mathbb{N}_{\geqslant 2}$ as such Given a poly-sized successor circuit $S : \mathbb{B}^n$ and a set of predecessor poly-sized circuits $\{P_i\}_{i \in [k]}$, we define the graph $G = (V, E)$ such that, $V = \mathbb{B}^n$ and E as:

$$\forall x, y \in V : (x, y) \in E \iff (S(x) = y) \wedge \bigvee_{i \in [k]} P_i(y) = x$$

We ensure that 0^n is as source, meaning $\deg(0^n) = (0, 1)$. A valid solution is a vertex v such that *in-deg*$(v) \neq$ *out-deg*(v).

In addition to the aforementioned adjustments, researchers were able to give a topological interpretation to the *EndOfLine* 1.2.1 problem, by creating embeddings of the graph into 2D or 3D spaces as demonstrated by several researchers [12, 29, 16]. We will mainly look at two of these problems, known as *RLeafD* and *2D-EOL* created by Chen et al. and Alexandros respectively.

Definition 1.2.7: *2D-EOL* problem [12, 29]

Input: Two poly-sized circuits $P, S : \{0,1\}^{2n} \to \{0,1\}^{2n}$ where $P(0^n, 0^n) = (0^n, 0^n) \neq S(0^n, 0^n)$.
Output: A point $x \in \{0,1\}^{2n}$ such that one of the following conditions are satisfied:

1. Sinks: $P(S(x)) \neq x$
2. Sources: $S(P(x)) \neq x \wedge x \neq 0^n$
3. Invalid successor: $x - S(x) \notin \{(0,0),(0,1),(0,-1),(1,0),(-1,0)\}$
4. Invalid predecessor: $x - P(x) \notin \{(0,0),(0,1),(0,-1),(1,0),(-1,0)\}$

The above description defines a superpolynomial such that an edge $(x, y) \in E$ if and only if $S(x) = y \wedge P(y) = x$ as well as the closeness constraint $\|x - S(x)\|_1 \leqslant 1$. Chen et al. [12] introduced a reduction on how one can parsimoniously reduce an *EndOfLine* problem to a *2D-EOL* problem (written as *RLeafD* in their paper) using planar graph embeddings. We find this construction useful for our results and therefore we give a full report on how it works.

Multi-Source vairiants Lastly we introduce the class of variants for which the *EndOfLine* problem which we extend the known number of sources. These multi source problems (formally defined using 1.2.1) were created by Hollender et al. [29] and were demonstrated to be **PPAD**-complete.

Definition 1.2.8: *PolyS-EoL* definition

Input: We are given triplet $(S, P, \bar{f})$, where S, P are successors, predecessor circuits of an *EndOfLine* problem with n input-output bits, and $\bar{f}$ is a poly-sized circuit that computes some polynomial function $f : \mathbb{N} \to \mathbb{N}$. In addition, we impose the following restriction:

$$\forall i \in [f(n)]_0 : S(P(i)) \neq i \wedge P(S(i)) = i$$

Output: A node $k \notin [f(n)]_0$ such that $S(P(x)) \neq x \oplus P(S(x)) \neq x$.

The PureCircuit problem

The *PureCircuit* problem is a discretised version of the ϵ-*GCircuit* which was first introduced by Chen et al. [21, 13]. General idea of ϵ-*GCircuit* is that we have a directed graph $T = (V, G)$ where V denote value nodes and G denote gate nodes. Each value node is assigned a value in $[0, 1]$ and the gates denote can denote the standard set of arithmetic operations $\{+, -, *\}$ with other nodes or some constant, or boolean operators $\{\wedge, \vee, \neg, <, >, =\}$ with other nodes or some constant. The goal of the problem is to assign every problem with a value such that the assingment is correct up to a factor of $\pm\epsilon$. This problem was crucial to demonstrate *PPAD*-hardness for constant $\epsilon > 0$ by Rubinstein [21, 46].

PureCircuit is the problem that we get when taking the limit $\epsilon \to \infty$ of the ϵ-*GCircuit* problem. This results in a circuit which uses the set of values $\{0, \bot, 1\}$ to denote the range of values $(0), (0, 1), (1)$. Incindentally, the above triplet of values coincides with *Kleene algebra* which extends the traditional binary logic [37]. We will explore the above algebra in greater depth in section 1.2.3. This problem was created by Deligkas et al. [21] to demonstrate the hardness of approximating **PPAD** problems and was shown to be **PPAD**-complete.

Definition 1.2.9: PureCiruit Problem Definition [21]

An instance of *PureCircuit* is given by vertex set $V = [n]$ and gate set G such that $\forall g \in G : g = (T, u, v, w)$ where $u, v, w \in V$ and $T \in \{\text{NOR}, \text{Purify}\}$. Each gate is interpreted as:

1. *NOR*: Takes as input u, v and outputs w
2. *Purify*: Takes as input u and outputs v, w

And each vertex is ensured to have in-deg$(v) \leqslant 1$. A solution to input instance (V, G) is denoted as an assignment $\mathbf{x} : V \to \{0, \perp, 1\}$ such that for all gates $g = (T, u, v, w)$ we have:

1. *NOR*:

$$\begin{gathered} \mathbf{x}[u] = \mathbf{x}[v] = 0 \implies \mathbf{x}[w] = 1 \\ (\mathbf{x}[u] = 1 \vee \mathbf{x}[v] = 1) \implies \mathbf{x}[w] = 0 \\ \text{otherwise} \implies \perp \end{gathered}$$

2. *Purify*:

$$\begin{gathered} \forall b \in \mathbb{B} : \mathbf{x}[u] = b \implies \mathbf{x}[v] = b \wedge \mathbf{x}[w] = b \\ \mathbf{x}[u] = \perp \implies (\mathbf{x}[v], \mathbf{x}[w]) \in \{(0, \perp), (\perp, 1), ((0, 1))\} \end{gathered}$$

In addition to the gates above, we will make use of the standard set of gates $\{\vee, \wedge, \neg\}$, whose behaviour is depicted in the tables 1.2.1. Moreover, we will make use of the *Copy* gate which we can define as $Copy(x) = \neg(\neg x)$. These gates are well defined based on the *NOR* gate as shown by Deligkas et al. [21] and do not directly affect the complexity of the problem. Furthermore, with respect to the *Purify* gate, the original problem states that one of two outputs must contain a pure value. We restrict the solution set to the one written in the definition as Deligkas et al. [21] showed that the only *Purify* solutions essential for **PPAD**-completeness are $\{(0, \perp), (\perp, 1), (0, 1)\}$, and the addition of more solutions does not affect its complexity.

not	
0	1
$\perp$	$\perp$
1	0

(a) not gate

and	0	$\perp$	1
0	0	0	0
$\perp$	0	$\perp$	$\perp$
1	0	$\perp$	1

(b) and gate

or	0	$\perp$	1
0	0	$\perp$	1
$\perp$	$\perp$	$\perp$	1
1	1	1	1

(c) or gate

Table 1.2.1: Three-valued logic [37]

Topological problems

This section refers to types of problems which take advantage of some combinatorial property of their topology in order to guarantee a solution.

Sperner Problems Below we will refer to the notion of Sperner problems, which are types of problems that are based around *Sperner's Lemma* [53]. The idea is as follows, given a triangle that has subdivided into smaller subtriangles, we first define the index of each point using the following set:

$$\forall n \in \mathbb{N} : \mathcal{T}_n \triangleq \{(x, y, z) \in [n] \mid x + y + z = n\}$$

We say $\lambda : \mathcal{T}_n \to [3]$ is a valid colouring if and only if it satisfies the following conditions:

$$\forall (x_1, x_2, x_3) \in \mathcal{T}_n : \lambda(x) \in \begin{cases} \{2,3\} & \text{if } x_1 = 0 \\ \{1,3\} & \text{if } x_2 = 0 \\ \{1,2\} & \text{if } x_3 = 0 \end{cases}$$

Given a valid colouring λ, we can always find a small triangle τ where all 3 colours are assigned to one of its vertices. A visual example can be seen in figure 1.2.3.

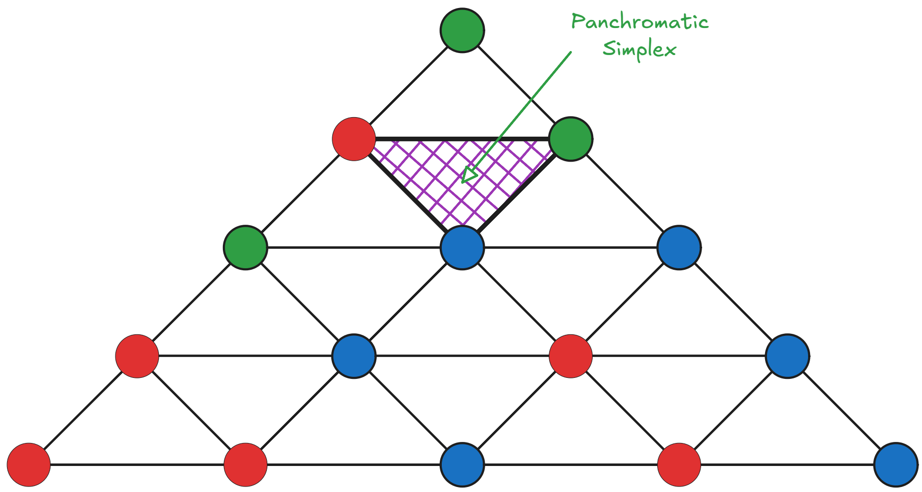


Figure 1.2.3: A colouring example of a subdivided simplex. As we can observe, no matter how we modify colourings, there will always be a small simplex that contains all colours

This type colouring will be referred to as the **linear** colouring where for dimension d, assign $d+1$ distinct colours to each point [16, 12], and it was shown that one can define a **PPAD**-complete variant of this lemma property. There several variants of the problem, where we can either work in a triangle with an exponential amount of subdivisions or polynomial subdivision but on higher dimensional simplexes. Both of these problems are **PPAD**-complete [45, 13, 12].

We introduce an alternative variant of the Sperner problems, introduced by Daskalakis et al. [19], known as *StrongSperner* or *BiSperner*. Instead of assigning $d+1$ colours to find a panchromatic simplex, each vertex is assigned d colours rather than one. We refer to the figure 1.2.4 for a visual interpretation of the colouring. Essentially for each point in some hypercube $[N]^d$, we assigned d colours, where for colour i have choices either $\{i^+, i^-\}$. The goal is to find a cube where in every dimension there exists at least two points that cover both colours. This has been used in many problems throughout literature as seen in the papers [13, 21, 19] to demonstated **PPAD**-hardness of many problems.

We will refer to this type of colouring as **bipolar** colouring, which has been used in grid-like topologies of the Sperner property. We note that these terms are not standard in literature; we adopt this naming convention for of clarity.

Definition 1.2.10: Bipolar colouring

Given a set S^d, where S is some arbitrary set, we refer to bipolar colouring C as:

$$\forall v \in S^d, j \in [d] : [C(v)]_j \in \{-1, 1\}$$

We say that a set of points $A \subseteq S^d$ **cover all the labels** if:

$$\forall i \in [d], \ell \in \{-1, +1\}, \exists x \in A : [\lambda(x)]_i = \ell$$

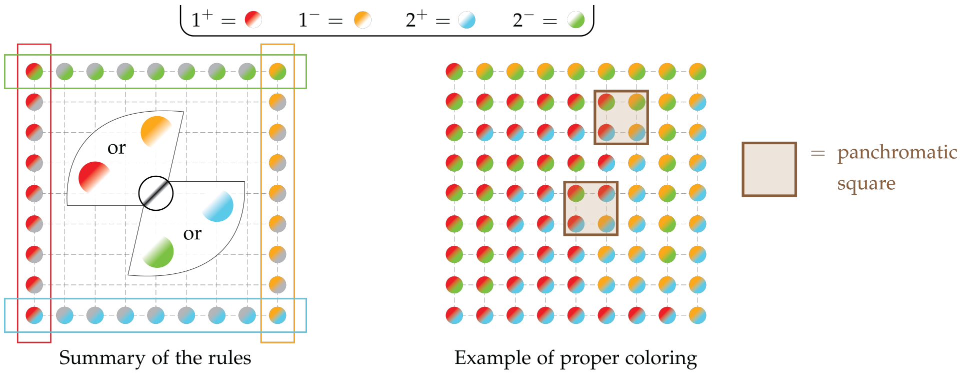


Figure 1.2.4: Visual interpretation of the *StrongSperner* problem. Figure from paper [18]

Compared to the linear colouring variant, Deligkas et al. [21, 20] showed that $\forall A \subseteq S^d : |A| \geqslant d+1$ and satisfies the colouring, $\exists D \subseteq A : |D| \leqslant d$ where D also satisfies the colouring condition, which implies that we need at most d separate points to achieve a colouring.

Definition 1.2.11: *StrongSperner* problem

Input: A boolean circuit that computes a bipolar labelling $\lambda : [M]^N \to \{-1, 1\}^N$ 1.2.1 satisfying the following boundary conditions for every $i \in [N]$:

- if $x_i = 1 \implies [\lambda(x)]_i = +1$
- if $x_i = M \implies [\lambda(x)]_i = -1$

Output: A set of points $\{x^{(i)}\}_{i \in [N]} \subseteq [M]^N$, such that:

- *Closeness condition*: $\forall i, j \in [N] : \|x^{(i)} - x^{(j)}\|_\infty \leqslant 1$
- *Covers all labels* as defined in 1.2.1

Throughout literature the same variants of the problem or specifications have been defined using the names Sperner or discrete Brouwer [13, 12, 16, 21]. This is very similar to a different type of problem known as *StrongTucker*, which can be thought of as the *PPA* analog of *StrongSperner* [45, 2]. The two problems satisfy different boundary conditions but due to their similar labelling nature, we will borrow some techniques used in later sections.

It is important to observe that the *StrongSperner* problem is defined for hypercubes of dimensionality n and width $m \in n^{O(1)}$. Chen et al. [13] demonstrated that for a fixed dimensionality with an exponentially large width the problem remains **PPAD**-complete, as defined in 1.2.1.

Definition 1.2.12: *nD-StrongSperner* problem

Input: A tuple $(\lambda, 0^k)$ such that $\lambda : (\mathbb{B}^k)^n \to \{-1, +1\}^n$, and λ abides the *boundary conditions* of the *StrongSperner* problem.
Output: A point $\alpha = (a_1, \ldots, a_n) \in (\mathbb{B}^k \backslash \{1^k\})^n$ such that

$$\{\alpha + x \mid x \in \mathbb{B}^n\} \text{ covers all the labels } 1.2.1$$

We assume dimensionality $n \geqslant 2$.

The authors refer to both colouring schemes interchangeably when discussing their reductions, so we can assume that all reductions (not necessarily counting reductions) work similarly for either colouring [13, 21, 16, 12].

Other than topological colouring, we will also give a small introduction, to a tiling problem known as mutilated chessboards. Tiling spaces has been used throughout literature [29, 10]. We will use as reference the **PPAD**-complete *1-MC* problem. We refer to the problem introduced by Blank [5] as follows: assume we have a $2n \times 2n$ chessboard. If we are provided with 1×2 pieces, we can trivially tile the board by filling each row with n horizontal pieces. The idea is when we remove the bottom left corner, the board becomes impossible to tile. The argument can be thought of as such: each tile piece covers an even square and an odd square, which implies that it has to be the case that one of the squares has to remain untilled. An example can be seen in the figure 1.2.5. More formally we use the following definition 1.2.1, and Hollender et al. [29] demonstrated that this problem

Definition 1.2.13: *1-MC* [29]

Input: Given a circuit $C : \{0,1\}^{2n} \to \{0,1\}^{2n}$, such that $C(0,0) = (0,0)$, such that tiling between two squares is indicated such that:

$$\forall x \in \{0,1\}^{2n} : C(C(x)) = x \wedge x - C(x) \in \{(0,1),(1,0),(0,-1),(-1,0)\}$$

Output: We want to find a point $x \in \{0,1\}^{2n} \backslash \{(0,0)\}$ such that we have no tiling.

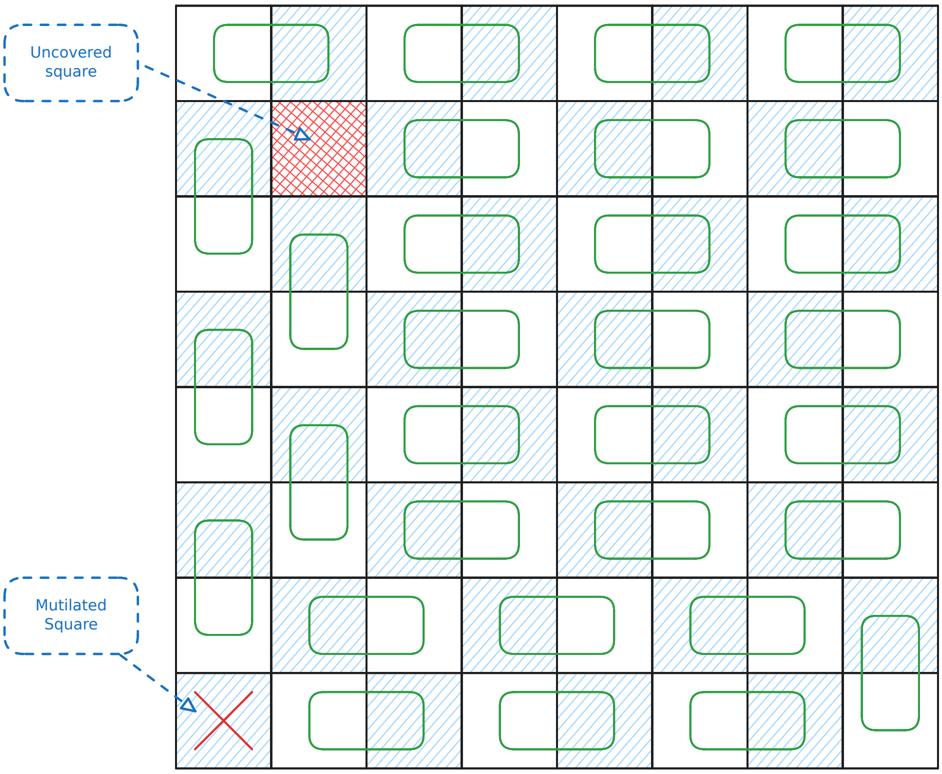


Figure 1.2.5: Example on an 8×8 chessboard with a missing square. As we can observe, there will always be one square we cannot tile

Lastly, we will introduce *Tarski* 1.2.15 which uses the Knaster-Tarski fixed point theorem 1.2.1 and belongs in **CLS**, the class that convers the intersection of parity graphs (**PPAD**) and local search (**PLS**) [45, 35, 23].

Definition 1.2.14: Monotone functions

Given two partially ordered sets $(L_1, \preceq_{L_1})$ and $(L_2, \preceq_{L_2})$, a function $f : L_1 \to L_2$ is **monotone** if and only if:

$$\forall x, y \in L_1 : x \preceq_{L_1} y \implies f(x) \preceq_{L_2} f(y)$$

Theorem 1.2.1: Knaster Tarski Fixed point theorem[7, 25]

Given a *monotone* function $f : L \to L$ 1.2.14 of a complete lattice $(L, \wedge, \vee)$, $\exists c \in L : f(c) = c$.

Definition 1.2.15: *Tarski* problem definition [25]

Given a *monotone* function $f : L \to L$ 1.2.14 of a lattice $(L, \wedge, \vee)$ we define solutions to the problem as:

1. Find $x \in L : f(x) = x$
2. Find $x, y \in L$ such that $x \preceq y$ and $f(x) \npreceq f(y)$

We assume that f is represented by boolean circuits.

The above argues the existence of either finding points that are maximum in the partial order chain, or points that violate the monotonicity constraint.

1.2.2 Counting Complexity and Combinatorial Interpretations

Search problems as we observed earlier ask what is a valid solution given the current input instance like what is a valid colouring of a graph, satisfying assignment to a 3-SAT problem etc. Counting complexity, or counting problems take it a step further and usually ask how many valid solutions does a problem have. Valiant formalised this idea with the help of **#P** complexity class 1.2.2, in which he demonstrated that even if a problem is polynomially solvable in its decision/search variant, can be much harder. A much simpler example can be found in the book by Barak et al. [3] where they provide a reduction from *Ham-Cycle* to *#Cycle*.

Definition 1.2.16: #P Complexity Class [51]

A function $f : \mathbb{B}^* \to \mathbb{N} \in$ **#P**, if there exists a poly-function $p : \mathbb{N} \to \mathbb{N}$ and a poly-time TM M such that:

$$f(w) = \left|\left\{v \in \mathbb{B}^{p(|w|)} \mid M(w, v) = 1\right\}\right|$$

Alternative but equivalent definition is as follows: A function $f \in$ **#P**, if there exists a non-deterministic poly-time TM N such that:

$$f(w) = \mathbf{\#acc}_N(w)$$

Where $\mathbf{\#acc}_N(w)$ denotes the number of accepting paths of N on w.

In hindsight, we can observe that functions in **#P** express functions of the form $\forall w \in \{0,1\}^* : f(w) = |A_w|$, where A_w is a set of verifiers or accepting paths of size polynomial to the input. Given the above definition, Pak et al. [43, 34] used **#P** to decide whether a class of numbers or objects have or do not have a combinatorial interpretation. Moreover, they argue that the usage of **#P** has several benefits:

1. By polynomially bounding objects, we avoid cases such as: $f(w) = |\{1, \ldots, f(w)\}|$. In other words, we create a set of combinatorial objects.
2. We can work with $f(\cdot)$ even if its direct computation is hard. For example, counting the number of cycles in a graph is believed to be an **NP**-complete problem but verifying individual cycles is trivial.
3. **#P** is a formal system that is highly flexible. Specifically, we are able to use results in complexity theory and NTMs to find interpretations or prove their non-existence.

Lastly, we introduce the idea of correlating two counting problems with the help of *parsimonious reductions*. Simply, we say that a problem A can be parsimoniously reduced to a problem B, if for inputs of A, one can create instances of B such that the set of solutions between the two problems have the same cardinality. More formally we refer to the defintion 1.2.17. An example

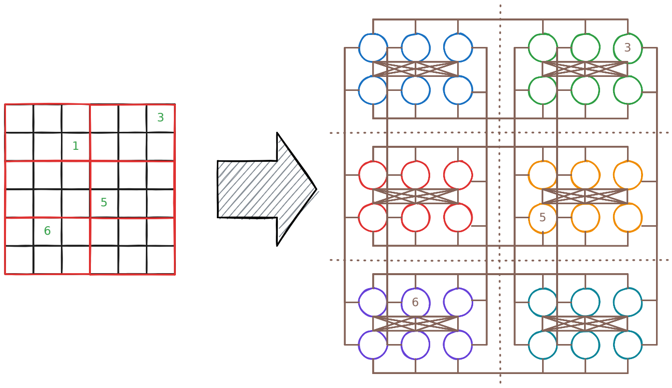


Figure 1.2.6: Example of reducing a 6x6 sudoku puzzle, into a graph colouring problem

of how such reduction may look like, can be seen in figure 1.2.6, where we relate the difficulty of sudoku solving to a graph colouring problem.

Definition 1.2.17: Parsimonious reductions

Let R, R' be search problems, and let f be a reduction of $S_R = \{x \mid R(x) \neq \emptyset\}$ to $S_{R'} = \{x \mid R'(x) \neq \emptyset\}$. We say f is **parsimonious** if:

$$\forall x \in S_R : |R(x)| = |R'(f(x))|$$

Where for relation R we say $R(x) \triangleq \{y \in \{0,1\}^* \mid xRy\}$

1.2.3 Kleene Logic and Hazard-Free Circuits

Kleene logic uses the boolean system with an additional *unstable* value [37]. The study of Kleene logic assisted in the development of robust physical systems [26] and aiding in the development of lower bounds for monotone circuits [22, 30, 31, 8]. The current section offers a brief summary to relevant notions and concepts in Kleene logic.

Definition 1.2.18: Kleene value ordering [42]

The *instability ordering* or *information ordering* for $\preccurlyeq^u$ is defined as: $\perp \preccurlyeq^u 0, 1$. The n-dimensional extension of the ordering is:

$$\forall x, y \in \mathbb{T}^n : x \preccurlyeq^u y \implies \forall j \in [n] : (x_j \in \mathbb{B} \implies x_i = y_i)$$

We can also refer to an alternative type of ordering, known as *voltage* ordering, such that:

$$0 \preccurlyeq^v \perp \preccurlyeq^v 1$$

Definition 1.2.19: Kleene Resolutions [42, 30]

Given $x \in \mathbb{T}^n$, we define the *resolutions* of x, using the following notation:

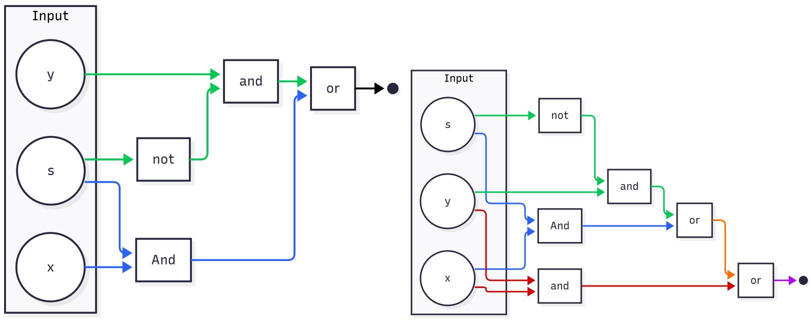
$$\textsc{Res}(x) \triangleq \left\{y \in \mathbb{B}^n \mid x \preccurlyeq^u y\right\}$$

Definition 1.2.20: Natural functions [30]

A function $F : \mathbb{T}^n \to \mathbb{T}^m$ for some $n, m \in \mathbb{N}$ is *natural* if and only if it satisfies the following properties:

1. *Preserves stable values*: $\forall x \in \mathbb{B}^n : F(x) \in \mathbb{B}^m$
2. *Preserves monotonicity* 1.2.14 1.2.3

Unless otherwise specified, we assume that all Kleene functions are *natural* 1.2.20, since they can be represented by circuits 1.2.1. Natural functions essentially indicate that a point with its resolution should respect the information order, or more simply if $f(00) = 1$ then $f(0\bot) \neq 0$ as then it will not preserve stability. Moreover we can think resolutions of a point x as all the values it "could be". For example resolutions of $0\bot$ could be either 00 or 01. Naturally, if $f(00) = f(01) = 1$ then naturally we should also expect that $f(0\bot) = 1$. This yields the concept of hazards or points in $x \in \mathbb{T}^n$ where, its resolutions have the same value but the point itself does not more formally defined in 1.2.3. Detecting hazards is a concept that is analysed heavily when talking about Kleene logic and it is an **NP**-hard problem [30]. An example of hazard values can be seen in the figure 1.2.7.



(a) Multiplexer with hazard at $(1, 1, \bot)$ (b) Hazard free multiplexer

Figure 1.2.7: Hazard circuit and hazard-free circuit. Figure by [30]

Proposition 1.2.1: Natural functions and Circuits [42, 30]

A function $F : \mathbb{T}^n \to \mathbb{T}^m$ can be computed by a circuit *iff* F is *natural* 1.2.20

Definition 1.2.21: Hazard [30, 22]

A *circuit* C, on n inputs has **hazard** at $x \in \mathbb{T}^n \iff C(x) = \bot$ and $\exists b \in \mathbb{B}, \forall r \in \textsc{Res}(x) : C(r) = b$. If such value does not exists then we say that C is hazard-free.

Definition 1.2.22: K-bit Hazard [30]

For $k \in \mathbb{N}$ a circuit $C : \mathbb{T}^n \to \mathbb{T}$ has a *k-bit hazard* at $x \in \mathbb{T}^n \iff C$ has a hazard at x and $\bot$ appears at most k times.

We also wish to introduce the following corollary by Ikenmeyer et al. [30] which allows us to construct k-bit hazard free circuits

Corollary 1.1: K-bit Hazard Construction [30]

Given circuit $C : \mathbb{T}^n \to \mathbb{T}$, one can construct a k-bit hazard free circuit of C denoted as C' such that:

$$\begin{aligned} size(C') &= \left(\frac{ne}{k}\right)^k (size(C) + 6) + O(n^{2.71k}) \\ depth(C') &= depth(C) + 8k + O(k \log n) \end{aligned}$$

Where $size(P)$ denotes the number of gates of P and $depth(P)$ denotes the longest path from the input bit to the output bit

Kleene logic and Complexity Theory

Lastly we want to introduce the notion of *Alternating Turing Machines*, which are meant to represent a generalisation of non-deterministic Turing machines. These have been introduced by Chandra et al. [11] and the idea is that we introduce universal and existential quantifiers to alternate the course of computation. The informal presentation can be thought as follows: In a non-deterministic TM, given a single state q or configuration c, we execute $\beta_1, ..., \beta_k$ separate processes. We say that q will accept if at least one of the processes $\{\beta_i\}_{i\in[k]}$ accepts. Alternating Turing machines introduce additional conditions where one impose the condition that all subprocesses $\{\beta_i\}_{i\in[k]}$ must accept accept or in case of a negation state, if β_1 accepts, then q rejects or vice versa. More formally, we offer the definition in 1.2.3.

Definition 1.2.23: Alteranting Turing Machines(ATM) [38, 11]

An alternating TM is a seven tuple

$$M = (k, Q, \Sigma, \Gamma, \delta, q_0, g)$$

Where each term denotes:

1. $k \in \mathbb{N}$: The number of work tapes
2. Q: Finite set of states
3. Σ: Finite input alphabet. Moreover, $\star \notin \Sigma$ denotes the end of the input
4. Γ: Finite work tape alphabet. Moreover $\# \in \Gamma$ to denote blank symbol.
5. δ transition relation which is defined as:

$$\delta \subseteq \Big(Q \times \Gamma^k \times (\Sigma \cup \{\star\})\Big) \times \Big(Q \times (\Gamma \setminus \{\#\})^k \times \{L, R\}^{k+1}\Big)$$

   Important to observe that for a single state, input symbol and work tape symbols, we can take multiple types of transitions.

6. $q_0 \in Q$: Is the initial state
7. $g : Q \to \{\vee, \wedge, \neg, \mathbf{1}, \mathbf{0}\}$: Where g denotes the type of state and $\mathbf{1}, \mathbf{0}$ denote whether it is accepting or not.

The machine essentially contains a read-only input tape with endmarkers and k work tapes which are initially blank. A *step* of M consists of reading a symbol from the input tape, writing a symbol to each work tape, moving the writing head in each head left or right while transitioning into a new state.

Definition 1.2.24: ATM configurations [38, 11]

A configuration $\mathcal{C}$ of an ATM M is a element of the set $\mathcal{C}_M$

$$\mathcal{C}_M \triangleq Q \times \Sigma^* \times (\Gamma \setminus \{\#\})^k \times \mathbb{N}^{k+1}$$

Which represent, the current state, the input, the current contents of each work tape, and the $k+1$ head positions.

We say that $\mathcal{C}$ that *configuration* is universal (existential, negating, accepting or rejecting) if q is also universal (existential, negating, accepting or rejecting). The initial configuration of M on input x is

$$\sigma_M(x) = (q_0, x, \lambda^k, 0^{k+1})$$

Where λ is a null string.

Definition 1.2.25: Successor of configurations [11]

Given ATM M, we write $\mathcal{C} \vdash \mathcal{C}'$ to say that $\mathcal{C}'$ is the successor of $\mathcal{C}$, if we can go from one configuration to another with a single δ rule. We require for δ that, for accepting or rejecting configurations to have no successors, for universal or existential to have at least one successor configuration, and for negating configurations to have only one. We can define the **computation** of a path on x as a sequence $a_0 \vdash a_1 \vdash ... \vdash a_n$ where $a_0 = \sigma_M(x)$

Denoting Accepting or Rejecting computations As mentioned previously, we start from the original configuration. If a process is in an existential configuration $\mathcal{C}$ and $\beta_1, ..., \beta_m$ are all possible successors, then we spawn a process for each configuration and run each one concurrently. If at least one process is accepting we report back to the parent and terminate the rest. Same line of reasoning can be done with the universal quantifiers. Ideally we would like a labelling function f which is defined as

$$f(\mathcal{C}) = \begin{cases} \bigwedge_{\mathcal{C}\vdash\beta} f(\beta) & \text{if } \mathcal{C} \text{ is universal} \\ \bigvee_{\mathcal{C}\vdash\beta} f(\beta) & \text{if } \mathcal{C} \text{ is existential} \\ \neg f(\beta) & \text{where } \mathcal{C} \vdash \beta \text{ if } \mathcal{C} \text{ is negating} \\ \mathbf{1} & \text{if } \mathcal{C} \text{ is accepting} \\ \mathbf{0} & \text{if } \mathcal{C} \text{ is rejecting} \end{cases}$$

But some processes may never halt or require a much greater computation than necessary. We therefore use Kleene logic $\mathbb{T}$ where $\perp$ denotes configurations that we do not know if they are accepting or not. The quantifiers use the same logic as described in the tables 1.2.1. We now say that a labelling of configurations is a map $\ell : C_M \to \{0, \perp, 1\}$. We define an operator τ to be an operator of mappings such that:

$$\tau(\ell)(\mathcal{C}) = \begin{cases} \bigwedge_{\mathcal{C}\vdash\beta} \ell(\beta) & \text{if } \mathcal{C} \text{ is universal} \\ \bigvee_{\mathcal{C}\vdash\beta} \ell(\beta) & \text{if } \mathcal{C} \text{ is existential} \\ \ell(\beta) & \text{where } \mathcal{C} \vdash \beta \text{ if } \mathcal{C} \text{ is negating} \\ \mathbf{1} & \text{if } \mathcal{C} \text{ is accepting} \\ \mathbf{0} & \text{if } \mathcal{C} \text{ is rejecting} \end{cases}$$

We can check that τ is monotone with respect to $\leq$ since if $l \leq l'$, then $\tau(l) \leq \tau(l')$. By 1.2.1 there must exist maximum labelling such that

$$\ell^* = \sup_{m \in \mathbb{N}} \tau^m(\Omega)$$

where the supremum is with respect to $\preceq$, where

$$\tau^0(\ell) = \ell$$
$$\tau^{m+1}(\ell) = \tau(\tau^m(\ell))$$

where Ω is denoted as the minimal labelling function $\lambda x.\perp$. In a way that the above procedure describes is the "propagation" process from the leaf nodes up to the root of the tree where the leaf nodes are assigned a label of $\perp$ or accepting. Lastly given the above we can finally give a definition of halting states

Definition 1.2.26: Halting states[11]

We say that M halts on x iff $\lambda^*(\sigma_M(x)) \in \{0,1\}$ and it accepts or rejects respectively iff $\lambda^*(\sigma_M(x)) = 1$ or $\lambda^*(\sigma_M(x)) = 0$

We aim for this ATM to work for only recursively enumerable sets and therefore we aim to limit to the depth of the recursion. Therefore, we define operator τ_C where $C \subseteq C_M$ such that

$$\tau_C(\ell)(a) \triangleq \begin{cases} \tau(\ell)(a) & \text{if } a \in C \\ \perp & \text{otherwise} \end{cases}$$

Where we can say that C is the set of configuration that are reachable within some $f(|x|)$ steps or require space of at most $f(|x|)$. An example of a computation tree can be observed in the figure 1.2.8.

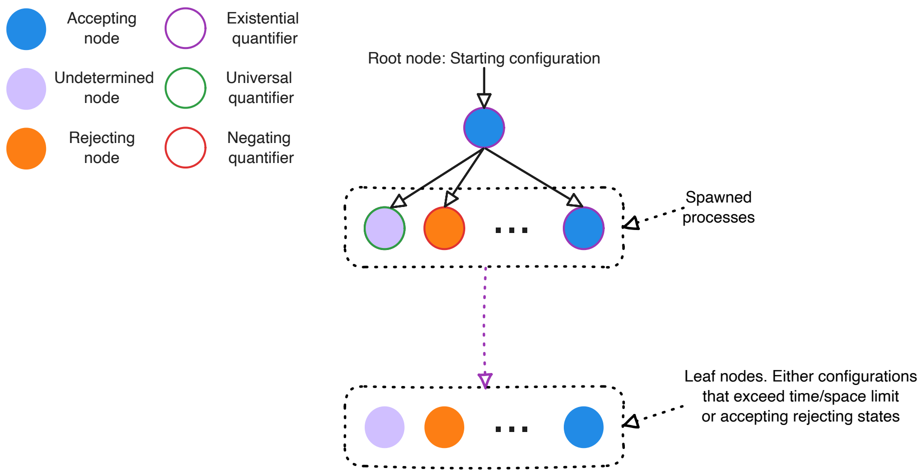


Figure 1.2.8: Computation tree of an ATM with labellings

We can observe that one can encode Kleene logic into the computation of Turing machines. We use this framework as one of the main motivations of the project.

1.2.4 Bridging Counting Complexity, Search problems and Kleene Logic

Our work is primarily focused around the paper by Ikenmeyer et al. [33] and the parsimonious reductions discovered across various **PPAD** problems. We introduce the notation **#PPAD**(L), where this indicates the class of problems which are parsimoniously reducible to L. In fact, we extend this notation with $f($**#PPAD**$(L))$ to indicate reductions of the form, given L', we say that $L' \in f($**#PPAD**$(L))$ if

$$\forall x \in S_{L'} : |R_{L'}(x)| = f(|R_L(g(x))|)$$

We demonstrate an example of the above notation using a finding from the paper by Ikenmeyer et al. [33]: **#PPAD**$(\mathit{EndOfLine} - 1)/2 =$ **#P**. To show inclusion it is easy to see that one can accept all the non-zero sources, since each source is paired with a sink, and we are able to ignore the sink paired with the 0^n source. To show the other direction, they demonstrate an example where they reduce $\#\mathit{CircuitSAT} \subseteq ((\#\mathit{EndOfLine}) - 1)/2$ by creating a graph, where for each input x, we create a source and sink pair $(0x, 1x)$ if and only if $f(x) = 1$.

In the paper by, Ikenmeyer et al. [33], they developed tools to demonstrate how one can show whether a problem is in **#P** or not, with the help of relativising techniques. The idea of relativisation is that we can equip our circuits or TMs with calls to arbitrary oracles. They demonstrated that for several problems in **TFNP** as seen in figure 1.2.1, some can be parsimoniously reduce to each other under all oracles, where as for some directions (the ones with the warning sign) this is not the the case. From the figure we can observe that for seveal **PPAD**-complete, **CLS** and **PPADS** problems one can indeed find parsimonious redutions. For other problems, even **PPAD**-complete ones like **#PPAD**$(\mathit{SourceOrExcess}(k, 1)) - 1$, this does not seem to be the case.

1.3 Project Question and Objectives

As mentioned in the previous section, we observed how different **PPAD** problems can behave differently. The main motivation comes from the observation of how *Kleene logic* can be used to simulate the computation paths of NTMs. We can parallelise this to the boolean problem of *SAT* which given by the *Cook-Levin* theorem [15] one can convert every **NP** problem into a *SAT* problem. The general idea is to create a boolean expression for each snapshot of the Turing Machine such that a satisfying assingment corresponds to an accepting path. As corollary the *Cook-Levin* also demonstrates that every **NP** problem can be parsimoniously reduced to the *SAT* problem, meaning:

$$\forall L \in \mathbf{NP} : \#L \subseteq \#\mathit{SAT}$$

A reasonable conjecture would also be if one can exploit Kleene-circuit nature of *PureCircuit* to create a *Cook-Levin* type equivalent reduction for **PPAD** problems, or more formally:

$$\forall L \in \mathbf{PPAD} : \mathbf{\#PPAD}(L) \subseteq \mathbf{\#PPAD}(\mathit{PureCircuit})$$

If the above statement were indeed true, then it would give us insights as to how counting problems behave in **PPAD** $- 1$. Unfortunately, there is limited research with respect to the combinatorial nature of the problem. We wish to offer a new perspective of *PureCircuit* and establish its *parsimonious relations* with other problems. To break down our conjecture we offer the following set of objectives:

1. Establish a chain of parsimonious reductions from the *EndOfLine* to the *PureCircuit* problem.
2. Establish a chain of parsimonious reductions from the $\mathit{SourceOrExcess}(2, 1)$ to the *PureCircuit* problem. Ideally, we would like to extend this to the entire hierarchy of $\mathit{SourceOrExcess}(k, 1)$ problems for some $k \in \mathbb{N}$ or ideally. We will introduce these problems more formally in the upcoming sections *PolySourceOrExcess*.
3. Establish a *Cook-Levin* type theorem for the all **PPAD** problems or more formally

$$\forall L \in \mathbf{PPAD} : \mathbf{\#PPAD}(L) \subseteq \mathbf{\#PPAD}(\mathit{PureCircuit})$$

The above milestones were created in mind to tackle the main conjecture in incremental steps. Demonstrating a parsimonious reduction from the *EndOfLine* is an important baseline as every **PPAD** problems reduces to the *EndOfLine* even if its not parsimoniously. The second milestone is meant to demonstrate that we can use *PureCircuit* for problems with no combinatorial interpretations. Our last objective is the overall main conjecture that we are trying to prove.

Our main findings sections focuses as to how we tackled each problem and the efforts we did. Even though, the project did not manage to yield any positive results, we provides several tools, new problems and conjectures that may help in future research. To assist with the theory development, we set out to develop a visualisation tool that will be able to visualise instances of the problem as well as find solutions or enumerate through solutions for small instances. We therefroe provide this additional list of software deliverables:

1. Establish a program that in which the user is able to construct a *PureCircuit* instance. The tool should be capable of verifying instance or highlighting unsatisfied nodes.

2. Establish algorithms to find solutions or at least approxiamte solutions.
3. Establish algorithms to enumerate solutions for small graphs.

We will a give a greater in-depth overview of the tool in the section 3 and the algorithms used to fulfil the aforementioned requirements.

Chapter 2

Theory Analysis

In this section, we explore on the approaches and findings we uncovered while tackling each of our theory objectives. More specifically, while we have not achieved a complete proof for each objective, we highlight several theorems that may assist in their full resolution. The core idea that we employed comes from following set of observations: when working with *PureCircuit*, one cannot fully work with binary logic, meaning there is no trivial method for detecting $\perp$ values and removing them, as made clear by Marino's continuity argument [40]. We therefore make use of topological problems with the idea that resolutions of our input corresponds to some set nearby points in a space. If all points abide some property then we can use that to detect solutions. This is an extension of the non-parsimonious reduction used by Deligkas et al. [21] when showing **PPAD**-hardness.

With respect to the first objective, we provide the following list of theorems.

1. *#nD-StrongSperner* $\subseteq$ *#HF-nD-StrongSperner* 2.1.1
2. *#Hf-DiscreteStrongSperner* $\subseteq$ *#PureCircuit* 2.1.2
3. *#EndOfLine* $\subseteq$ *AntipodalStrongSperner* 2.1.3

The problems mentioned are defined in the section 2.1, where we also discuss their implications with respect to the current objective and with **PPAD** as a whole

Following the same thought process for our second objective, we demonstated how to reduce *SourceOrExcess* to a topological problem. Specifically in section 2.2 we argue the following result

$$\#\textit{SourceOrExcess}(2,1) \subseteq \#\textit{3D-CountingStrongSperner} \ 2.2.1$$

Where we define a different combinatorial version of the *3D-StrongSperner* and demonstrate how it can be used for parsiomonious reductions between the *SourceOrExcess*$(2,1)$ problem.

In the third section, we offer an combinatorial extension to the **PPAD** class with regards to directed graphs and the number of known sources. Specifically given Hollender et al.'s work [29] on directed graphs with multiple known sources, we make on observations as to the combinatorial nature of the problems that can be created as well as important problems to analyse for a full *PureCircuti* Cook-Levin type theorem.

Lastly we provide two supplementary findings with regards to the *PureCircuit* reductions. We first demonstrate how we can relate to *Tarski* with *Kleene theory* and show how to create parsimonious reductions using *PureCircuit*. Second, we show how simplifications of the *PureCircuit* problem can lead to parsimonious reductions to the *EndOfLine*. Since these proofs to not offer any indication of the hardness of the problem, we provide their full proofs in the appendix.

2.1 PureCircuit and EndOfLine

As mentioned previously, the current project focuses on using topological expressions for reductions between problems. We built upon the *StrongSperner* idea that Deligkas used by expressing how we can redefine the problems under *Kleene logic* [21]. We then demonstrate how we can relate *PureCircuit* and the *EndOfLine* problem to each of them as well as highlighting the current limitations of these approaches.

Hazard Free StrongSperner

First we begin with the idea that solutions of *StrongSperner* problems revolve around the notion of panchromatic hypercubes. We can use the idea and make the following definitions

Definition 2.1.1: *StrongSperner* anchor points

Given a hypercube $H \subset \mathbb{Z}^n$ of side lengths 1, we denote the $\textbf{Anc}(H)$ as

$$\textbf{Anc}(H) = \min_{p \in H} \|p\|_1$$

Alternatively, we will refer to these hypercubes using the notation $\textbf{Cube}(\cdot)$ such that

$$\textbf{Cube}(p) \triangleq \{(p+b) \mid b \in \{0,1\}^n\}$$

Ideally, we would like to express a cube as a coordinate of Kleene numbers, implying the resolutions of the point will result in a hypercube, where each point is within $\|\cdot\|_\infty \leqslant 1$ of each other. This leads to a natural interpretation of the *StrongSperner* problem in *Kleene logic*. We will start with two definitions, one for *nD-StrongSperner* and the other for *StrongSperner*. We remind that the main difference of the two problems is the former has fixed dimensions n of exponentially large cubes, whereas the other can have varying dimensions of polynomially large widths.

Definition 2.1.2: *HF-nD-StrongSperner* problem

Input: A circuit $\lambda : (\{0,1\}^k)^n \to \{-1,+1\}^n$, that describes a *nD-StrongSperner* labelling, as explained in 1.2.1, such that it is hazard free on inputs $(p_i\bot)_{i\in[n]}$ where $p_i \in \{0,1\}^{k-1}$
Output: A point $p \in \bigtimes_{i\in[n]} \left(\{0,1\}^{k-1} \times \{0,1,\bot\}\right)$ if and only if: $\lambda(p) = \bot^n$.

We argue that the above problem is **PPAD**-complete for all *HF-nD-StrongSperner* problems. First we can observe that all *HF-nD-SS* are reducible to their *nD-SS* counterparts since hazard-free transformations have no effect on binary inputs and therefore our circuits acts like any other boolean circuit. To show the hardness, we claim that

Theorem 2.1.1: *HF-nD-SS* hardness

$$\textbf{\#PPAD}(nD\text{-}SS) \subseteq \textbf{\#PPAD}(HF\text{-}nD\text{-}SS)$$

Proof. Given circuit $\lambda : (\{0,1\}^k)^n \to \{-1,+1\}^n$ from *nD-SS*, assume that S is the set of anchors points of a panchromatic cube. We create $\Lambda : (\{0,1\}^{k+1})^n \to \{-1,+1\}^n$ such that:

$$\Lambda((x_i)_{i\in[n]}) = \lambda\left(\left(\left\lfloor \frac{x_i+1}{2} \right\rfloor\right)_{i\in[n]}\right) \tag{2.1.1}$$

We prove the following claim

Claim 2.1.1: Transformation claim

Given S_{even} the set of even solutions of the transformed space or more formally:

$$S_{\text{even}} \triangleq \left\{(p_i 0)_{i\in[n]} \mid p \in (\mathbb{B}^{(k)})^n : (p_i 0)_{i\in[n]} \text{ is an anchor of a panchromatic cube}\right\}$$

We claim $|S| = |S_{\text{even}}|$

Proof. We argue that we can bijectively map between $S \leftrightarrow S_{\text{even}}$. To show the left to right direction, assume $p \in S$. Using figure 2.1.1 as reference, we can observe that each point becomes an n-dimensional hypercube. Essentially the hypercube that was anchored at $(p_1, ..., p_n)$ is mapped to the hypercube anchored at $(2p_1, ..., 2p_n)$. To prove that, we can observe that:

$$\begin{aligned}\forall (2p_i)_{i\in[n]} \in (\{0,1\}^{k+1})^n, b \in \{0,1\}^n : \Lambda((2p_i + b_i)_{i\in[n]}) \\ &= \lambda\left(\left(\left\lfloor\frac{2p_i + b_i + 1}{2}\right\rfloor\right)_{i\in[n]}\right) \\ &= \lambda\left(\left(\left\lfloor p_i + \frac{b_i + 1}{2}\right\rfloor\right)_{i\in[n]}\right) \\ &= \lambda\left((p_i + b_i)_{i\in[n]}\right)\end{aligned}$$

Which implies that the cube anchored at p must also be panchromatic. □

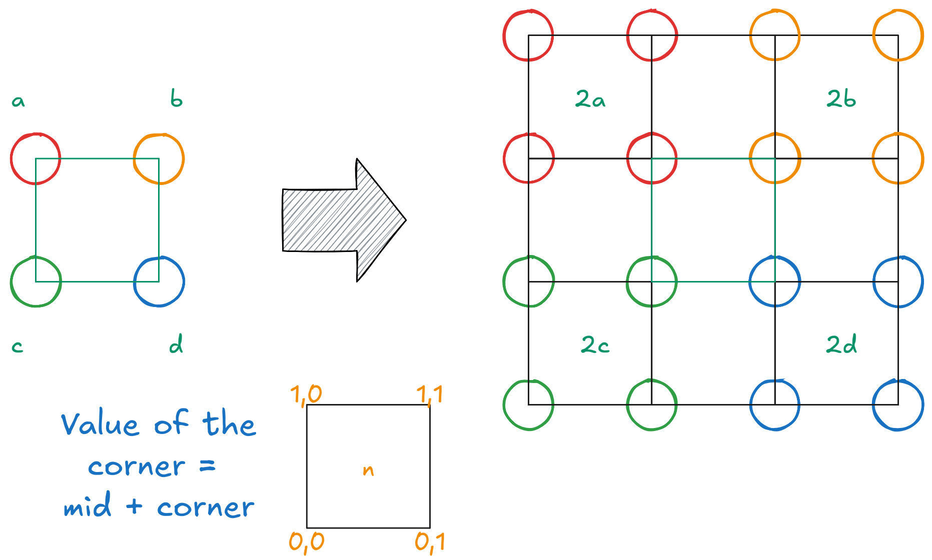


Figure 2.1.1: Dimension doubling. We demonstrate the example in two dimensions.

To ensure that Λ is hazard free we can use the corollary **??**, where we can create an n-bit hazard-free circuit. One must observe that even though, the size of the circuit increases exponentially with respect to the dimensions, but since all the input instances scale with respect to the width of the hypercube space, that would imply that we are able to keep the size of the circuit polynomial. We therefore claim that this reduction is polynomial and can parsimoniously find all panchromatic cubes. □

To introduce *HF-StrongSperner*, we will first introduce how to work with kleene unary 2.1 numbers. We will then demonstrate how to extend this notation to define the full *HF-StrongSperner* problem.

Definition 2.1.3: Kleene Unary Numbers

Given a number $p \in M$, its unary representation can be depicted as a binary number $\{0,1\}^M$ such that: $k \in M \rightarrow 1^k \in \bar{M}$. Under Kleene algebra, we mainly refer to notations M_k^n where $k, n, M \in \mathbb{N}_0$ such that:

$$M_k^n \triangleq \bigtimes_{i \in [n]} \{1^j \perp^k \mid \forall j \in \mathbb{N}_0 : j + k \leqslant M\}$$

For a cleaner notation, given $1^j \perp^k \in M_k$ we denote it as $j \perp^k$.

For examples of the above notation we refer to the following table:

Kleene Unary Sets	Member examples
3_1^2	$(11\perp, 1\perp 0) = (2\perp, 1\perp)$
5_0^3	$(11000, 11110, 11111) = (2, 4, 5)$
7_2^1	$11\perp\perp 000 = 2\perp^2$

Table 2.1.1: Kleene Unary Examples

Definition 2.1.4: *HF-StrongSperner* problem

Input: A hazard-free circuit $\lambda : [M]^n \rightarrow \{-1, +1\}^n$, where $M \in n^{O(1)}$ that describes a *StrongSperner* labelling, as explained in 1.2.1, where the inputs are represented using Kleene Unary numbers 2.1.
Output: A point $p \in M_{\leqslant 1}^n$ if and only if: $\lambda(p) = \perp^n$.

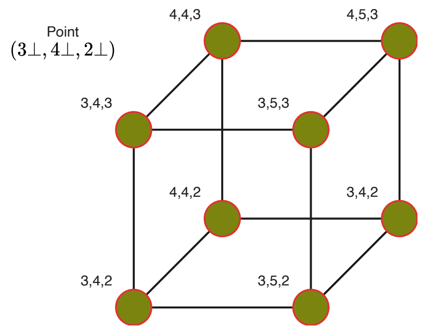


Figure 2.1.2: Kleene Hypercube using unary notation

We will mainly focus on subproblems of the *HF-StrongSperner* where two cubes do not overlap with each other. We will use figure 2.1.3 for reference, but essentially if two cubes overlap, that means a subspace of the cube (face or edge) is panchromatic. This means that for the same cube, we can have multiple solutions, such as the cube it self, the face that contains the panchromatic edge or the edge it self. To simplify the counting arguments, we define the problem *HF-DiscreteStrongSperner* 2.1.5, such that all solutions are *discrete*, meaning there are no overlapping panchormatic cubes,

Definition 2.1.5: *HF-DiscreteStrongSperner* problem

An *HF-StrongSperner* instance λ such that $\forall p \in M_1^n$, if $\lambda(p) = \perp^n$, then

$$\forall y \in M_{\leqslant 1}^n : p \preceq y \implies \lambda(y) \neq \perp^n$$

Essentially, any subspace of the cube should not yield a panchromatic solution.

Lastly we introduce a specific subset of the above solutions where we enforce antipodal panchromatic cubes. In the specific type of solution, we argue that every two opposite corners of a cube contain opposite colourings. More formally we say

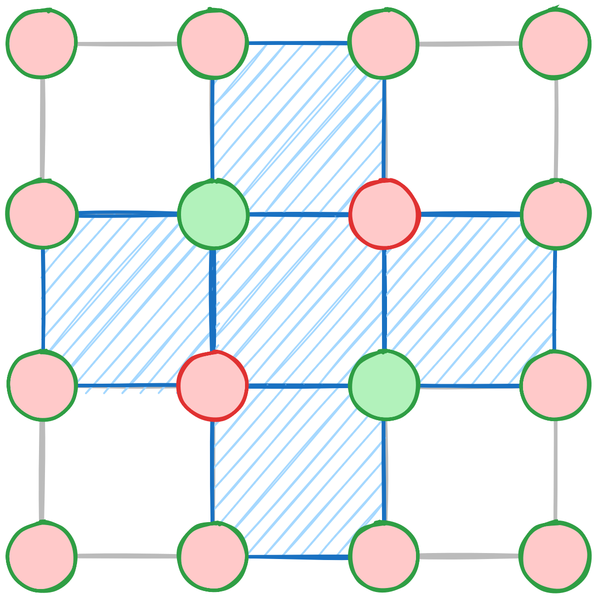


Figure 2.1.3: Overlapping Sperner Solutions

Definition 2.1.6: MinRes min and max resolution functions

We use the **MinRes**, to denote the smallest resolution of a kleene string. More formally we can use the following equivalent definitions:

$$\forall x \in \{0,1,\perp\}^*, j \in [|x|] : [\mathbf{MinRes}(x)]_j \triangleq \begin{cases} x_i & \text{if } x_i \in \{0,1\} \\ 0 & \text{otherwise} \end{cases}$$

For the maximum resolution we will use the **MaxRes** function where we instead replace all $\perp$ with 1 instead.

Definition 2.1.7: *HF-AntipodalStrongSperner* problem

An *HF-DiscreteStrongSperner* instance λ such that $\forall p \in M_1^n$, if $\lambda(p) = \perp^n$, and denote $\mathbf{MinRes}(x) = \hat{x}$ as the *anchor* of the cube, then:

$$\forall b \in \{0,1\}^n, j \in [n] : [\lambda(\hat{x} + b)]_j = -[\lambda(\hat{x} + 1^n \oplus b)]_j$$

We emphasize that the above class considers the subset of the *Hf-DiscreteStrongSperner* as the antipodality condition cannot ensure discreteness. We can also define a non-HF variant of the problem which we can refer to *AntipodalStrongSperner* where for each cube, we only need to check its two opposite ends to verify if its a valid solution.

Claim 2.1.2: Antipodal discrete squares

Given a panchromatic solution of a *HF-AntipodalStrongSperner* p, it must be the case then that:

$$\forall j \in \{-1,1\}^n, \exists! x \in \mathbf{Res}(p) : \lambda(x) = j$$

We essentially claim that a panchromatic solution of a *HF-AntipodalStrongSperner* contains all 2^d colours.

Proof. We are going to assume proof by contradiction. Assume there exists colour j that is duplicated on two points $p_1 = p + b_1, p_2 = p + b_2$ such that p is the anchor of a panrchomatic solution and $b_1, b_2 \in \mathbb{B}^n$. Since the cube is antipodal, that means that we have to be wary of $\bar{p}_2$ which is at the position $p + \overline{b_2}$. Since the cube in discrete, it implies that $\bar{p}_2$ and p_1 cannot belong in any $(n-1)$-subspace, otherwise our cube won't be discrete. But that would imply that $\forall i \in [n] : |[\bar{p}_2 - p_1]_i| = 1$. We know the only position in the hypercube that satisfies that is $\bar{p}_1$ which implies $p_1 = p_2$. Therefore every corner of an antipodal discrete cube must contain a different colour. □

All the problems above are clearly in **PPAD** as all of them reduce to the *StrongSperner* problem, since we assume by construction that they *natural* and therefore behave as normal binary circuits. For the purposes of our current research, we demonstrate a parsimonious reduction from *HF-DiscreteStrongSperner* to *PureCircuit*.

Hazard Free StrongSperner To PureCircuit

Theorem 2.1.2

$$\mathbf{\#PPAD}(\textit{HF-DiscreteStrongSperner}) \subseteq \mathbf{\#PPAD}(\textit{PureCircuit})$$

Proof. Given an input instance Λ of a *HF-DiscreteStrongSperner*, we create a *PureCircuit* instance which we break down into the following parts: *Input nodes*, *Output nodes*, *Bit generator gadget* and *Circuit application gadget*. To visualise our construction we will refer to the figure in 2.1.4. Each of these parts describe the following:

1. **Input nodes**: Vector of n value nodes which will be represented as s_i for element i.
2. **Bit generator gadget**: Given a value node s_i, apply the *Bit generator* gadget such that we create Kleene unary number as defined in 2.1. We will refer to the bit generator circuit as C_i and the Kleene unary numbers as (u^i_j) for $i \in [n]$ and $j \in [M]$.
3. **Circuit application gadget**: Since the above numbers form coordinates in a $M^n_{\leqslant 1}$ space, we apply the Λ function to extract a set of output nodes $(o^i)_{i \in [n]}$.
4. **Output nodes**: We use the Copy gate to copy $o^i \to s_i$, for all $i \in [n]$.

Bit generator gadget In order to create Kleene unary numbers we make use of a binary tree of Purify gates as seen in the figure 2.1.5. We now claim that the current construction has certain desirable properties.

Claim 2.1.3: Purification Claim

Given the description of the bit generator gadget, we make the following two claims:

1. $b \in \{0,1\}^n : C_i(b) = b^n$
2. $C_i(\bot) \in M_n^{\leqslant 1}$

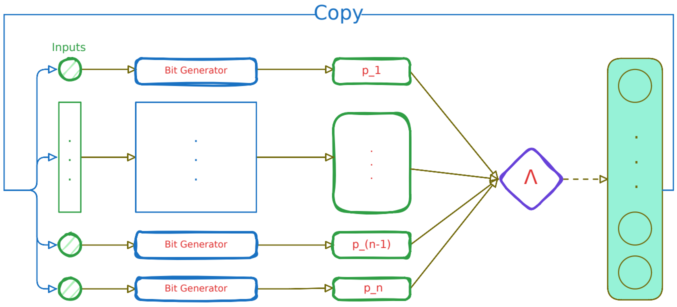


Figure 2.1.4: Pure Circuit construction visualisation

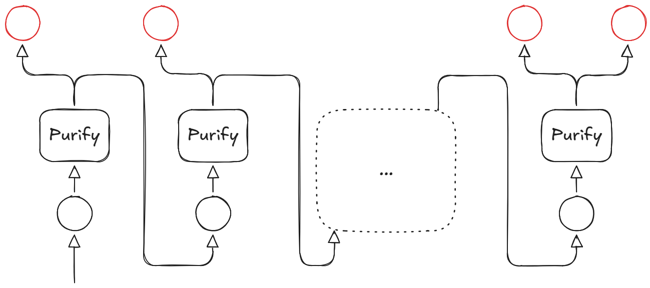


Figure 2.1.5: Purification gadget. The red nodes denote the outputs of the purification gadget.

Proof. From the definition of the PURIFY gate we know that $\forall b \in \{0,1\}$: PURIFY$(b) = (b,b)$, and therefore any subtree that takes a pure input, will copy its value to all its leafs, which follows our first claim directly. To the second part of our lemma, we can observe that the only valid outputs are $(0,\perp),(\perp,1),(0,1)$. Based on our previous observation, if a purify gate outputs a $(0,\perp)$ value, it will propagate $\perp$ to the rest of the tree and return 0 in the current level, if the value is $(\perp/0,1)$, the tree will return $\perp/0$ at the specific level, and then propagate 1 to the rest of the leaves. This shows that all possible strings generated are of the form $0^*\perp^{0|1}1^k \in M_{\leqslant 1}$. □

Correctness and parsimonious argument One can observe that we have a valid construction since the instace is of polynomial size, all gates have the correct arity and every value node has in-degree of at most 1. It suffices to demonstrate that every assingment corresponds to a solution of *HF-DiscreteStrongSperner* using lemma 2.1.

Lemma 2.1.1: Correctness Lemma

A correct assignment corresponds to panchromatic cube in the original problem instance.

It suffices to show that every vector assignment is of the form $\forall i \in [n] : \mathbf{x}[o^i] = \perp$. Assume that we have a correct assignment, and WLOG $\mathbf{x}[o^i] = 0$ for some $i \in [n]$. By our copy that, we know that $\mathbf{x}[o^i] = \mathbf{x}[s_i]$. By our bit generator lemma 2.1 we know that $\mathbf{x}[u^i] = 0^i$. Since our circuit is hazard-free and applies the boundary conditions of the sperner problem, we know that $[\Lambda(x_{-i}(0))]_i = 1 \neq \mathbf{x}[o^i]$. We can make a similar argument for $\mathbf{x}[o^i] = 1$. Therefore, a satisfying assignment corresponds to some panchromatic solution. Given that our instance is in *HF-DiscreteStrongSperner*, it has to be the case that all $u^i \in M_1^n$, otherwise that would imply that there exists a $(n-1)$-subspace that also covers all the labels. This also shows that there exists a 1-to-1 mapping between a panchromatic cube and a satisfying assignment. □

Although the above reduction also works for the general *HF-StrongSperner*, it will not be parsimonious since the *PureCircuit* will accept solutions for all $(n-1)$ subspaces that contain a covering set as explained previously. Since all *AntipodalStrongSperner* are also discrete, it directly follows The corollary 2.1.

Corollary 2.1

$$\textbf{\#PPAD}(\textit{HF-AntipodalStrongSperner}) \subseteq \textbf{\#PPAD}(\textit{PureCircuit})$$

EndOfLine to AntipodalStrongSperner

Provining hardness for *HF-StrongSperner* is a much harder task. Chen et al. [13] showed that every Sperner problem of space $[f(n)]^d$ where $d = \left\lceil \frac{n}{f(n)} \right\rceil$ and $f(n) \leqslant n/2$ for large enough n, it is **PPAD**-complete and parsimonious from the *EndOfLine* problem, using *Snake Embedding* technique. The idea is that we start from the *2D-Sperner* problem and repeatedly fold and pad dimensions until we reach the desired space.

Ideally we would like to use the same method to go from *2D-DiscreteStrongSperner* to *HF-DiscreteStrongSpe* as that would demonstrate a full parsimonious reduction chain from the *EndOfLine*. The problem is that we cannot guarantee, the hazard-freeness property after every fold. If we try to use the corollary 10 by **??**, we observe that the circuit size will increase exponentially. This implies the best we can hope is the following space:

$$M = n^k, f(n) = \log_2(n)^k \implies d = \frac{n}{log(n)^k}$$

But that wound imply that the size of our circuit is $\Omega\left(2^{n/log^k(n)}\right)$. In the upcoming sections, we propose a chain of parsimonious reductions from the *EndOfLine* to *AntipodalStrongSperner*. We emphasize that this method will still not be hazard-free, but we hope that the antipodal properties of resulting problem may enable for a full reduction in the future since by 2.1, we know that its hazard-free variant can be parsimoniously reduced. We will start with a formal description of the planar embedding technique in order to define *2D-EoL* instances.

Planar Graph Embedding We define the graph embedding of a graph $G = (V, E)$ as $G_n = (V_n, E_n)$ where $|V| = n$ and V_n is defined as

$$\forall n \in \mathbb{N} : V_n \triangleq \left\{ \mathbf{u} \in \mathbb{N}_0^2 \mid \mathbf{u}_1 \leqslant 3(n^2 - 1), \mathbf{u}_2 \leqslant 6(n-1) + 3 \right\}$$

G can have any in- and out-degree. For a vertex $i \in V$, we correspond it to the node $(0, 6i)$ in the embedded version. To define the edge set, we will first introduce the following notations.

Definition 2.1.8: $E(\mathbf{p}^1, \mathbf{p}^2)$

Given two points $\mathbf{p}^1, \mathbf{p}^2 \in V_n$ we define the set of edges $E(\mathbf{p}^1, \mathbf{p}^2)$ as:

$$E(\mathbf{p}^1, \mathbf{p}^2) \triangleq \begin{cases} \{(\alpha, k) \mid k \in \mathbb{N} : \min\{\mathbf{p}_2^1, \mathbf{p}_2^2\} \leqslant k \leqslant \max\{\mathbf{p}_2^1, \mathbf{p}_2^2\}\} & \text{if } \mathbf{p}_1^1 = \mathbf{p}_1^2 = \alpha \\ \{(k, \alpha) \mid k \in \mathbb{N} : \min\{\mathbf{p}_1^1, \mathbf{p}_1^2\} \leqslant k \leqslant \max\{\mathbf{p}_1^1, \mathbf{p}_1^2\}\} & \text{if } \mathbf{p}_2^1 = \mathbf{p}_2^2 = \alpha \\ \varnothing & \text{otherwise} \end{cases}$$

Definition 2.1.9: E_{ij}

Given an edge of the original graph $(i, j) \in E$, we define the set of edges E_{ij} as:

$$\begin{aligned} \mathbf{p}^1 &= (0, 6i) \\ \mathbf{p}^2 &= (3(ni + j), 6i) \\ \mathbf{p}^3 &= (3(ni + j), 6j + 3) \\ \mathbf{p}^4 &= (0, 6j + 3) \\ \mathbf{p}^4 &= (0, 6j) \\ E_{ij} &\triangleq E(\mathbf{p}^1\mathbf{p}^2) \cup \cdots \cup E(\mathbf{p}^4\mathbf{p}^5) \end{aligned}$$

To formally define E_n, we first split E_n into two disjoint sets E_n^1 and E_n^2. We define E_n^1 as $\bigcup_{ij \in E} E_{ij}$. The graph generated by (V_n, E_n) has some special properties such as, $\forall v \in V_n$: in-deg(v) + out-deg$(v) \leqslant 4$ and $\forall v \in V_n$: in-deg(v) + out-deg$(v) = 4 \implies$ in-deg$(v) = 2 =$ out-deg(v). Moreover, for every 4-degree node, every incoming edge has an opposite outgoing edge. We use this fact to add 4 additional edges denoted in blue based on the cases denoted in the figure 2.1.6.

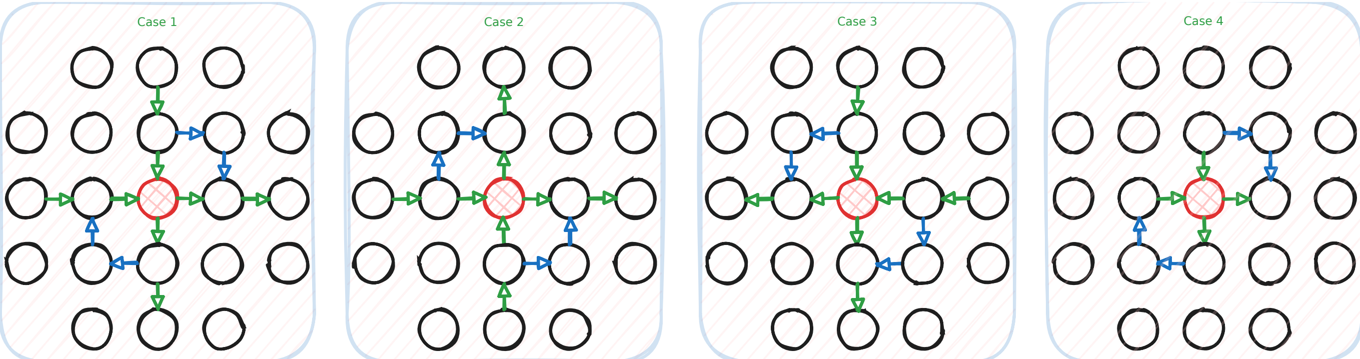


Figure 2.1.6: Figure redrawn from [12]

Given the above we have a full description of G_n. A visual explanation can be seen in figure 2.1.7 and an example of the K_3 complete graph in figure 2.1.8.

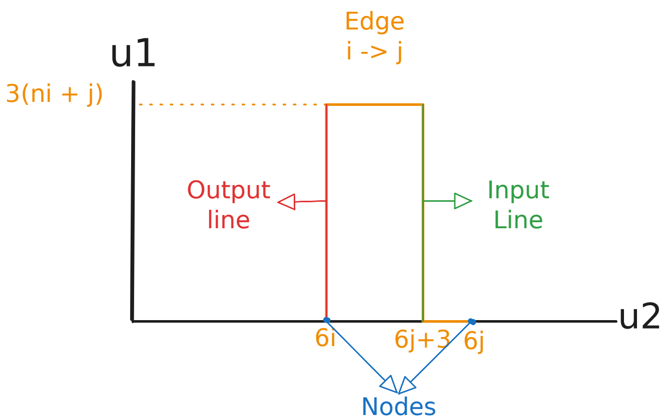


Figure 2.1.7: Edge between points i, j of the original graph. Main observation is the fact that each point comes with a pair of exponentially long lines, which we refer to as the input output lines.

Embedding of *EndOfLine* Chen et al. [12] demonstrated how every *EndOfLine* can be reduced to a *2D-EoL* problem. Key insight, is the observation that every planar embedding of an *EndOfLine*

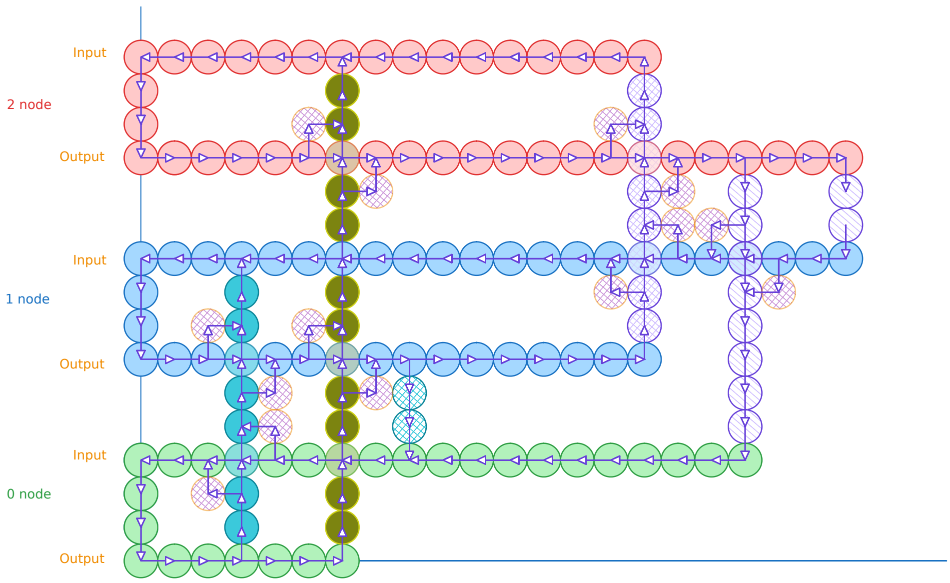


Figure 2.1.8: K_3 graph construction. Combinations of colours indicate an edge between the two nodes. Solid dots indicate edges from 0, crossed lines indicate edges from 1 and stripped lines indicate edges from 2. We Add two coloured stripped circles on the 4 degree nodes to indicate the E_n^2 edges. Image reconstructed from [12].

graph has the following property:

$$\forall v \in V_n : deg(v) \in \{(0,0),(1,1),(1,0),(0,1),(2,2)\}$$

Using as reference the figure 2.1.7, we remove all the edges of nodes with degree of $(2,2)$. This in a way creates a different graph but the key insight is that the number of sources or sinks remain the same as we are essentially redirecting the lines as we can see from the figure 2.1.9. We will use these reductions in later sections to demonstrate reductions with excess problems as well.

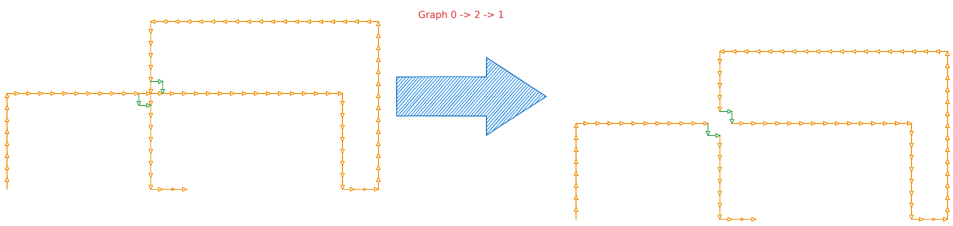


Figure 2.1.9: Depiction of the edge removal. Main idea is that this we keep the number of sources or sinks as well as invertible (we can trace the source or sink from the original instance), despite losing a lot of graph information.

Assume that $C_n \subset G_n$ are the sets of all planar graphs such that $\forall v \in V_n : deg(v) \in \{(1,0),(0,0),(0,1),(1,1)\}$

Clearly we can observe that by the above connstruction of edge removal, all *EndOfLine* problems parsimoniously reduce to the *2D-EoL* problem.

Bipolar colouring of *2D-EoL* We give bipolar colouring variant of Chen et al. [12] colouring technique for *2D-EoL* instances. Given an *2D-EoL* we construct space :

$$B_n \triangleq [n]_0 \times [n]_0$$

We will use points $\mathbf{u},\mathbf{v},\mathbf{w} \in V_{2^k}$ and $\mathbf{p},\mathbf{q},\mathbf{r} \in B_{2^{4k+10}}$. Moreover a panchromatic square will be denoted if all the points in the neighbouring squares satisfy the bipolar colouring. Below we

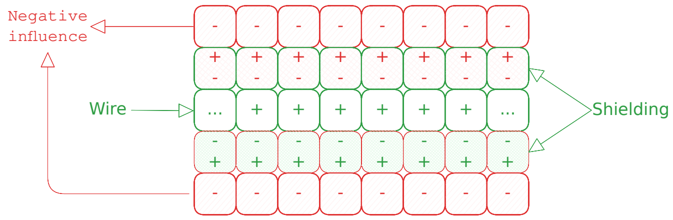


Figure 2.1.10: Shielding method on 2D-EoL embeddings.

will denote our colouring scheme for clarity:

$$\begin{aligned} + &= (+1, +1) \\ \pm &= (+1, -1) \\ \mp &= (-1, +1) \\ - &= (-1, -1) \end{aligned}$$

First, we define mapping $\mathcal{F} : V_{2^k} \to B_{2^{4k+10}}$ such that

$$\mathcal{F}(\mathbf{u}) = \begin{pmatrix} 6\mathbf{u}_1 + 6 \\ 6\mathbf{u}_2 + 6 \end{pmatrix} \in B_{2^{4k+10}}$$

Since $G \in C_{2^k}$, its edge-set can be decomposed into a collection of paths and cycles $(P_i)_{i \in [m]}$. By using $\mathcal{F}$, every P_i is mapped to a set $I(P_i) \subset B_{2^{4k+10}}$ We will colour every point on $I(P_i)$ with $+$

$$P = \mathbf{u}^1, \ldots, \mathbf{u}^t, \text{ if P is a cycle } \mathbf{u}^1 = \mathbf{u}^t$$
$$I(P) \triangleq \bigcup_{i \in [t-1]} I(u^t u^{t-1})$$

It is common to use the method of shielding when dealing with line reductions*Line reductions*, as seen in [12, 20, 2, 13, 16] Essentially we embed our lines and cycles in a grid, everything not a line is denoted as a negative charge $-$. All lines are denoted as positive charge $+$. To protect our lines from creating solutions, we cover them with $\pm, \mp$ lines. A visualisation can be seen in figure 2.1.10. We define the following set to denote the set of shielding points around our paths.

$$O(P) = \{\mathbf{p} \in B_{2^{4k+10}} \backslash I(P) :|: \exists \mathbf{p}' \in I(P), \|p - p'\|_\infty = 1\}$$

If P is a simple path we can decompose $\{\mathbf{s}_p, \mathbf{e}_p\} \cup L(P) \cup R(P)$ such that: $\mathbf{s}_p = \mathcal{F}(\mathbf{u}^1) + \Delta + \Delta^\perp$, where $\Delta = (\mathbf{u^1} - \mathbf{u}^2)$, $\mathbf{e}_p = \mathcal{F}(\mathbf{u}^1) + \bar{\Delta} + \bar{\Delta}^\perp$, where $\bar{\Delta} = (\mathbf{u}^t - \mathbf{u}^{t-1})$ and $x^\perp$, denotes the vector perpendicular to x. Choice of which perpendicular vector we consider is arbitrary. Starting from $\mathbf{s}_p$, enumerate vertices in $O(P)$ clockwise as $\mathbf{s}_p, \mathbf{q}^1, \ldots, \mathbf{q}^{n_1}, \mathbf{e}_p, \mathbf{r}^1, \ldots, \mathbf{r}^{n_2}$ such that

$$L(P) = \{\mathbf{q}^i\}_{i \in [n_1]} \, ,$$
$$R(P) = \{\mathbf{r}^i\}_{i \in [n_2]}$$

If P is a simple cycle, then can can decompose $L(P) \cup R(P)$, where $L(P)$ contains all the vertices on the left side of the cycle and $R(P)$ all the vertices on the right side If G is specified by $(S, P, 0^k) \subset C_{2^k}$, we can uniquely decompose the edge set into $P_1, \ldots P_m$. Every path is either a maximal path, or a cycle in G. We can use the following to describe our colouring algorithm:

Algorithm 2.1.1 Turing machine F with input $(p_1, p_2) \in B_{2^{4k+10}}$

```
if p1 = 0 then
    if p2 ⩽ 6 then
        return +
    else
        return ±
else if p1 < 6 thenk
    if p2 = 6 then
        return +
    else if p2 < 6 then
        return ∓
    else if p2 = 7 then
        return ±
    else
        return −
else
    return M_K(p)
```

Where $M_k(\mathbf{p})$ is can be described with the following description:

$$M_k(\mathbf{p}) = \begin{cases} + & \text{if } \exists i, \mathbf{p} \in I(P_i) \\ \pm & \text{if } \exists i, \mathbf{p} \in L(P_i) \vee p_1 = 0 \\ \mp & \text{if } \exists i, \mathbf{p} \in R(P_i) \vee p_2 = 0 \\ - & \text{otherwise} \end{cases}$$

We will call the path of nodes $(0,0) \to (0,6) \to (6,6)$ as the tail, which ensures that the 0^n node of our original graph is not a solution. An example of how such reduction looks like can be seen in figure 2.1.11.

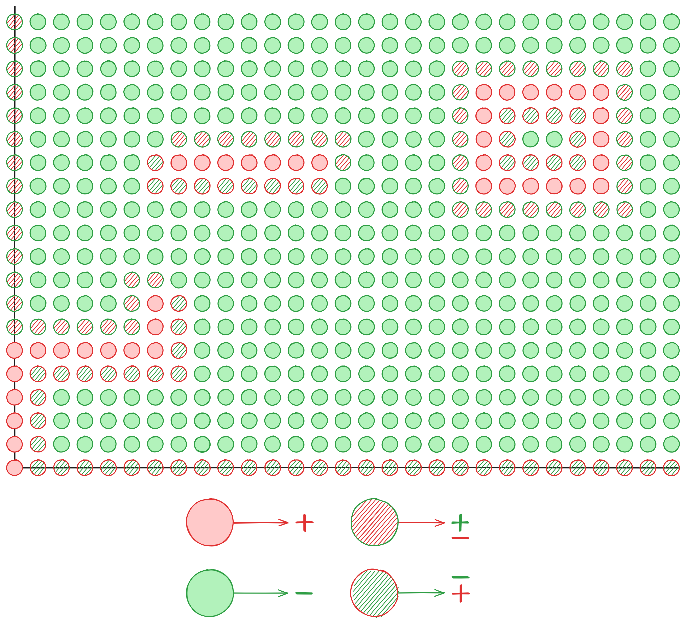


Figure 2.1.11: *2D-EoL* Bipolar Colouring

Lemma 2.1.2: Correctness Lemma

The following statements hold correct:

1. $\forall i, j : \min_{x \in P_i, y \in P_j} \|\mathcal{F}(x) - \mathcal{F}(y)\|_\infty > 4$
2. All panchromatic squares represent solutions

To prove the first statement, assume we have $a, b \in V_{2^k} : \|a - b\|_\infty \geqslant 1$. WLOG assume $a - b = (x, y)$ for some $x, y \in \mathbb{N}$ which implies the following:

$$\mathcal{F}(a) = \mathcal{F}\begin{pmatrix} b_x + x \\ b_y + y \end{pmatrix} = \begin{pmatrix} 6b_x + 6x + 6 \\ 6b_x + 6y + 6 \end{pmatrix}$$

$$\mathcal{F}(a) - \mathcal{F}(b) = \begin{pmatrix} 6x \\ 6y \end{pmatrix}$$

Since $x + y > 0 \implies \|\mathcal{F}(a) - \mathcal{F}(b)\|_\infty \geqslant 6 > 4$. This statement is meant to show that all lines and their shieldings are far enough from each other such that the colourings will not create additional solutions. Visually what this looks like is, if we had to parallel lines in V_{2^k}, then the mappings with the colourings, will look like the figure in 2.1.12.

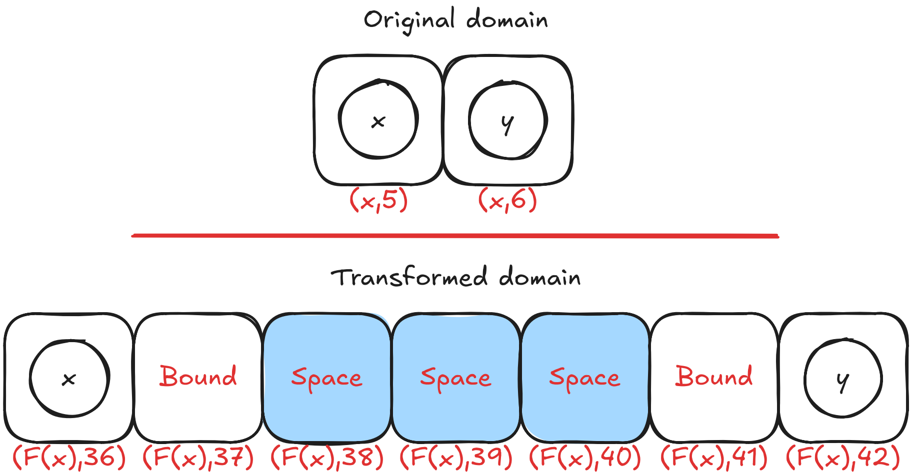


Figure 2.1.12: Transformed domain mapping example.

To show our second part of our solution, we can observe that: All nodes that are not boundary and not in $\bigcup_i I(P_i) \cup L(P_i) \cup S(P_i)$ are coloured $-$. We will use figure 2.1.11 for reference. We can observe that all squares on the left or the right of the line do not result in solutions as they contain only labels $\{\pm, +\}$ and $\{\mp, +\}$ respectively as seen in figure 2.1.11. One can observe that the square that contains s_{P_i} or e_{P_i} and a point of the line will be panchormatic as the two points will be antipodal. Therefore, we have a valid and parsimonious reduction since we only cubes and the beginnings or ends of lines can be solutions. We call this coloured instances as *2D-StrongEoL*.

Going from 2D to higher dimensions We now demonstrate how to go from *2D-StrongEoL* to *AntipodalStrongSp* We employ a slightly adjusted method than one proposed by Chen et al. [13], which closely resembles the *Snake Embedding* technique used by Deligkas et al. [20] on the *StrongTucker* version of the problem, which reduced the space from $2^n \times 2^n \to [7]_0^m$, where $m = O(n)$. We will show how to expand their method for the *StrongSperner* problems as well as demonstrate that such reduction will resolve in an *AntipodalStrongSperner* instance.

Theorem 2.1.3

$$\mathbf{\#PPAD}(\textit{2D-StrongEoL}) \subseteq \mathbf{\#PPAD}(\textit{AntipodalStrongSperner})$$

Proof. We are given as an input a $2^n \times 2^n$ grid representing any *2D-StrongEoL* problem. The procedure is summarised as follows: Given dimension to fold i where width of i is $n > 8$, apply the *padding operation* such that $n' = 3 \cdot s + 1$ for some $s \in \mathbb{N}$. Then apply the *folding operation*, which reduces "width" of the space to $s + 3$ and compresses the space into a higher dimension. This resembles how we can take a sheet of paper and fold it half way such that it now has a third dimension, but a smaller width on the folded one. We will now go into a greater detail as to how each operation works. To make notation simple and consistent, we define M^t as

$$M^t \triangleq \bigtimes_{i \in [n_t]} [m_i^t]$$

Where m_i^t denotes the width of dimension i at iteration t, and n_t as the number of dimensions.

Padding Dimensions We will use the definition 2.1 to define our operation.

Definition 2.1.10: Padding dimension $P(T, i, u)$

Given labelling function $\Lambda^t : M^t \to \{-1, +1\}^{n_t}$, we define the padding operation $P(\Lambda^t, i, u)$ for $i \in [n_t]$ and $u \in \mathbb{N}$ where $n_{t+1} = n_t$ and

$$\forall j \in [n_{t+1}] : m_j = \begin{cases} m_i & \text{if } i \neq j \\ m_i + u & \text{otherwise} \end{cases}$$

The above operation extends dimension i with u additional layers. How the operation work is that we append a *null* dimension such that, given $\bar{x} = x_{-i}(m_i^t + 1)$ where $x \in M^t$

$$\forall j \in [n] : [\Lambda^{t+1}(\bar{x})]_j = \begin{cases} -1 & \text{if } \bar{x}_i > 0 \\ +1 & \text{otherwise} \end{cases}$$

We can be sure that the above will not result in any new solutions since in $[\Lambda^t(\cdot, m_i^t, \cdot)]_i = -1$, and therefore any solution between $(\cdot, m_i^t, \cdot), (\cdot, m_i^t + 1, \cdot)$ will not yield in a solution. We can repeat the above procedure u times, if we need to pad the dimension by u.

Snake Folding below we will give a overview of the snake folding operation. Before giving a formal definition, we will first give an informal overview of what the technique actually does based on the figure 2.1.13 where we view our folding with respect to the folded and new dimension. The y-axis denotes the new dimension, denoted as n_{t+1} and the x axis denotes the folded dimension referred as $\bar{i}$ with width $m_{\bar{i}}^{t+1} = s + 3 = m_{\bar{i}}^{t'}$, given that the original width was $3s + 1$ for some $s \in \mathbb{N}$. Moreover, we will use notation (a, b) to refer to a as the labels in the n_t subspace and b as the label of the $n_t + 1$ dimension. From the figure we observe that we have orange, red, green and blue lines. Blue and orange lines denote $(\omega, -)$ and $(\omega, +)$ where ω is the n_t-dimensional *null* space, where for each point > 0, we use label $-$ and for 0 we use label $+$. These lines are meant to satisfy the boundary conditions. For the red and green lines, these are denoted as $(\star, -), (\star, +)$ where $\star$ is the n_t-dimensional copy of the lower space with some additional points. The core idea is that all solutions, must belong in between the red and green lines, such that if we have a solution in the n_t-space, we should also have a solution in the $n_t + 1$ space, as shown in the figure 2.1.14.

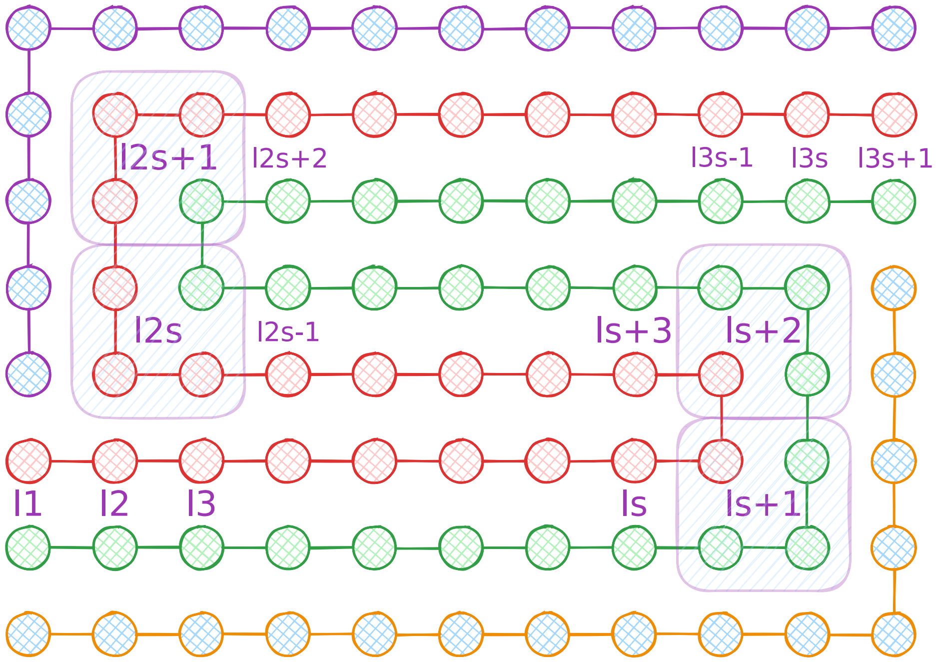


Figure 2.1.13: Snake Embedding technique view between the folded dimension j and the new dimension. The x-axis denotes the folded dimension and the y-axis denotes the new dimension. The point ℓ_i denote the indexes that correspond the j dimension of the pre-folded space.

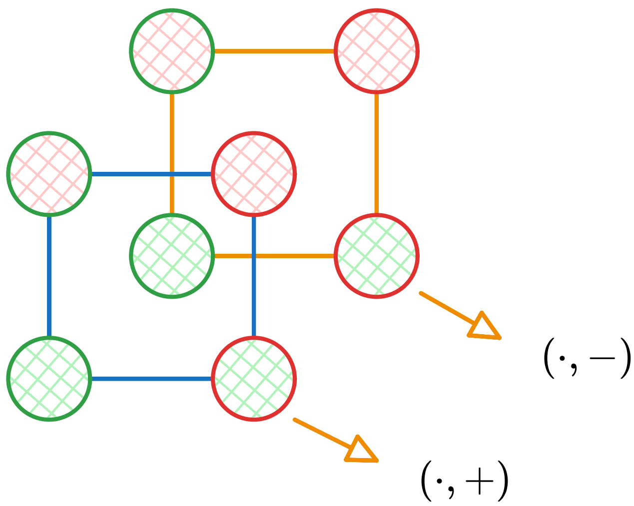


Figure 2.1.14: Snake Embedding technique between the folding dimension and the new dimension

To define the above more formally, we define sets $B^U, B^L \subseteq M^{t'}$ which denote the set of boundary indices of the orange and blue lines. In addition, we define functions $\ell^P, \ell^N \subseteq M^{t'}$ to denote the points on the green and red lines with functions ϕ^P, ϕ^N that are the inverse maps to the original original space. Given our new circuit Λ^{t+1}, we define the following:

$$\forall x \in M^{t+1}, j \in [n_t] : [\Lambda^{t+1}(x)]_j = \begin{cases} -1 & \text{if } x \in B^U \cup B^O \wedge x_j > 0 \\ +1 & \text{if } x \in B^U \cup B^O \wedge x_j = 0 \\ [\Lambda^t(\phi^P(x))]_j & \text{if } x \in \ell^P \\ [\Lambda^t(\phi^N(x))]_j & \text{if } x \in \ell^N \end{cases}$$

For the new dimension, we apply the labelling rules as described previously

$$\forall x \in M^{t+1} : [\Lambda^{t+1}(x)]_{n_t+1} = \begin{cases} -1 & \text{if } x \in B^U \cup \ell^N \\ +1 & \text{if } x \in B^L \cup \ell^P \end{cases}$$

Using the above we formally define the operation the snake embedding operation $S(T, i)$.

Definition 2.1.11: Snake Embedding operation $S(\Lambda^t, i)$

Given labelling function $\Lambda : M^t \to \{-1, +1\}^{n_t}$, apply the snake embedding operation on dimension $i \in [n]$, where $m_i \equiv 1 \mod 3$, to a get a new labelling function $\Lambda^{t+1} : M^{t+1} \to \{-1, +1\}^{n_{t+1}}$ such that $m'_i = s + 3, n_{t+1} = n_t + 1, m_{n_{t+1}} = 8$, where Λ^{t+1} is defined based on the above description.

Claim 2.1.4: Antipodality Solution preservation

Given operation $S(\Lambda^t, i)$, we argue the following hold true:

1. If Λ^t is a discrete antipodal colouring function, Λ^{t+1} will preserve its properties.
2. The number of solutions between the two spaces will be preserved

Proof. We observe to the labelling conditions by comparing the original with its folded state. Assume we apply the snake folding operation on dimension j. We can observe that all solutions in the embedded space will be anchored between the red or green lines (using figure 2.1.13), otherwise the new dimension will not be covered. Moreover, one can observe that a panchromatic hypercube can never be anchored in any of the turning points since our labelling is discrete and we know that any $n_t - 1$ subspace cannot yield any panchromatic solutions.

Assume we have a point in $p \in M^{t+1}$, which is an anchor of some panchromatic solution for some $p_j \in [m_j^{t+1}]$ and $p_x \in [m_{n_{t+1}}^{t+1}]$. We can observe from the diagram 2.1.13 that the coordinates $(\cdot, p_j, p_x)$ correspond to some point $(\cdot, j)$ for some point in the pre-folded space. We can also observe that the point $(\cdot, j + 1)$ may correspond to the point $(\cdot, p_j + 1, p_x)$. or $(\cdot, p_j, p_x + 1)$, depeding on whether $j \in \{2s, 2s + 1\}$ as it implies whether its on the vertical portion of the space or the horizontal portion. We denote this observation as the effective direction, meaning if $(\cdot, j) \to (\cdot, j + 1) \implies (\cdot, p_j, p_x) \to (\cdot, p_j + 1, p_x)$, then we say that the jth dimension is the effective direction, otherwise the n_{t+1}th will be the effective dimension

WLOG we assume the effective dimension is j Since $(\cdot, p_j, p_x)$ are the anchor of a panchromatic cube it implies that the points $(\cdot, p_j, p_x), (\cdot, p_j + 1, p_x)$ cover all the n_t labels and p_x will cover the n_{t+1}th dimension. This shows that we can map bijectively between each space and therefore preserve the cardinality of the solution sets.

We can also observe that our solutions remain antipodal. WLOG assumet that the effective dimension is j. We know that if p is a solution of the prefolded, then $\forall b : \Lambda^t(p + b) = -\Lambda^t(p + \bar{b})$. This implies that if $\hat{p}$ is its corresponding solution in the folded space then:

$$\forall b \in \{0, 1\}^{n_{t+1}} : \Lambda^{t+1}(\hat{p} + b_{-n_{t+1}}(0)) = -\Lambda^{t+1}(\hat{p} + \bar{b}_{-n_{t+1}}(1))$$

since the n_{t+1}th dimension will not affect the labelling of the other dimensions. □

We observed that our operations do not affect the cardinality of solutions. We now outline our construction of our new circuit

1. Check if there is a dimension $j$with $m_j^t > 8$. If yes, first pad the dimension such that $m_j^t \equiv 1 \mod 3$. Afterwards fold the dimension on j to get new folded circuit

2. If the folding results in a width of $m_j^t < 8$, then we can pad the dimension to have a width of 8
3. If there are no operations that can be applied then we can terminate

We can observe that in each step of the algorithm we preserve our solutions. Moreover, each fold divides the dimension by some factor of 3. Since we start with space of $2^n \times 2^n$, we can conclude that the number of operations needed will be on the factor or $O(n \log_3 2)$. This concludes our proof. $\square$

This is the closest that we manage to bring the reduction chain across *EndOfLine* and *PureCircuit*. We can observe two interesting directions: if one can create a bit generator for kleene binary numbers as presented previously, then we can a similar construction to the proof we did in 2.1.2 for a full parsimonious reduction between *nD-SS* to *PureCircuit*. Alternatively one can attempt to reduce *AntipodalStrongSperner* to *HF-AntipodalStrongSperner* or *PureCircuit* parsimoniously. We argue that the antipodality property allows us to check for solutions much more efficient than trying to find $d+1$ points in some hypercube. In hindsight, we argue that if one could construct a gadget to generate the maximum and the minimum resolution of a *Kleene unary*, it would offer a major breakthrough towards the full reduction.

2.2 SourceOrExcess analysis

The current section focuses on how we can extend our reductions to *SourceOrExcess*. In the previous sections, we gave an overview on how we can go from an *EndOfLine* problem to a topological *StrongSperner* problem. By providing a topological interpretation of the *SourceOrExcess* problem, we hope to re-use the construction for future reductions with the *PureCircuit* problem.

We will first show how we can topologically reduce the *SourceOrExcess*$(2,1)$ problem to a *2D* space. We will demonstrate some limitations with *2D* topological problems and then we will demonstrate how using a *3D* counting version of the *StrongSperner* problem will allow for a full parsimonious reduction.

2.2.1 Topological representations

Given a *SourceOrExcess*$(2,1)$ input we first eliminate all excess nodes, meaning nodes with degree $(2,1)$ by separating one of the lines as shown in the figure 2.2.1. This simplifies our counting argument to only 3 scenarios. Moreover for double sinks, we enforce that the MSB value of the bit-string for one of the two values is one. We can observe that these modifications can be done in polynomial time since they only require to analyse the local neighbouring of nodes, and they preserve the number of solutions

Planar Transformation Our transformation 2.1 is identical to the one that done when reducing *EndOfLine* to *2D-EoL* with a slight change on the space and the paths the nodes take. First we double V_n as such:

$$\forall n \in \mathbb{N} : V_n \triangleq \left\{ \mathbf{u} \in \mathbb{N}_0^2 \mid \mathbf{u}_1 \leqslant 6(n^2-1), \mathbf{u}_2 \leqslant 12(n-1)+6 \right\}$$

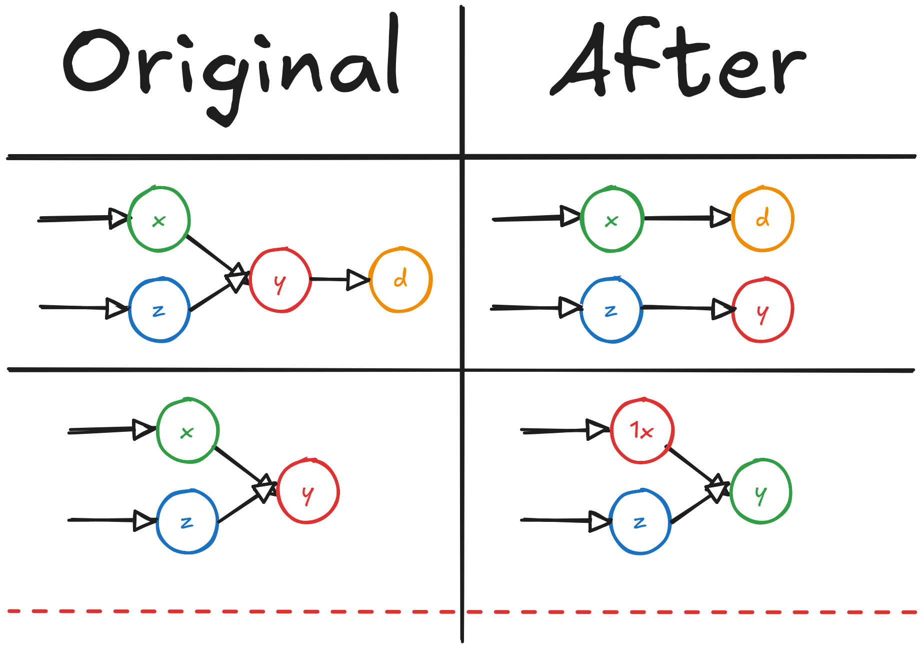


Figure 2.2.1: The current table shows how we deal with excess nodes. All other nodes such as sources, sinks, lines or isolated vertices stay the same.

Such that now every node is represented as $(0, 12i)$. Morevover for edge in $(i, j) \in E_n$ create path and MSB of i is 0, then we define path:

$$\begin{aligned}
\mathbf{p}^1 &= (0, 12i) \\
\mathbf{p}^2 &= (6(ni + j), 12i) \\
\mathbf{p}^3 &= (6(ni + j), 12j + 6) \\
\mathbf{p}^4 &= (0, 12j + 6) \\
\mathbf{p}^5 &= (0, 12j) \\
E_{ij} &= E(\mathbf{p}^1\mathbf{p}^2) \cup \cdots \cup E(\mathbf{p}^4\mathbf{p}^5)
\end{aligned}$$

If j is a double sink, attach a small tail:

$$\begin{aligned}
\mathbf{p}^6 &= (3, 12j) \\
E'_{ij} &= E_{ij} \cup E(\mathbf{p}^5\mathbf{p}^6)
\end{aligned}$$

If the MSB value of i is 1, then we create path

$$\begin{aligned}
\mathbf{p}^1 &= (0, 12i) \\
\mathbf{p}^2 &= (6(ni + j), 12i) \\
\mathbf{p}^3 &= (6(ni + j), 12j) \\
\mathbf{p}^4 &= (3, 12j) \\
E_{ij} &= E(\mathbf{p}^1\mathbf{p}^2) \cup E(\mathbf{p}^3\mathbf{p}^4) \cup E(\mathbf{p}^3\mathbf{p}^4)
\end{aligned}$$

In the original problem, we argued that for every node, we have I/O lines that denote the successor and predecessors of the current node, which can be exponentially large. This would make it computationally difficult to differentiate between double sinks and excess nodes and we would end up double counting solutions. By removing excess nodes, we only have to focus on double sinks. Since there is no output line in a double sink, we take advantage of that by sending one inputs from the traditional input path, and the other from the output path.

We can use similar arguments as in the original proof, to argue that: $\forall v \in V : degree(v) \in \{(0,1),(1,0),(1,1),(2,2),(2,0)\}$. We can observe that, if we do not have any double sinks, our graph follows the original transformation where only 4 out of the 5 possible degrees are possible. The only difference occurs on double sinks. We can observe that if the MSB of a node is 1, it can only ever be incident to double sink, meaning $(3,12i)$ can never have an outwards edge. We know that for every node in its path, besides $(3,12i)$, its degrees can only ever be one of the 4 cases outlined. Similarly for the other node with LSB 0 its main path stays the same except if its incident to a double sink, where we add 3 additional edges. For all edges, we know that the height of the path $E(\mathbf{p}^2,\mathbf{p}^3)$ can only be at height greater or equal to 6, and therefore no path other than the two incoming edges will ever intersect $(3,12i)$.

We follow the same construction for the E_n^2 and apply the same removal transformation. The resulting graph is the same as in the traditional *2D-EoL* problem but with the addition of double sinks and we call this *2D-SourceOrExcess*$(2,1)$. Moreover we can observe that *2D-SourceOrExcess*$(2,1)$ is parsimonious to the *SourceOrExcess*$(2,1)$ problem as sources and sinks are preserved, and any removal of edges can never affect our doubled sinks.

Topological Difficulty We observe that in the *2D-StrongEoL* bipolar colouring, we use the left handed rule to colour the shielding of our lines in a specific way, such that we always end up with panchromatic solutions at the ends of a line. The issue is that when we try to extend the same logic to the *2D-SourceOrExcess*$(2,1)$ problem, we end up with the problem shown in the figure 2.2.2. Any permutation of colours we choose will always result in 2+ solutions. Even if we relax the problem to allow squares incident to panchormatic edges to count as a single solution, we still get multiple solutions. The problem occurs due to the chirality of the colouring since we are connecting a left handed line with another left handed line at opposite ends.

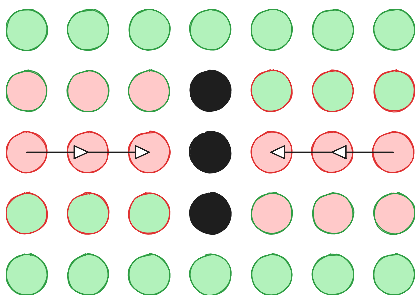


Figure 2.2.2: *2D-StrongSperner* with *2D-SourceOrExcess*$(2,1)$

The chirality problem extends *StrongSperner* instances. Hollender et al. [29], demonstrated how one can reduce parsimoniously the *2D-EoL* problem to *1-MC*. To avoid the heavy details, given an edge between two nodes, we can add 3 domino tiles to represent the edge. When two opposite paths meet we always end up with two squares that cannot be covered as seen in the figure 2.2.3. We can observe that again due to chirality, we can never find appropriate tiling.

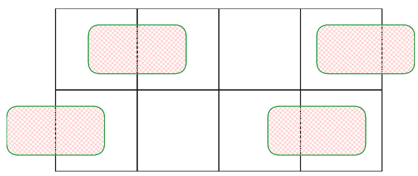


Figure 2.2.3: *1-MC* with *2D-SourceOrExcess*$(2,1)$

2.2.2 Topological colouring

The key insight of our objective is showing how going to 3 dimensions and allowing panchromatic edges, allows us to achieve parsimonious reductions with *StrongSperner*. We first define the problem *3D-CountingSS*, which is a combinatorial friendly version of the *3D-StrongSperner*.

Definition 2.2.1: *3D-CountingSS* problem definition

Input: A colouring circuit $\lambda : (\{0,1\}^k)^3 \to \{-1,+1\}^3$ which follows the boundary conditions of the *StrongSperner* problem as explained in 1.2.1.
Output: A coordinate $p = (i, j, k) \in (\{0,1\}^k)^3$ such that the points in $\mathbf{Cube}(p)$ cover all labels. To reduce duplicated solutions, assume there exists $H \subset \mathbf{Cube}(p)$ which also covers the labels. Assume $\mathcal{D} = \{A \subseteq H \mid A \text{ covers all labels}\}$ such that $A = \arg\min_{A\in\mathcal{D}} |A|$. Accept only if $p = \mathbf{Anc}(A)$.

The above idea is a natural extension of the *StrongSperner* interpretation, where its labelling represents the direction of a continuous function. If a face of a cube is panchromatic, then that implies that there has to be a fixed point somewhere, as seen in the figure 2.2.5. If an edge were to be panchromatic, then that means that a fixed point could lie on the edge. If we were treating this as a fixed point problem instead of a colouring one, it would make sense to argue that we have one solution. We abuse this notion to count only the anchor of the cube which is incident to the panchromatic subspace. An example of the counting argument is seen in figure 2.2.4.

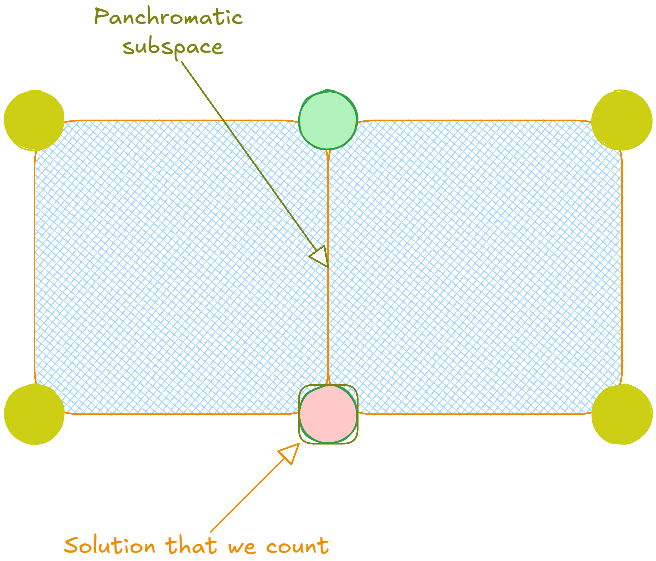


Figure 2.2.4: Panchromatic edge counting example

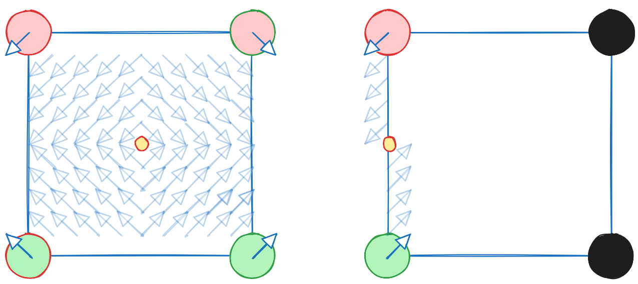


Figure 2.2.5: Panchromatic edge for 2 dimensions

Trivially *3D-CountingSS* can be reduced to *3D-StrongSperner* even if we are overcounting the cubes. To show hardness, we make the following theorem:

Theorem 2.2.1

$$\#\mathbf{PPAD}(\textit{2D-SourceOrExcess}(2,1)) \subseteq \#\mathbf{PPAD}(\textit{3D-CountingSS})$$

Proof of the theorem

Our first reduction is transforming our $2D$ grid, into a $3D$ with grid points:

$$B_n^3 = \left\{\mathbf{p} \in \mathbb{N}_0^3 \mid \mathbf{p}_3 \leqslant 6, \mathbf{p}_{-3} \in [n]_0 \times [n]_0\right\}$$

We denote the transformation $\mathcal{F} : V_{2^k} \to B^3_{2^{4k+10}}$ such that

$$\forall \mathbf{p} \in B_n : \mathcal{F}(\begin{pmatrix}\mathbf{p}_1\\ \mathbf{p}_2\end{pmatrix}) \to \left\{\begin{pmatrix}6\mathbf{p}_1+6\\ 6\mathbf{p}_2+6\\ 0\end{pmatrix}, \begin{pmatrix}6\mathbf{p}_1+6\\ 6\mathbf{p}_2+6\\ 1\end{pmatrix}\right\}$$

We can observe that the above essentially is our *2D-SourceOrExcess*$(2,1)$ but copied twice in the 3D space. We note of the terms $I^b(P_i), L^b(P_i), R^b(P_i), \mathbf{s}^b_{P_i}, \mathbf{e}^b_{P_i}$ that we defined in the paragraph 2.1 and say use index b to denote the first copy and the second copy. Lastly, we introduce m^b_{ij} to denote the "midpoint" or the point with degree $(2,0)$ that meets path i and path j with b to denote the upper and lower face. The ends of the paths P_1, P_2 are placed collinearly to the "midpoint" m_{ij} as seen in the figure 2.2.6.

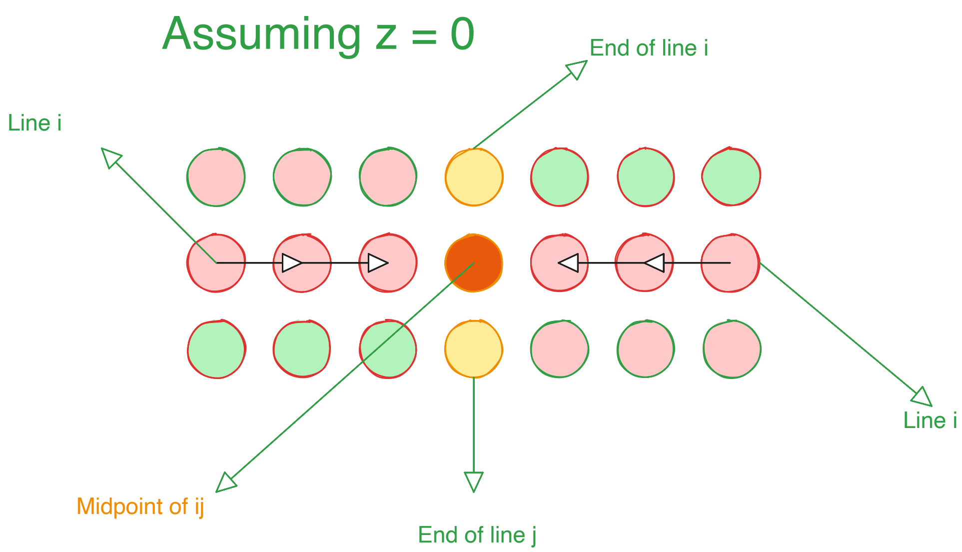


Figure 2.2.6: Double sink depiction from a 2D point of view

We now propose our colouring circuit $\mathcal{M}$:

$$\mathcal{M}(\mathbf{p}) \triangleq \begin{cases} (+,+,(-1)^b) & \text{if } \exists i,b : \mathbf{p} \in I^b(P_i) \\ (-,+,(-1)^b) & \text{if } \exists i,b : \mathbf{p} \in R^b(P_i) \\ (+,-,(-1)^b) & \text{if } \exists i,b : \mathbf{p} \in L^b(P_i) \\ (-,-,(-1)^b) & \text{if } \exists i,b : \mathbf{p} \in \{\mathbf{s}^b_{P_i}, \mathbf{e}^b_{P_j}\} \\ (-1)^b * (-,-,+) & \text{if } \exists i,j,b : \mathbf{p} = \mathbf{m}^b_{ij} \\ ((-1)^{\mathbb{1}(\mathbf{p}_1>0)}, (-1)^{\mathbb{1}(\mathbf{p}_2>0)}, (-1)^{\mathbb{1}(\mathbf{p}_3>0)}) & \text{otherwise} \end{cases}$$

An example of how the above colouring for simple lines can be seen in figure 2.2.7.

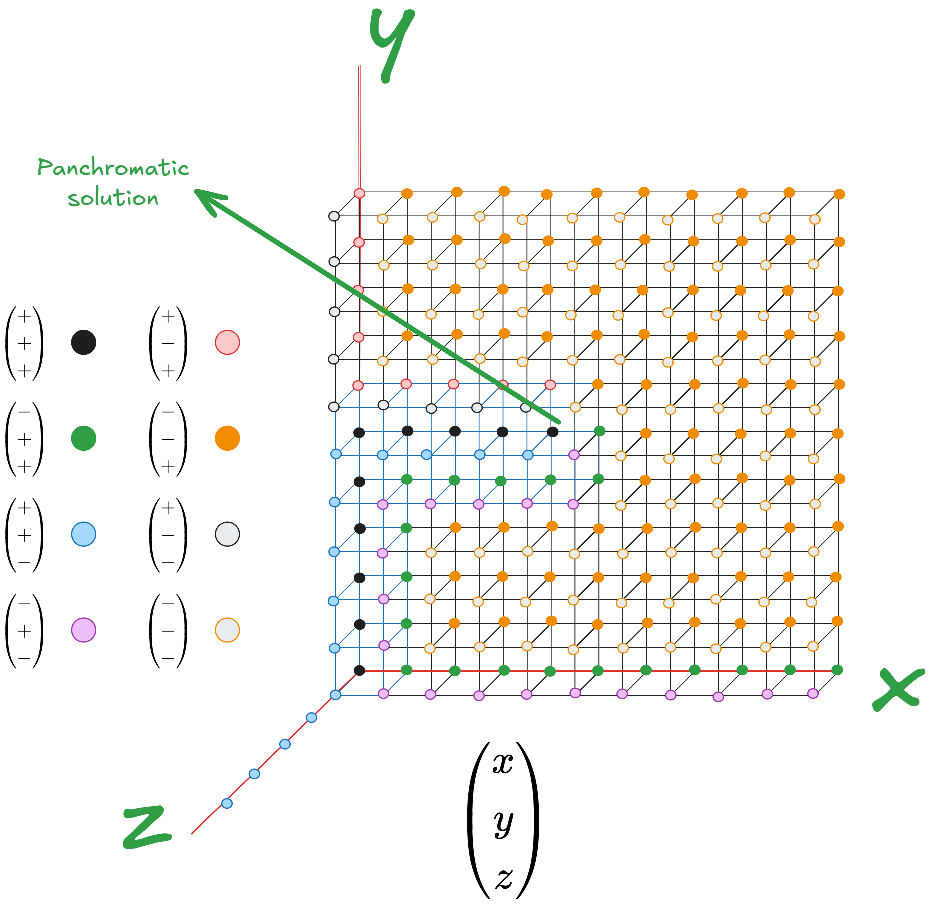


Figure 2.2.7: Colouring example based on the proposed circuit

Correctness proof

We now prove that our reduction is correct. We use figure 2.2.8 and assume that we have an edge $\mathbf{p}^t, \mathbf{p}^{t+1}$ of some path. We use red to represent the points on the left of the edge, blue the points on the right, and brown for the surrounding. The red cube will not be panchromatic as x-coordinate is $+$ on all corners and vice versa for the blue cube on the y-coordinate. In addition we can observe that the brown cubes surrounding each face will also not result in a panchromatic cube as: for the brown cube on the left of red will be $-$ on all y coordinates, with the same argument for blue. And for the brown cubs on top, we can observe that z-coordinate will be $-$ on all points. Therefore points in between lines or cycles will not result in solutions. As we can observe from the figure 2.2.7, we can observe that the cubes create discrete antipodal solutions.

For the double sink case, we use the figure 2.2.9, to observe what happens around the double sink area. For points m^0_{ij}, m^1_{ij}, we can observe that their labelling is antipodal, meaning given our counting defintion, all surrounding cubes that contain the side (m^0_{ij}, m^1_{ij}) count as one solution. We will refer to all the cubes incident to the edge as the combined panchromatic cube. We can manually observe that for each outer face of the combined panchromatic cube there will always exist a dimension whose colouring is not covered. For example we can see that for all faces on top of the combine cube never cover the $-$ dimension. We can make similar arguments for the left and right outer faces. Since the outer faces do not yield a solution, that means we have only one solution and therefore we can observe that the set of panchromatic solutions correspond to a 1-to-1 mapping with the *SourceOrExcess*$(2, 1)$ problem.

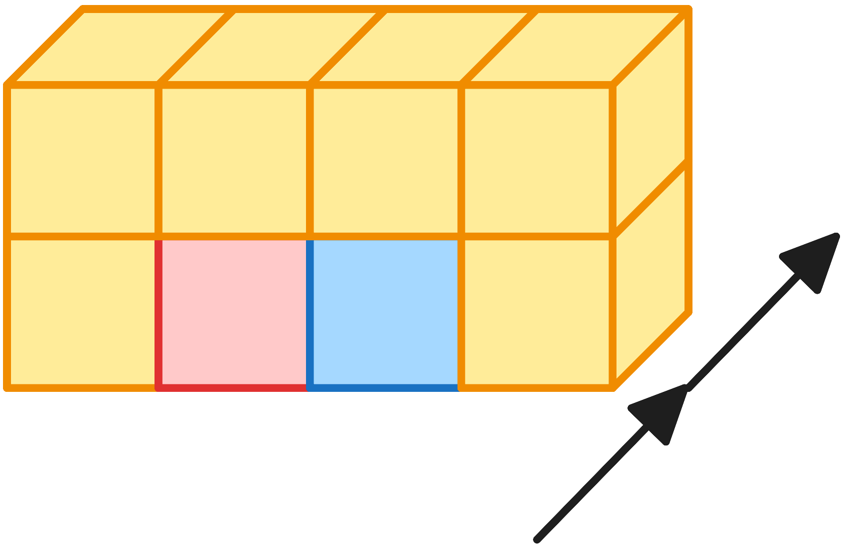


Figure 2.2.8: Example colouring where the arrow denotes the direction. In between the red and blue cubes in the bottom corners we have the positive line. The brown cubes around denote the negative labels

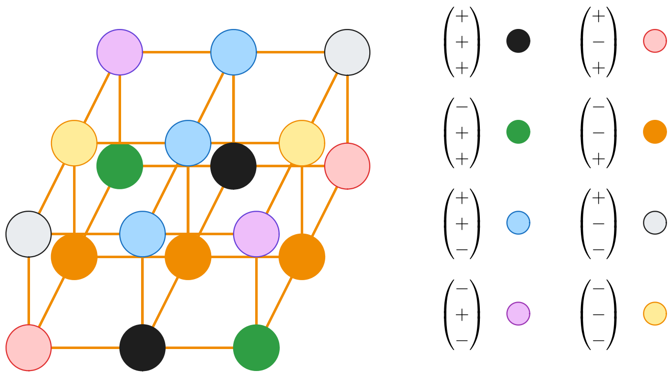


Figure 2.2.9: Colouring around the double sink points. The colours correspond to the colour legend of 2.2.7

2.2.3 Analysis of our results

We observed that in order to make the current reduction work, we had to remove excess nodes and focus only on double sinks. Moreover, we had to introduce antipodal edges in order to group solutions as one. As a result, the current method cannot be generalised directly to other *SourceOrExcess*$(k, 1)$ problems unless we generalise the planar embedding technique. In addition, our snake embedding method, the cardinality of solution is not guaranteed to be the same the duplication of turning points may increase the number of cubes. On the contrary, our result raises questions as to the relation between the degree of a graph and its the dimension of its topological embeddings. We analyse this in greater detail in our future works section 4.2.

Additionally we observed that a Kleene interpretation of the problem may not be as natural. As we have seen in *nD-HF-SS*, we showed that we parsimoniously reduce from its non-hazard-free variant by doubling dimensions and applying the corollary 1.1. The same methodology may not work in the current counting variant. Moreover, generalising our solution to *Counting-SS* may not be possible using our previous methods since we allow for panchromatic subspaces, we cannot guarantee that the cardinality of solutions will be preserved over each folding.

2.3 PPAD Counting Hierarchies

As previously mentioned, we want to extend the work done by Ikenmeyer et al. [33], in order to analyse the combinatorial nature of **PPAD** as a whole. Although the project did not manage to achieve the desired theorem, we hope to demonstrate interesting findings that can allow us to reveal more intricate properties of **#PPAD**. Our first step is to extend the hierarchy between *EndOfLine* and *SourceOrExcess*$(2,1)$. We can observe that

$$\forall a,b \in \mathbb{N} : a \leqslant b \implies \mathbf{\#PPAD}(\mathit{SourceOrExcess})(a,1) \subseteq \mathbf{\#PPAD}(\mathit{SourceOrExcess})(b,1)$$

In fact, we can extend the degree of graphs to polynomial degrees. For that we provide a formal definiton of the *Unbalanced* problem 2.3.1

Definition 2.3.1: *Unbalanced* problem definition

Input: We are given tuple $(\mathbf{S},\mathbf{P},f,g)$, such that $\mathbf{S} = \{S_i\}_{i\in[f(n)]}$ and $\mathbf{P} = \{P_i\}_{i\in[g(n)]}$ and $f,g : \mathbb{N} \to \mathbb{N}$ are poly-time computable polynomial functions. This describes a graph $G = (V,E)$ such that $V = \{0,1\}^n$ and

$$\forall(x,y) \in E, \exists i \in [f(n)], j \in [g(n)] : S_i(x) = P_j(y)$$

We enforce the condition that $\forall P_i : P_i(0^n) = 0^n$ and $\exists S_j : S_j(0^n) \neq 0^n$
Output: A node $v \in \{0,1\}^n \smallsetminus \{0^n\}$ such that *in-deg*$(v) \neq$ *out-deg*(v).
If $f(\cdot) = p$ or $g(\cdot) = k$ for some $p,k \in \mathbb{N}$, we will denote the problem as *Unbalanced*(p,k)

Lemma 2.3.1

Unbalanced is **PPAD**-complete [29]

The proof of the above lemma is presented in the appendix 5.6.1 As previously mentioned, Hollender et al. [29], demonstrated how extending the number of known sources of the *EndOfLine* problem does not affect its **PPAD**-completeness. Moreover, combinatorially we can observe that if a *EndOfLine* problem has k known sources, it implies that there must exist k at least k solutions. This raises an interesting question as to what happens when we combine the number of known sources of a graph with its degree. For starters, we can observe that by lemma 2.3.2, the *Unbalanced* problem is not affected.

Lemma 2.3.2

$$\mathbf{\#PPAD}(\mathit{PolyS\text{-}Unbalanced}) \subseteq \mathbf{\#PPAD}(\mathit{Unbalanced})$$

For simpler instances, such as the *EndOfLine*, the reduction demonstrated by Hollender is non parsimonious, meaning there is a possibilitiy of an oracle separation betwwen *EndOfLine* and *2S-EoL*. In fact one can observe that we can create a 3D-matrix of problems such that

$$\forall i,j,k \in \mathbb{N} \cup \{\mathit{Poly}\} : M_{i,j,k} = \mathit{kS\text{-}Unbalanced(i,j)}$$

Where ω denotes polynomial values. Morover, these hierarchies seem to reveal some hidden combinatorial structures in **PPAD** and in **TFNP** as a whole. For example, one can demonstrated the following claim 2.3.1

Claim 2.3.1: SourceOrExcess with fixed degrees and sources

Given $\alpha,\beta \in \mathbb{N}$, the problem α*S-SourceOrExcess*$(\beta,1)$ will have at least $\mu = \lceil \alpha/\beta \rceil$ solutions.

The proof our statement will be provided in the appendix 5.6.1. This implies there exists subclasses in **PPAD** which we can denote as **PPAD**(k) where we have **PPAD** problems with at least k solutions. This may lead in future directions where we can ask about combinatorial interpretations of **PPAD**$(\alpha) - \beta$ for $\alpha \geqslant \beta$.

2.4 Supplementary Findings

In this section we would like to present an additional finding with respect to the parsimonious nature *PureCircuit* and **CLS** problems. We define the following subclass of *Tarski* 1.2.15:

Definition 2.4.1: *KleeneTarski* problem definition

Given $F : \mathbb{T}^n \rightarrow \mathbb{T}^n$, where F is a *natural* function 1.2.20, find $x \in \mathbb{T}^n$ such that $F(x) = x$. We assume F will be represented as a circuit that uses set of gates $\{\wedge, \vee, \neg, \mathbf{0}, \mathbf{1}\}$.

Proposition 2.4.1: Parsimonious Reduction between *KleeneTarski* and *PureCircuit*

$$\#KleeneTarski \subseteq \#PureCircuit$$

Proof. Given an input circuit $C : \mathbb{T}^n \rightarrow \mathbb{T}^n$, construct *PureCircuit* instance:

1. Initiate a vector of n nodes, which we denote as s.
2. Construct the C using the standard set of gates and apply $C(x)$ to create output vector o.
3. Copy the results back into s.

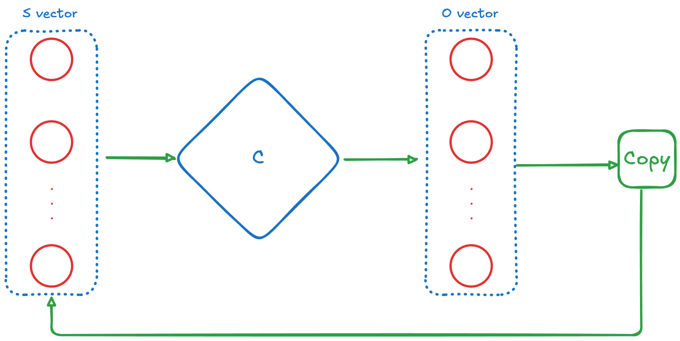


Figure 2.4.1: TARSKI construction

We can visualise the above construction in figure 2.4.1. We can observe that, for any valid assignment $\mathbf{x} \implies \mathbf{x}[s] = \mathbf{x}[o]$, which implies $\mathbf{x}[o] = F(x) = \mathbf{x}[s]$. □

The above problem demonstrates how we can use *PureCircuit* to find fixed points in *Tarski*. We argue that this may relate with ATMs as the existence of the assignment function depends on the creation of partially ordered chains of assignment functions. One could investigate the possibility of extending *KleeneTarski* to incorporate the configuration functions of ATMs.

Lastly in our appendix we demonstrate how one can simplify the *PureCircuit* problem in order to find parsimonious reductions to *EndOfLine* problem. We mainly demonstate, how certain conditions such as the cyclicity of *PureCircuit* or nature of *Purify* can affect the counting complexity of the problem.

Chapter 3

Software Deliverable

In the current section, we give an extensive overview of the software tool we developed to help with our theory crafting. Core idea is to be able to draw *PureCircuit* gadgets, verify correctness and find solutions. We split this chapter giving an overview on the tooling we used, the architecture of the software and the algorithms we deployed for solutions finding.

3.1 Tooling Selection

3.1.1 Rust Programming Language

The development language used for is Rust [1]. Rust is a systems-level compiled programming language that can achieve high speeds due to its low-level compilation, while maintaining memory safety. Moreover, it offers high level features such as its strong algebraic and trait type system which allows us to represent complex states.

3.1.2 Bevy and the ECS architecture

`Bevy` [2] is a `Rust` library, primarily used for game development. It uses the Entity-Component-System architecture where entities with varying components exist in world state. Each entity interacts with each other based on systems. A visual reference is provided in the figure 3.1.1.

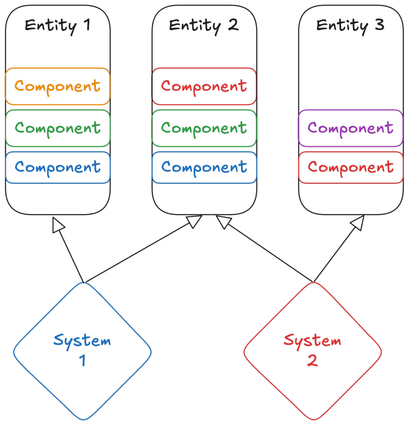


Figure 3.1.1: ECS visualisation implementation

The current tool comes with several benefits. It offers flexibility when it comes to drawing and interacting with shapes on a screen. We wanted to give our tool and editor-like aesthetic where

[1]https://www.rust-lang.org/
[2]https://bevy.org/

the user can move the screen and individual nodes themselves as well as interact with them. Moreover, we are able to decouple the logic from entities to their individual components and express more intricate relations. Regarding the current application, we chose to represent the graph nodes as separate entities which interact with each other based on their edges.

3.1.3 Software Testing methodology

The core philosophy of our testing is Test-Driven-Development. This methodology was used to develop and validate our critical parts of our code, such as graph validation or solution enumeration. The main idea of this development is to first write the unit test, and then develop the code to resolve them. This allows us for the code to be self-documenting as well as ensuring that we are acting within the expected scope.

Another testing technique that we heavily took advantage of is *Property-based Testing*, which we implement using the `Proptest` library [3]. This idea was inspired by the `Haskell` library `QuickCheck` [14], where we test the properties of a pure function across a range of inputs. The idea is, the more complex, the input of the function, the greater is the input space and therefore harder and cumbersome to check all possible combinations or edge-cases. With `Proptest`, we check whether a property of our function is maintained across all possible inputs, such as a sorted output, or a validating assignment.

For our purposes, we use value-based property testing in which: We define a set of types where in each one we define a *generator* and a *shrinking strategy*. The *generator* is responsible for generating significant values that represent the specific type. The *shrinking strategy* is responsible for finding the smallest case in which our case fails. We then define test cases that automatically run the generator and shrinker procedures until we have tested that our function passes across all generated inputs, or finds the smallest test case it fails on.

3.2 Program Visualisation Summary

We provide a figure of our application in the figure 3.2.1. Our application contains a UI legend and a drawing board. The UI legend provides the user with the information about the current state of the app. Moreover, we provide the user with the key bindings that we assigned, in order to modify the state and lastly buttons that will execute the solution finding algorithm. We will dive deeper into the functionalities in the upcoming sections.

Our application is composed of 5 different states which are summarised in the table 3.2.1. Use the aforementioned states, we are able to add/remove edges as well as move them around or change their values. We added visual indicator so if gate has an incorrect arity or incorrect assignment, the user will be notified. Moreover, we attached numbers to our edges so the user we will be able to differentiate between inputs and outputs. This is mostly useful for the PURIFY gate as each output behaves differently. In our appendix we will attach user stories that explain in greater detail how one can perform the above operations.

State	Functionality
Mouse Mode	Determines whether we want to work with edges or nodes
Node Mode	Determines whether we want to add a value node or a gate node
Value Mode	The type of value node we want to add
Gate Mode	The type of gate node we want to add
Edge Mode	Determines whether we want to add an edge or delete an edge

Table 3.2.1: Application states

[3]https://github.com/proptest-rs/proptest

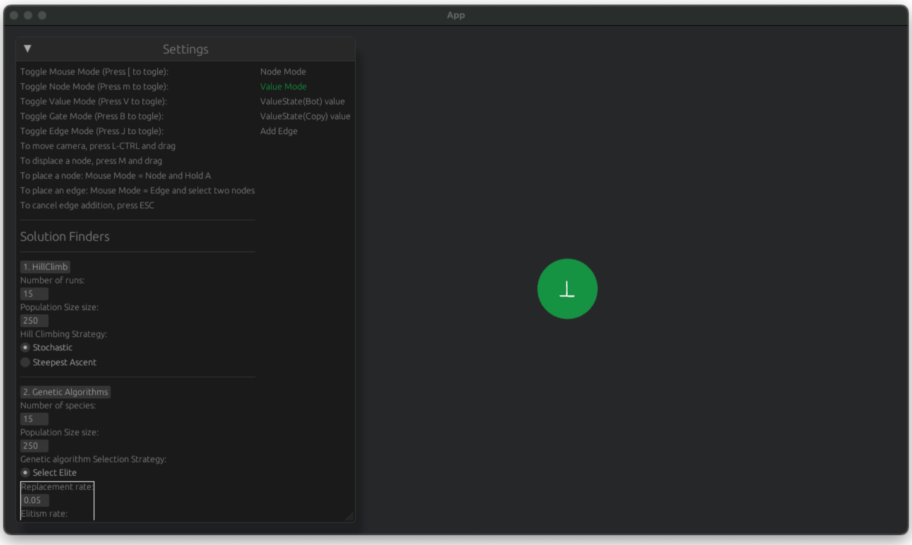


Figure 3.2.1: Application Layout

3.2.1 Solution Selection

For the solution selection we opted with metaheuristic algorithms. Since these problems are conventionally hard to find solutions especially as the number of nodes increase, we focused on the idea of approximation algorithms in order to find mostly satisfying assignments in large graphs or gadgets. With the help of the `genetic_algorithm`[4] library, we use the *HillClimbing algorithm* and *Genetic algorithm* in order to find solutions. We will give a brief explanation on how they work in the upcoming sections but in general we express our graph as a chromosome of *value* nodes which correspond to nodes in the original graph. Moreover, we formulate our problem as series of gates with references to the location of their incident value nodes, and we count the of unsatisfying gates. The goal of either algorithm is to minimise the number of errors. Once an assignment is found, we pass the solutions back into our graph and display them.

Hill Climbing Algorithm

The idea of hill climbing algorithms is to start by some base solution and repeatedly improve the solution. We allow for 2 strategies, one of them strictly always chooses the best neighbour whereas one allows for stochastic choice. Moreover, we allow for additional option such as multiple re-runs in order to increase the odds of finding a successful assignment.

Overall this algorithm, one of the simplest and faster algorithms to run and can work well for convex problems as explained in [47]. It may significantly struggle against large graphs or instances that multiple changes need to occur in order to reach an optimal point. In fact we can further argue that hill climbing may struggle on the *Purify* gate due to the non-linearity that they present.

In our figure, we use the arguments as shown in figure 3.2.2. The number of runs determine how many times we can re-run the algorithm until an optimal solution is found. Population size determines the number of assignments to work with in parallel. The strategies are the one that we described previously.

[4]https://crates.io/crates/genetic_algorithm

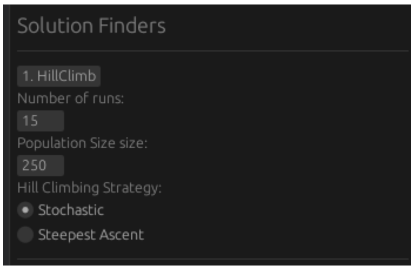


Figure 3.2.2: Hill Climb parameters

Genetic Algorithm

Genetic algorithms are based of the idea of evolution. They were introduced by [28], it is a useful algorithm has found several usages in computer science, particularly in optimisation problems such as *TSP* problems [41] or *Neural Networks* [50, 49].

We summarise the steps of the algorithm as follows, using figure 3.2.4 for reference. We start with a population of chrosomomes. We define a function that evaluates how good a specific chromosome is, also known as a *fitness function*. We then apply the *selection method* which picks the parents for the next generation. The *crossover method* determines how to combine the genes of two parents in order to create the offspring. And lastly we have the *mutation step*, where we randomly mutate the genes of the new population in order to escape from local minima. We can repeat the above series of steps until we reach the target value or until we pass the maximum timestep limit. In our application, using figure fig:chap-4:ga-params, we use refer to the following list as a summary of parameters:

1. **Number of species**: We allow for our algorithm to evolve population independently and then merge each population at the end to find the optimal solution. Increase the odds of avoiding local minima.
2. **Population size**: Number of chromosomes per species.
3. **Selection step**: We allow for two selection methods: *Tournament based* and *Elitism selection*. *Elitism selection* selects the based elements of each population whereas *Tournament based* selection creates random sets tournaments across our chrosomomes and picks the best ones from each.
4. **Crossover step**: We allow for two main types of crossover methods:
   (a) *Crossover Uniform*: Each gene of the child is randomly selected from either parent. This allows for diverse solutions to be explored. Can reduce diversity but performs well on large population sizes [36].
   (b) *Crossover MultiPoint*: Using figure 3.2.5 as reference, we select crossover points that determine cutoff points from each parent. For each even sequence we use the sequence of the other parent. This method can be useful if consecutive pairs of genes are needed to improve the fitness.
5. **Mutation step**: Given the offspring of the **Crossover step**, we stochastically modify each gene in order to introduce diversity, which allows our chromosones to explore the solution space and escape local optima [36].

Algorithm comparison

Both algorithms are capable of finding an assignment. The *HillClimbing* is much cheaper to run but more prone to be stuck at local minima [6]. The *Genetic algorithm* is much more

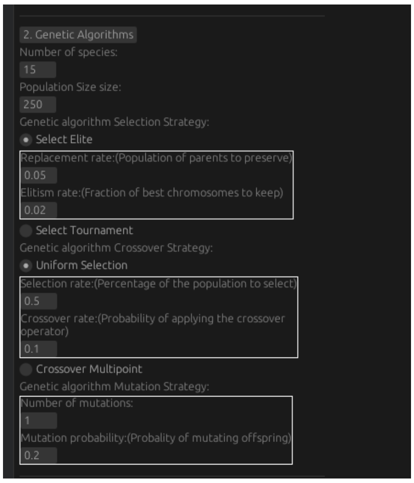


Figure 3.2.3: Genetic Algorithm parameters

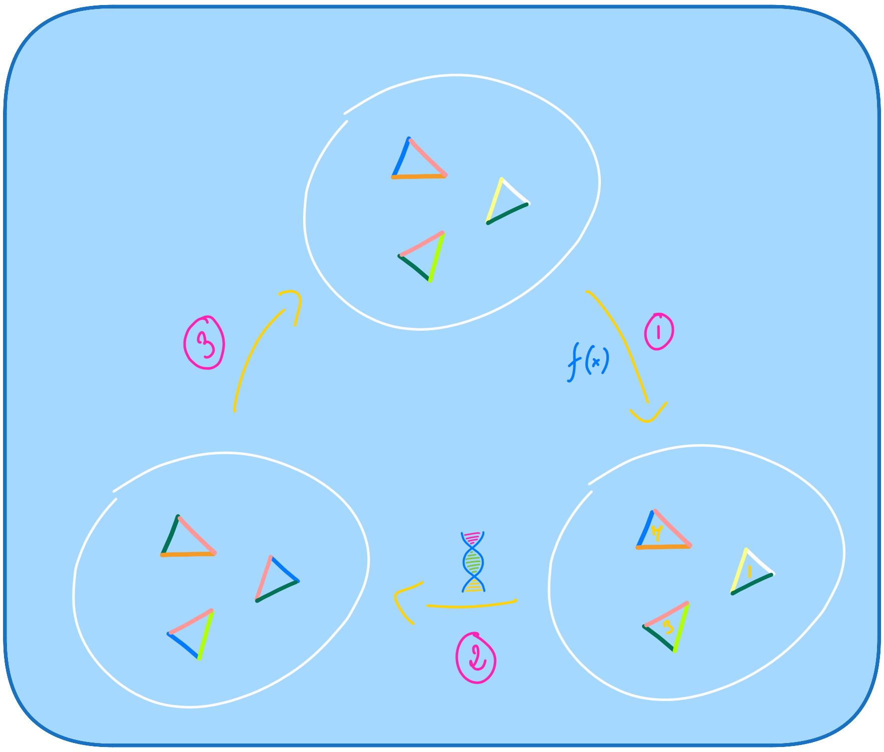


Figure 3.2.4: Visual representation of the genetic algorithm cycle. Image source [52]

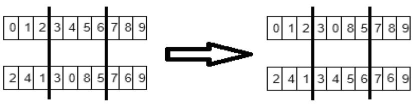


Figure 3.2.5: Crossover with multiple cutoff points. Image source [36]

computationally expensive out of the two but can be fine-tuned sufficiently to almost always find a solution. In general, there are many metaheuristic algorithms that can be applied as listed in [1] but we for now we consider them as future extensions.

3.2.2 Solution Enumeration

To enumerate through solutions, we employed a traditional backtracking algorithm. Our idea is similar to the DPLL algorithm used for *SAT* solving, where we recursively assign values to our graph. If all the values have been assigned a value, then we return the current assignment. If at a specific node, there are no more values that we can assign then we reject the current solution. We reduce the computational burden by prioritising nodes that can take only one value and each time we limit the solution set of each gate based on the assignment of the surrounding nodes. We summarise our algorithm in 3.2.1 and we summarise our algorithm into the following components: **Backtrack Initialisation**, **Unit Propagation operation**, **Set Simplification operation**, and **Backtrack Operation**.

Algorithm 3.2.1 PureCircuit Backtracking Algorithm Enumerator

```
function BacktrackInstatiation(pc : PureCircuitInstance)
    V ← PCExtract(pc)                        ▷ Extract value nodes from the PureCircuit instance
    S ← Arr(|V|, KleeneSetInit({0, ⊥, 1}))  ▷ Create a state vector that contains the KleeneSet data-structure.
    Q ← PriorityQueueInit(∅)
    for el ∈ V do
        pq_key ← KeyStruct(3, el.is_source(), el.num_neigh())
        PushPQ(Q, pq_key, el.index)
    return S, Q
function BacktrackPureCircuit(pc :PureCircuitInstance)
    S, Q ← BacktrackInstatiation(pc)
    A ← Arr(|V|, ∅)
    r ← BacktrackIteration(pc, A, S, Q)
    return r
function BacktrackIteration(
     pc :PureCircuitInstance,
     A :Arr[Kleene],
     S :Arr[KleeneSet],
     Q :PriorityQueue
)
    if IsEmpty(Q) then ▷ If all nodes have been assigned a value, that means we found a solution and therefore we return the current assignment.
        return [A]
    min_graph_ind ← ExtractMin(Q)
    r ← []
    for v ∈ S[m] do
        S′ ← Clone(S)
        Q′ ← Clone(Q)
        A′ ← Clone(A)
        result ←UnitPropagate(pc, min_graph_in, v, A′, S′, Q′) ▷ For each assignment in each solution set, clone the datastructure and assign the value selected
        if result = ⊥ then ▷ If our algorithm return false, it means that the current assignment is invalid and we ignore it
            Continue
        r′ ←BacktrackIteration(pc, A′, S′, Q′)
        r ← ConcatArray(r, r′)
    return r
```

```
function UnitPropagation(
    pc :PureCircuitInstance,
    nod_ind :GraphIndex,
    nod_val :GraphValue,
    A :Arr[Kleene],
    S :Arr[KleeneSet],
    Q :PriorityQueue
)
  AssignNode(nod_ind, nod_val, A, S, Q)
  PropagateNode(pc, nod_ind, nod_val, A, S, Q)
  while ¬IsEmpty(Q)∧ PeekKey(Q).set_length = 1 do ▷ While the queue contains an element
  whose value can take only one element, assign its value
    ind ← ExtractMin(Q)
    val ← S[ind]
    AssignNode(ind, val, A, S, Q)
    PropagateNode(pc, nod_ind, nod_val, A, S, Q)
  if IsEmpty(Q) then                                ▷ If every element is assigned, accept
    return ⊤
  if PeekKey(Q).set_length = 0 then ▷ If the queue contains an element which does not have a
  satisfying assignment, discard the current assignment
    return ⊥
  return ⊤
function PropagateNode(
    pc :PureCircuitInstance,
    nod_ind :GraphIndex,
    A :Arr[Kleene],
    S :Arr[KleeneSet],
    Q :PriorityQueue
)
  for g ∈ GetAllGateNeighbours(pc, nod_ind) do
    I ← GetValueSets(Input)
    O ← GetValueSets(Output)
    (I′, O′) ← SetSimplification(g, I, O) ▷ Determine the available value sets for each input
    output element given the elements that have been already assigned
    PropagateSetValues(Q, S, I′, I) ▷ Propagate the changes to their value sets and to their
    queue keys for both the input and output elements
    PropagateSetValues(Q, S, O′, O)
```

Backtrack Initialisation We initialise two main structures, a priority queue Q, and a solution map S for each node. S the solution map contains the *KleeneSet* data structure which is the set of values a node can take. We assume that all nodes can take $\mathbb{T}$ in the beginning.

The priority queue will essentially dictate which nodes to select and evaluate. We want to reduce the recursive calls as much as possible and eliminate any partial assignments that lead to unsatisfying solutions. The queue stores the node references of the value nodes and uses as keys the structure KeyStruct(e). The KeyStruct is a special structrue that orders elements lexicographically based on the following order:

1. **In descending**: The number of possible value elements the current node can take
2. **In Ascending**: Whether the node has an *in-degree* of 0
3. **In Ascending**: The total number of neighbours

We argue that the above strategy minimises the breadth of the recursive calls. Moreover, for each node with *in-degree* of 0 as for each value of their assignment, a solution should exist. The heurestical argument for this can be observed in Deligkas et al. [21], where there construction for finding solutions allows for source nodes to be independent. Lastly, prioritising nodes with the highest neighbour count implies that we propagate changes to a greater number gates and therefore maximise the influence of the *Unit Propagation* procedure.

Unit Propagation operation The algorithm assigns the selected value to a node by modifying to the solution set and the value set. Then for each gate that is incident to the node apply the *Set simplification operation*. While the top element of the queue has one possible solution, extract the solution and repeat the *unit value propagation operation* until the queue is empty or the cardinality of the value set at the head of the priority queue is different than 1. If it's zero then current assignment is not valid and we return false, otherwise we return true.

Set Simplification operation Given a gate, we have a set of possible values for each of our nodes, as determined by our value set. The idea is that we want to reduce the set possible values by following the rules of the gate. For example, we given an **AND** gate. If the output value is assigned the value 1, then we know that the inputs can only take values $(1, 1)$ and therefore we restrict the values of the input sets to $\{1\}, \{1\}$. If the current solution set of a node is v_t, then its updated set is $v_{t+1} \leftarrow v_t \cap \omega$, where ω are the set of values denoted by the simplification operation. We provide the rules for **AND**, **OR** and **Purify** in the tables 3.2.2, 3.2.3 and 3.2.4. The rules for the other gates will be provided in the appendix 5.6.1. If a value is not assigned, we denote it as $\cdot$.

AND input	Output sets
$\forall v \in \mathbb{T} : (\cdot, \cdot, v)$	$(\mathbb{T}_{\geqslant v}, \mathbb{T}_{\geqslant v}, \{v\})$
$\forall v \in \mathbb{T} : (\cdot, v, \cdot)$	$(\mathbb{T}, \{v\}, \mathbb{T}_{\leqslant v})$
$\forall a, b \in \mathbb{T} : a < b \implies (\cdot, a, b)$	Invalid combination
$\forall v \in \mathbb{T} : (\cdot, v, v)$	$(\mathbb{T}_{\geqslant v}, \{v\}, \{v\})$
$\forall a, b \in \mathbb{T} : a > b \implies (\cdot, a, b)$	$(\{b\}, \{a\}, \{b\})$
$\forall a, b \in \mathbb{T} : (a, b, \cdot)$	$(\{a\}, \{b\}, \{\min(a, b)\})$
$\forall a, b, c \in \mathbb{T} : (a, b, c)$	$(\{a\}, \{b\}, \{c\})$, only if the triplet (a, b, c) is a correct **AND** assignment

Table 3.2.2: **AND** value propagation rules

OR input	Output sets
$\forall v \in \mathbb{T} : (\cdot, \cdot, v)$	$(\mathbb{T}_{\leqslant v}, \mathbb{T}_{\leqslant v}, \{v\})$
$\forall v \in \mathbb{T} : (\cdot, v, \cdot)$	$(\mathbb{T}, \{v\}, \mathbb{T}_{\geqslant v})$
$\forall a, b \in \mathbb{T} : a < b(\cdot, a, b)$	$(\{b\}, \{a\}, \{b\})$
$\forall v \in \mathbb{T} : (\cdot, v, v)$	$(\mathbb{T}_{\leqslant v}, \{v\}, \{v\})$
$\forall a, b \in \mathbb{T} : a > b(\cdot, a, b)$	Invalid combination
$\forall a, b \in \mathbb{T} : a > b(a, b, \cdot)$	$(\{a\}, \{b\}, \{\max(a, b)\})$
$\forall a, b, c \in \mathbb{T} : (a, b, c)$	$(\{a\}, \{b\}, \{c\})$, only if the triplet (a, b, c) is a correct **OR** assignment

Table 3.2.3: **OR** value propagation rules

Backtrack Operation The *backtrack* operation is summarised in 3 key steps. If the priority queue is empty then we found a satisfying assignment. Otherwise, pick a node from the priority queue. For each value in its set, create a copy of the current queue, current solution assignment and value set state and apply the *Unit value propagation* operation. After that create a recursive call apply the *backtrack opertation* again until the current recursion chain terminates with a solution or it rejects. At the root of every execution call, merge the solution assignments in each level and return the sets. The root call will return a set of every possible valid assignment set.

Purify input	Output sets
$\forall b \in \mathbb{B} : (b,\cdot,\cdot) \vee (b,\cdot,b) \vee (b,b,\cdot)$	$(\{b\},\{b\},\{b\})$
$(\perp,\cdot,\cdot)$	$(\{\perp\},\mathbb{T}_{\leqslant\perp},\mathbb{T}_{\geqslant\perp})$
$(\cdot,\cdot,0)$	$(\{0\},\{0\},\{0\})$
$(\cdot,\cdot,1)$	$(\mathbb{T}_{\geqslant\perp},\mathbb{T},\{1\})$
$(\cdot,\cdot,\perp)$	$(\{\perp\},\{0\},\{\perp\})$
$(\cdot,0,\cdot)$	$(\mathbb{T}_{\leqslant\perp},\{0\},\mathbb{T})$
$(\cdot,1,\cdot)$	$(\{1\},\{1\},\{1\})$
$(\cdot,\perp,\cdot)$	$(\{\perp\},\{\perp\},\{1\})$
$(\perp,0,\cdot)$	$(\{\perp\},\{0\},\mathbb{T}_{\geqslant\perp})$
$(\perp,\cdot,\perp)$	$(\{\perp\},\{0\},\{\perp\})$
$(\perp,\cdot,1)$	$(\{\perp\},\mathbb{T}_{\leqslant\perp},\{1\})$
$(\perp,\perp,\cdot)$	$(\{\perp\},\{\perp\},\{1\})$
$\forall a,b,c \in \mathbb{T} : (a,b,c)$	$(\{a\},\{b\},\{c\})$, only if the triplet (a,b,c) is a correct **PURIFY** assignment
–	Any unlisted combinations are considered invalid

Table 3.2.4: **Purify** value propagation rules

Correctness and Running Time analysis

For the running time analysis, we can observe that each time we assign a node, the unit propagation operation, restricts the solution set of the value nodes that are incident by one gate. Even in the worst case scenario where no reduction occurs, we can clearly see that each time, our priority queue will select a node and will recurse through each of its possible values, creating a call tree with depth of at most n. Therefore, we can argue that the running time of the algorithm is $O(3^n)$.

As for correctness, the algorithm essentially picks nodes in a certain order and selects a value that we believe that will give a satisfying assignment. We only have to look at what happens at the *unit propagation* stage. We can observe that in each stage, we are essentially limiting the set of solutions to only the ones that will give a satisfying assignment. The idea is that we are fast-forwarding the backtracking calls by taking advantage of the localised changes of each gate. Values that we discard are nodes that guarantee to give unsatisfying assingments, and therefore we are not discarding any solutions. This implies that our resulting array of assignments contains only valid solutions, and this array covers the maximal solution set.

We provide a table in 3.2.5 with respect to the number of backtracking calls. The smaller the number the less smaller the recursion tree and therefore less values to check. We can observe that the premature cancellations and the value set reductions allow us to keep a low number of calls despite the graph size. We will provide a list of the solutions in the appendix 5.6.1.

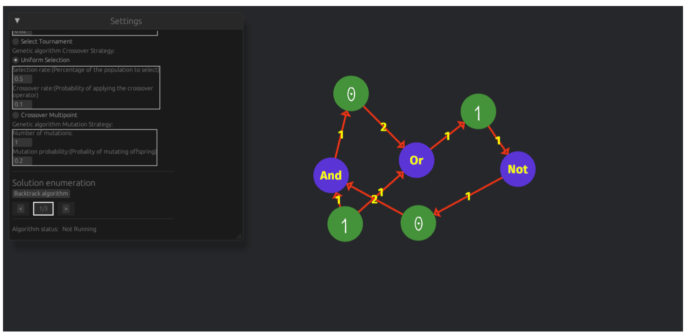


Figure 3.2.6: Small Circuit example

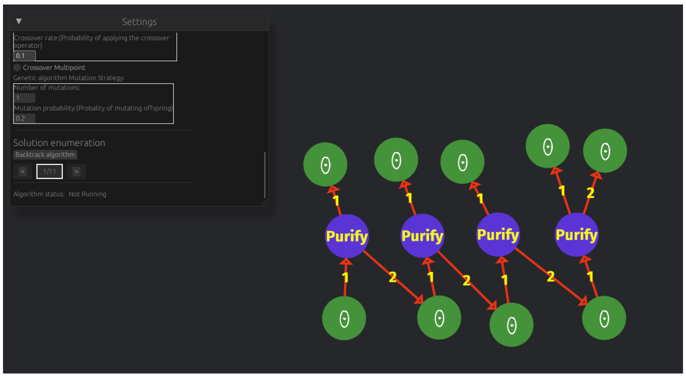


Figure 3.2.7: Bit Generator

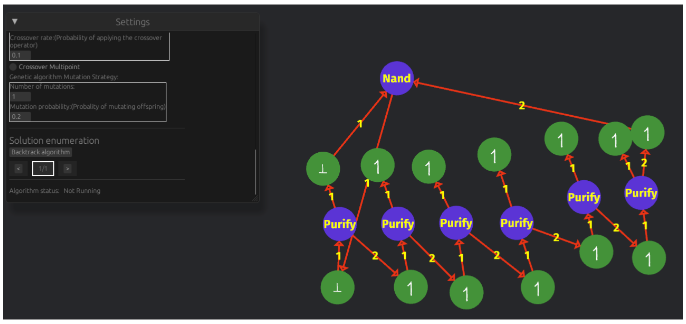


Figure 3.2.8: Complex Circuit

Circuit types	Number of nodes	Number of backtrack calls
Simple circuit 3.2.6	4	2
Medium circuit 3.2.7	9	9
Complex circuit 3.2.8	13	3

Table 3.2.5: Number of Backtrack calls

Chapter 4

Chapter Conclusions and Future Directions

4.1 Conclusions

In summary, we expanded upon the combinatorial nature of **PPAD**, under the lens of topolgical representations and the *PureCircuit* problem. First we showed how *Kleene logic* can be embedded into **PPAD** using as a primary example the *StrongSperner* problem and how certain problems can retain their **PPAD**-completeness. Morevover, we demonstrated that under hazard-free conditions and discreteness, how one can parsimoniously reduce *HF-DiscreteStrongSperner* to *PureCricuit*. We were also able to construct a parsimonious reduction from the *EndOfLine* problem to *StrongSperner* instrances with discrete and antipodal properties. Given that, we bridged the gap by demonstrating how the parsimonious reduction chain to *PureCircuit* from the *EndOfLine* depends on the hazard free nature of the *AntipodalStrongSperner*.

Extending the topological interpretation idea, we showed first hand how graph problems such as the *SourceOrExcess*$(2, 1)$ can also be encoded topologically and demonstrated how the dimensionality of the space we encode it, can affect the counting hardness of problems. We argued that this has some substantial consequences as first we found another **PPAD**-complete problem with no clear combintorial interpretation. More interestingly, we conjectured the possibility of a connection between the degrees of graph and the dimensionality of its topological embedding.

Moreover we investigated the different counting classes that can be represented in **PPAD** by expanding on Hollender et al.'s work [29]. Specifically we built upon their combinatorial argument by relating *SourceOrExcess* problems and argued about the potential infinite hiararchy of **PPAD** problems with respect to their minimum number of solutions. In addtion, we argued the importance of the *Unbalanced* problem as an important stepping stone to a total *PureCircuit* Cook-Levin theorem, due to its overall relation with all the other unbalanced digraph problems.

Lastly, we showcased our work on our *PureCircuit* visualisation software. We demonstrated its core functionalities such as *PureCircuit* editor tool with assignment verification. Moreover we implemented several metaheuristic algorithms for assignment finding assignments in large graphs, such as *Hill Climbing* and *Genetic Algorithms*. Most importantly, we also implemented a specialised backtracking algorithm whick takes advantage of the structure of the graph. Despite its high worst-case running time, we manage to use several greedy methods to cut down breadth and with of the tree making it ideal for verification of small circuits.

4.2 Future Work

4.2.1 Theory Directions

In our research were able to highlight several directions that one could take. We mainly highlight to directions: **HF-PPAD** in **PPAD** and topological representations of directed graphs

HF-PPAD in PPAD

In our sections 2.1 and 2.2, made references to how Kleene logic can be incroportated with **PPAD** problems. For the *nD-StrongSperner* problems we observed that *Kleene logic* provides a natural extension and is equivalent to its non hazard counterpart. For other problems such as *StrongSperner*, we observed that this is not clear by current complexity assumptions and indeed proving **PPAD**-hardness would be essential for establishing the counting hardness of *PureCircuit*. But one could ask further questions such as, could we define complexity classes such as **HF-PPAD** and could we show that **HF-PPAD** = **PPAD**. Along similar line, we can also ask the questions as to existence of problems *HF-nD-Counting-SS* and if they are parsimonious to their non hazard-free variants. We know that the non counting version can be resolved parsimoniously but we argue that it is not so simple when we omit panchromatic cubes as by doubling the cubes, we may introduce more solutions.

Even more naturally, one can also ask further questions such as, how does hazard-freeness affect **TFNP** as a whole. We conjecture that one can make similar observations to the *StrongTucker* problem [2], in order to create similar classes such as **HF-PPA** as it also uses a very similar colouring argument as the *StrongSperner* argument. Morevore we were able to observe how we can use *Kleene logic* with *PLS* subproblems like *Tarski*. We believe that this shift in perspective may also give insights to other areas such as the circuit complexity *Hazard-Free Circuits*.

Parsimonious Topological Representations of PPAD problems

Conjecture 4.2.1: *nD-CountingSS* and *Unbalanced*

There exist some polynomial computable function $f : \mathbb{N}^3 \to \mathbb{N}$, such that given $\alpha, \Delta_i, \Delta_o \in \mathbb{N}$ and $m = f(\alpha, \Delta_i, \Delta_o)$

$$\mathbf{PPAD}(\alpha\text{-}SUnbalanced(\Delta_i, \Delta_o)) \subseteq \mathbf{PPAD}(mD\text{-}CountingSS)$$

What the above conjecture essentially describes is, given a number of known sources and the maximum number of input/output vertices of graph, can we create some *nD-CountingSS* that is parsimonious to its graphical representation? Since we know that the problem *Unbalanced* can parsimoniously count every variant, can we also reduce *Unbalanced* to a topological problem? A reduction like that would be an important milestone for the usage topological *PureCircuit* constructions for enumerating solutions.

PureCircuit Gate Extension

The proof for the **PPAD**-inclusion can be summarised as expressing a *PureCircuit* instances as a *Brouwer* function from $[0,1]^n \to [0,1]^n$, given fixed point $x^\star$ then we denote the folloiwng assignment: if $x_i^\star \in \{0,1\}$ then $v_i = x_i^\star$, otherwise $v_i = \bot$. We argue that there may exist gates which exhibit any behaviour of the sort $f \in [0,1]^k \to [0,1]^m$ such that f is contiuous, where they may make our reductions easier and more straight forward. This could raise additional directions as to the behaviour of the additional functions and whether they may affect the combinatorial nature of the *PureCircuit* problem.

4.2.2 Software future Directions

Algorithmic Extensions

Although the aforementioned algorithms work as expected, there are several improvements that can be made. For all three of the algorithmic methods we mentioned, the first and most important extension is to treat weakly directly connected components individually. Trivially an assignment in one weakly connected component should not affect the solution set of the other one. We can therefore reduce the search spaces of our problems by finding the solution sets in each graph separately. Moreover, for the *backtracking* algorithm, we can extend the above idea by considering the cartesian product of the solution sets, meaning:

$$\mathbf{sol}(G) = \bigtimes_{k \in [k]} \mathbf{sol}(G_k)$$

Where G_i is the ith connected component.

We can also optimise the backtracking algorithm even further. First one can observe that if our graph was acyclic, then the permutation of the nodes with no *in-degree* solutions essentially dictate the assignments of the downstream nodes (excluding nodes incident to purify gates). With this in mind, we can apply the following sequence of steps to make the backtracking algorithm simpler:

1. Identify the strongly connected components (SCCs) of the circuit, using polynomial time algorithms such as Kosaraju's Algorithm [48] or Gabow's Algorithm [27].
2. Treating SCCs as an individual node, identify nodes by their distances from base nodes (nodes with no incoming edges)
3. Extend the backtracking algorithm so instead of the using the is start label, use the distance from base instead. The greater distance a node has from the base, the less impact it has and therefore should be postponed in later stages of the backtracking calls.

We argue that the above heuristic may result in improved average times but of course further research can be done in order to determine more suitable selection heuristics.

Chapter 5

Project Management

5.1 Project management methodology

Due to the hybrid nature of our project, we wanted to make sure were able to both the software portion and the theory portion of the project. With respect to our software management, we primarily used the waterfall approach while incorporating iteartive sprints, since expected the core set of requirements to change over time as we developed our theories and ideas.

With regard to the research management, we revolved our methodology around the *Zettelkasten method*. *Niklas Luhmann* developed this method to keep track of notes and information when dealing with a massive wave of publications [9]. The method embodies the following elements:

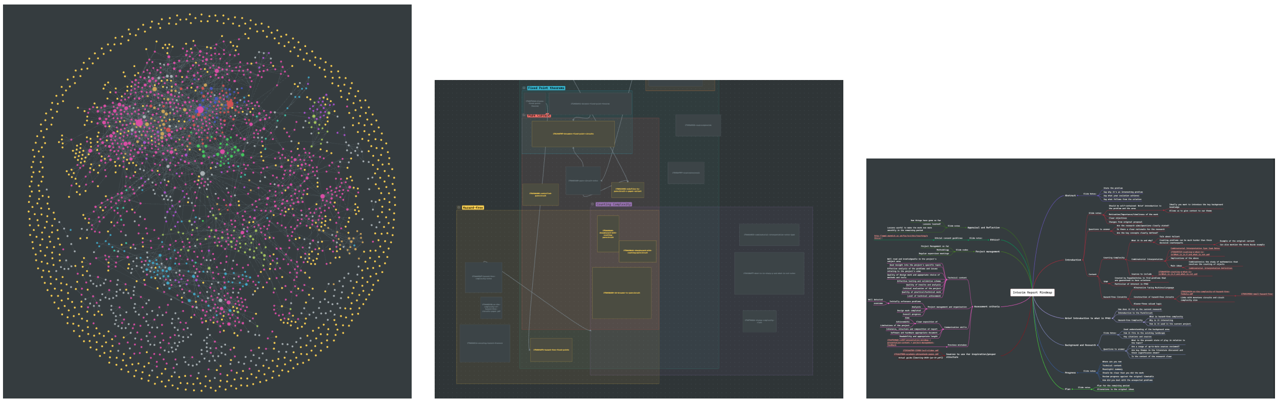
1. **Literature Notes**: Atomic notes that have been extracted from each paper which contain the ideas and techniques of the paper.
2. **Fleeting notes**: Ideas that pop up randomly but are not really linked with any particular source.
3. **Permanent notes**: Reviewed literature notes and fleeting notes which have been promoted to **permanent** ones.

Each of these notes can be annotated with a tagging systems and relate to each other with the use of references.

5.2 Tooling

To aid our research management, we made use of the `Obsidian` [1] tool. `Obsidian`'s graphical reprentation of the semantical relations between nodes as well as its smart tagging system, made it easier to keep track of our ideas and on-going theories. Moreover it incorperates additional features as seen in the figure 5.2.1, such as a canvas or a mindmap. All these features allowed us to streamline and refine thought processes, throughout the project. In additon, to keep version control, we used `Github` [2] as it one of the most well known version control applications in industry. This allowed us to keep constant backups of our code and notes, whilst being able to refer back in case something unexpected occured.

[1] https://obsidian.md/
[2] https://github.com/



(a) Obsidan graph. Each dot represents a node and eack link corresponds to a reference between two nodes

(b) Obsidian Canvas

(c) Obsidian mindmap

Figure 5.2.1: Usages of Obsidian

5.3 Ethical Considerations

Due to the nature to the theoretical nature of the project, there were no ethical consideration that were required. All the data and diagrams that were created were either made by the author, redrawn by the author or borrowed from another paper. The libraries used to build the software were under the MIT licence.

5.4 Risk Management

5.5 Timetable

We provide a figure of our final iteration of our gantt chart in figure. Overall the project followed the expected timetable, with some changes that had to be made after the interim report. First and foremost, we prioritised finishing the software. This was mainly due to ensure a final deliverable as well as to have a tool to verify our *PureCircuit* gadgets. Moreover, we shifted our focus to find supporting arguments for objectives, we faced a lot of difficulties achieving our goals.

5.6 Appraisals and Reflections

Unfortunately there were some things that did not go as planned. First, a minimum viable version of the software was completed towards a much later stage project. In an idea scenario, it would have been more valuable for us had we prioritised the development of the visualisation tool as it would have allowed for a faster verification of results. Moreover, one of our major results from the interim report, ended up having a major issues that halted our process. Luckily we were able to adjust to that incident by finding results that help or may help resolve our objectives.

On the hand, we were able handle some of the incidents. Through our intensive note taking procedure, we were able to tackle a difficult topic by breaking ideas down and combining ideas from people. In addition, a lot of our failed ideas came together in the end such as, the main proofs in our sections in our sections 2.1 and 2.2, For each proof or theorem in those sections, we have 10 other failed attempts or ideas, from all which we were able to extrapolate the missing piece of the next puzzle.

Morevover, our extensive testing gave us confidence in our resulting applciation.

Severity	Probability	Description	Mitigation	Address
High	Medium	Software may not be feasible within the remaining time frame.	We focus on the validation and creation of instances. The current project is primatoly research-based, and therefore we are mainly prioritising the theoretical development aspect.	Not yet
Low	Medium	Software not identifying a correct solution.	Usage of TDD techniques and property testing to ensure correctness. Comparison with handmade instances.	Not yet
Medium	Low	Difficulty of showing the parsimonious reduction between PureCircuit and nD-StrongSperner.	Consult with the researchers that actively work on these problems. Lower the requirements to polynomially bounded reductions instead.	Not yet
High	High	Inability to make significant progress on the SourceOrExcess problems or the generalised statement.	Establish reductions from other problems in order to demonstrate #P hardness by starting from nD-StrongSperner.	Not yet
High	High	Incorrect proofs or reductions.	Analyse the problem under different constraints. Apply the duck method, where you attempt to explain the solution to a person who is not necessarily an expert. Validate proof with the supervisor or use the tool to investigate sample cases.	On-going
High	Medium	Develop combinatorial-friendly variants of PureCircuit.	Apply robustness on the gate set of PureCircuit. Develop new gates or variants are easier to work with or focus on a subset of solutions as explained in 1.2.1.	✓

Table 5.6.1: Risk management table.

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Appendices

PolyS-Unbalanced with Unbalanced

Proof of the lemma 2.3.1

Proof. To show our reduction, that we have an instance of *PolyS-Unbalanced*, where have $f(n)$ known sources. Assume that $g(n)$ describes the input degree and $h(n)$ the output degree. We create a new *Unbalanced* instance with in- and out-degree polynomials $p(n) = (\max\{g(n), h(n)\}) \cdot f(n)$. We will use the figure .0.1 to illustrate our reduction. Essentially, we are

where s_i is the ith known source of the original problem. For each node k of the ith group, denoted as v_k^i, we are adding edge $v_k^i \to s_i$. Since each s_i element has *in-degree* = *out-degree*, we can observe that we are not creating any additional solutions. Since this transformation is local, any other unbalanced node will stay the same and therefore the set of Solutions is preserved.

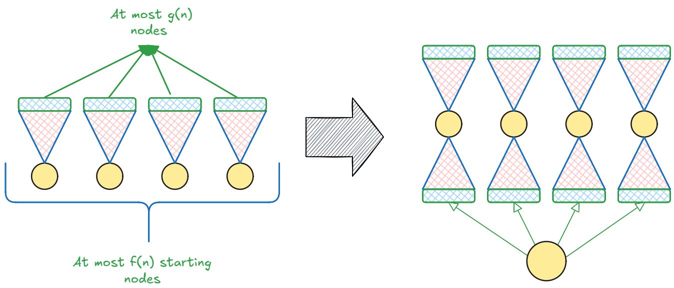


Figure .0.1: *PolyS-Unbalanced* transformation to *Unbalanced*

□

SourceOrExcess with number of solutions

We provide a proof of the 2.3.1.

Proof. Given αS-*SourceOrExcess*$(\beta, 1)$, in order to find the minimum number of solutions, we can assume that we there are no other sources other than the first α nodes. We argue that the minimum number of solutions is $\mu = \lceil \alpha/\beta \rceil$ since we can create μ sinks that connect to at most β nodes each. We will refer to these graphs as sink graphs.

We can demonstrate that any graph can be reduced to a sink graph. Our first observation is to convert every excess node to a sink node. To do that we can apply two simplification phases:

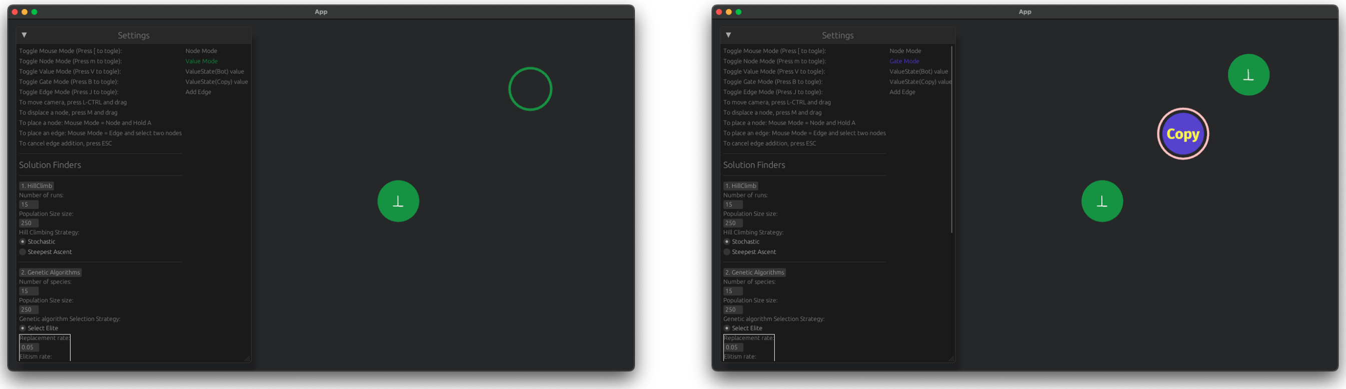
1. Removal phase: Remove all paths between two excess nodes or between excess and sink nodes. We can observe that after this steps our graph only contains sources and sinks.
2. Merging phase: If there are sinks with degrees $(a, 0)$ and $(b, 0)$ such that $a + b \leqslant \beta$ then we can merge the two to remove a solution. Keep repeating this process until no two sinks can be merged

We can observe that any graph will simplify to a sink graph and therefore we will always have at least μ solutions. □

Visualisation Program Demonstration

User stories

Node addition We demonstrate how we can add nodes. Essentially, ensure the user is in mouse mode, hold A and press left click to place the node, as seen in the figures .0.2. We observe in figure .0.2b that the Copy gate is circled in pink to denote invalid arity, meaning its missing its input and output edge. In addition, hovering over a node will highlight the node in yellow as seen in figure .0.3.



(a) Holding A gives an indicator of where the node will be placed

(b) Final graph after addition of values and gate nodes

Figure .0.2: Node addition

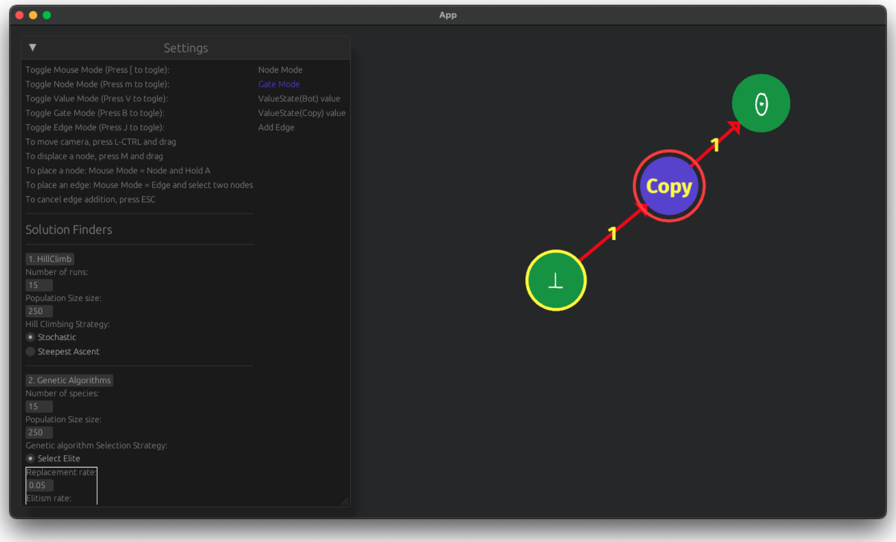


Figure .0.3: Node hovering

Edge addition To add an edge between two nodes, we select the source of the edge. We can observe from the figure .0.4, that we highlight the source node with light blue and the heterogeneous nodes with orange. If we select a gate, we highlight only nodes have an in-degree of 0. Pressing ESC or changing mode can cancel the selection operation. Selecting on any of the orange nodes completes the edge addition operation as seen in figure .0.5. We can also observe that since the Copy has no error circles since it has the correct arity and the input/output values satisfy the gate as seen in .0.6.

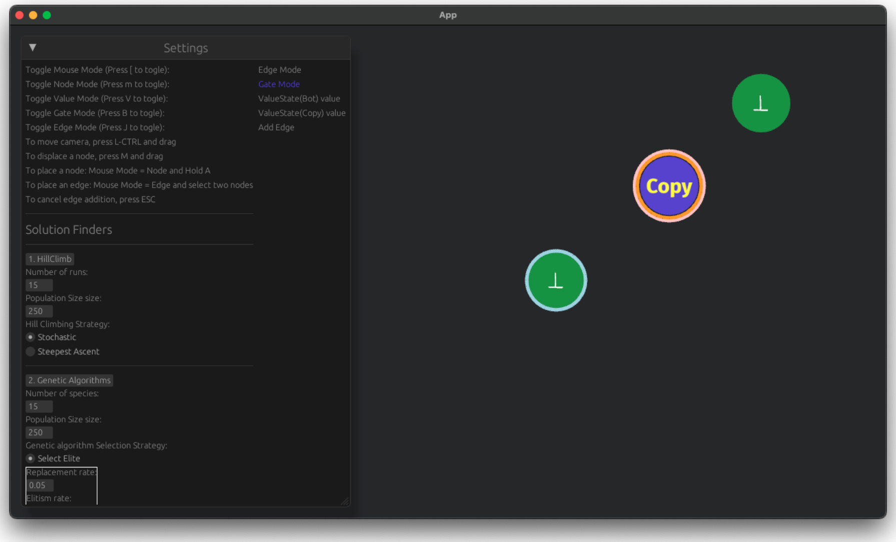


Figure .0.4: Edge addition: Select source node

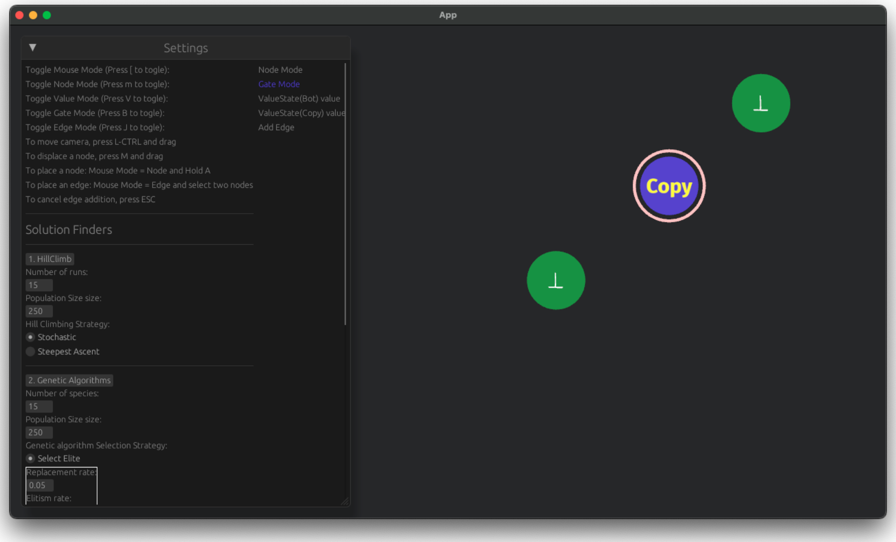


Figure .0.5: Edge addition: Select destination node

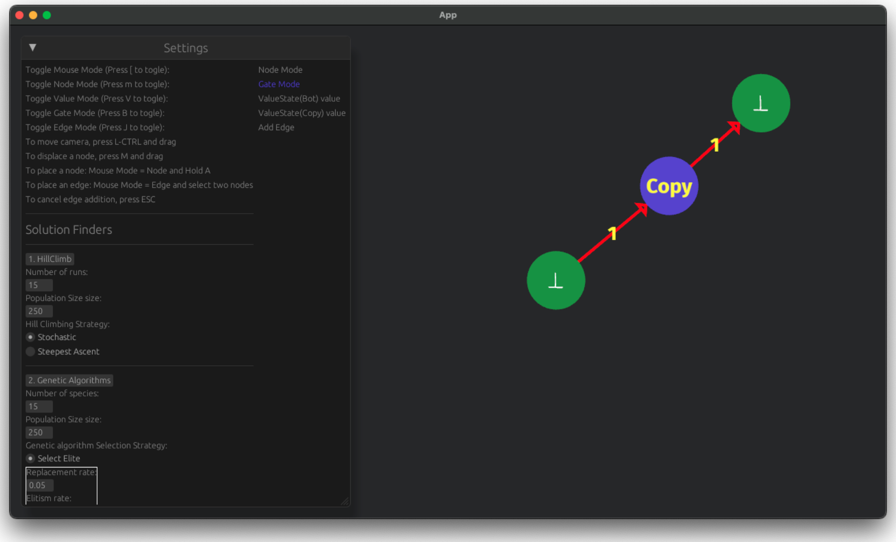


Figure .0.6: Full graph with edges

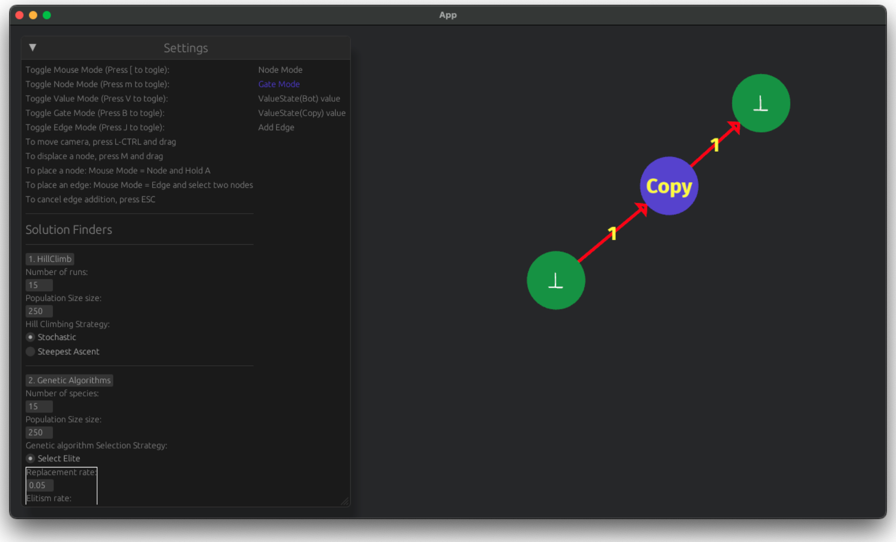
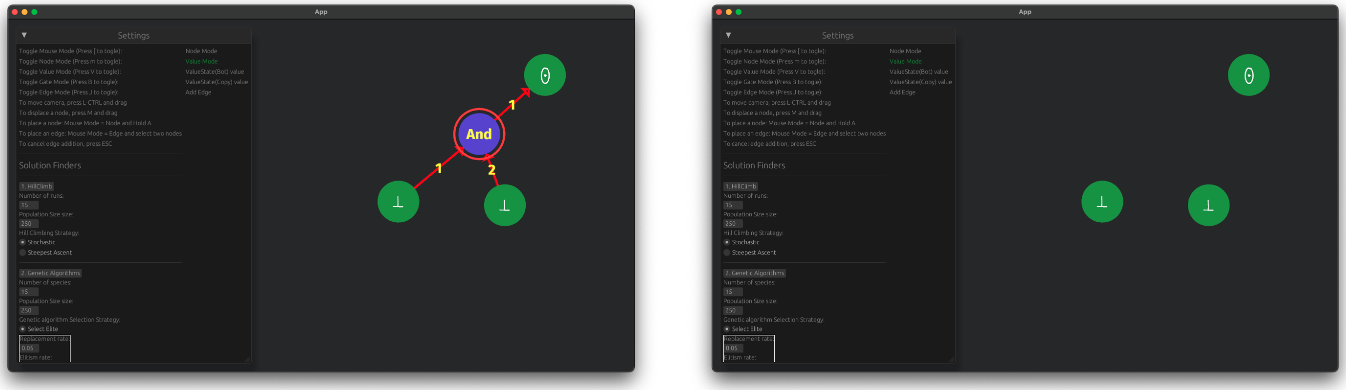


Figure .0.7: Value node toggle

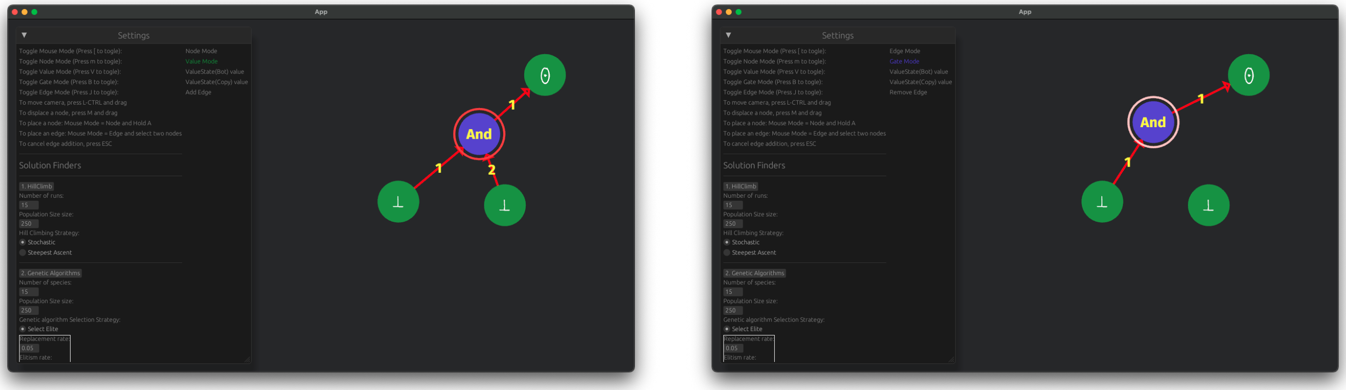
Node/Edge deletion To delete nodes we can simplify hover over a node and press D. For edge deletion the procedure is the same as the edge addition but we can only select nodes that are incident to our current node. We provide examples in .0.8 and .0.9



(a) Node removal: Before removal

(b) Node removal: After removal

Figure .0.8: Node removal



(a) Edge removal: Before removal

(b) Edge removal: After removal

Figure .0.9: Edge removal

Gate rules

NOR input	Output sets
$\forall v \in \mathbb{T} : (\cdot, \cdot, v)$	$(\mathbb{T}_{\leqslant \neg v}, \mathbb{T}_{\leqslant \neg v}, \{v\})$
$\forall v \in \mathbb{T} : (\cdot, v, \cdot)$	$(\mathbb{T}, \{v\}, \mathbb{T}_{\leqslant \neg v})$
$\forall a, b \in \mathbb{T} : a > b \implies (\cdot, a, b)$	$(\{\neg b\}, \{a\}, \{b\})$
$\forall v \in \mathbb{T} : (\cdot, v, \neg v)$	$(\mathbb{T}_{\leqslant v}, \{v\}, \{\neg v\})$
$\forall a, b \in \mathbb{T} : a < b \implies (\cdot, a, b)$	Invalid combination
$\forall a, b \in \mathbb{T} : (a, b, \cdot)$	$(\{a\}, \{b\}, \{\neg \max(a, b)\})$
$\forall a, b, c \in \mathbb{T} : (a, b, c)$	$(\{a\}, \{b\}, \{c\})$, only if the triplet (a, b, c) is a correct **NOR** assignment

Table .0.2: **NOR** value propagation rules

NAND input	Output sets
$\forall v \in \mathbb{T} : (\cdot, \cdot, v)$	$(\mathbb{T}_{\geqslant \neg v}, \mathbb{T}_{\geqslant \neg v}, \{v\})$
$\forall v \in \mathbb{T} : (\cdot, v, \cdot)$	$(\mathbb{T}, \{v\}, \mathbb{T}_{\geqslant \neg v})$
$\forall a, b \in \mathbb{T} : a < b(\cdot, a, b)$	$(\{\neg b\}, \{a\}, \{b\})$
$\forall v \in \mathbb{T} : (\cdot, v, \neg v)$	$(\mathbb{T}_{\geqslant v}, \{v\}, \{\neg v\})$
$\forall a, b \in \mathbb{T} : a > b(\cdot, a, b)$	Invalid combination
$\forall a, b \in \mathbb{T} : a > b(a, b, \cdot)$	$(\{a\}, \{b\}, \{\neg \max(a, b)\})$
$\forall a, b, c \in \mathbb{T} : (a, b, c)$	$(\{a\}, \{b\}, \{c\})$, only if the triplet (a, b, c) is a correct **NAND** assignment

Table .0.3: **NAND** value propagation rules

COPY input	Output sets
$\forall v \in \mathbb{T} : (\cdot, v) \vee (\cdot, v)$	$(\{v\}, \{v\})$
$\forall a, b \in \mathbb{T} : (a, b)$	$(\{a\}, \{b\})$, only if the tuple (a, b) is a correct **COPY** assignment

Table .0.4: **COPY** value propagation rules

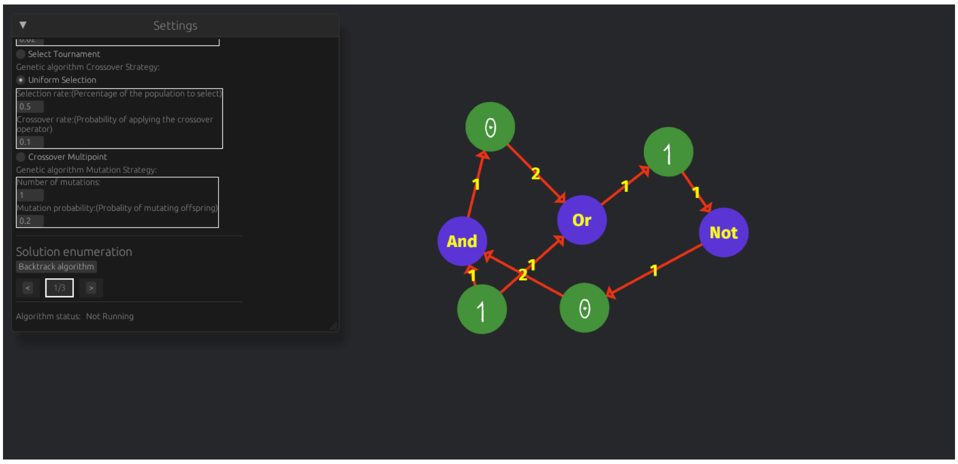
NOT input	Output sets
$\forall v \in \mathbb{T} : (\cdot, v)$	$(\{\neg v\}, \{v\})$
$\forall v \in \mathbb{T} : (v, \cdot)$	$(\{v\}, \{\neg v\})$
$\forall a, b \in \mathbb{T} : (a, b)$	$(\{a\}, \{b\})$, only if the tuple (a, b) is a correct **NOT** assignment

Table .0.5: **NOT** value propagation rules

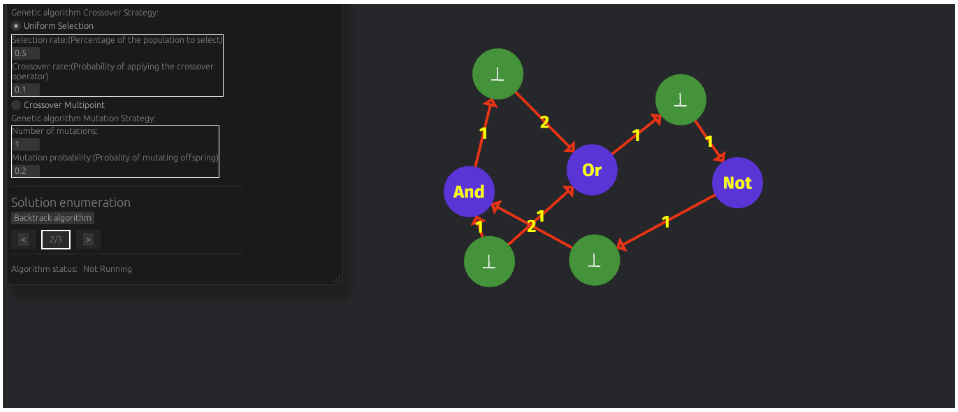
Software enumerators

Small circuit Example

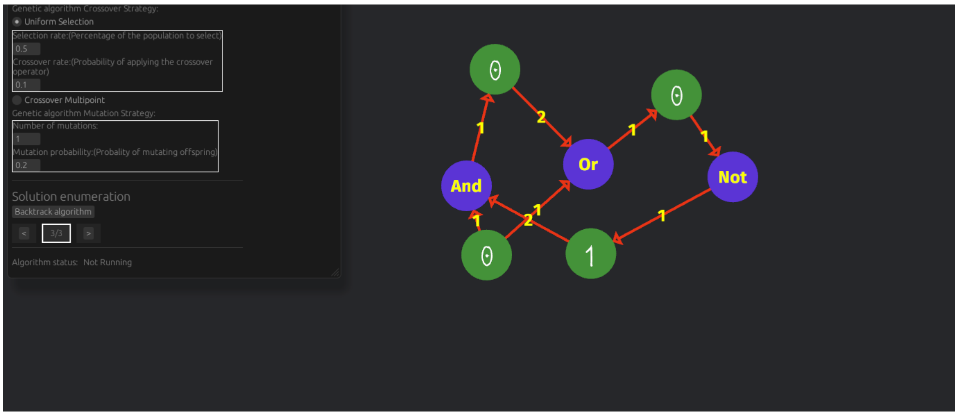
We provide the solution set for the small circuit from section 3.2.2.



Solution 1



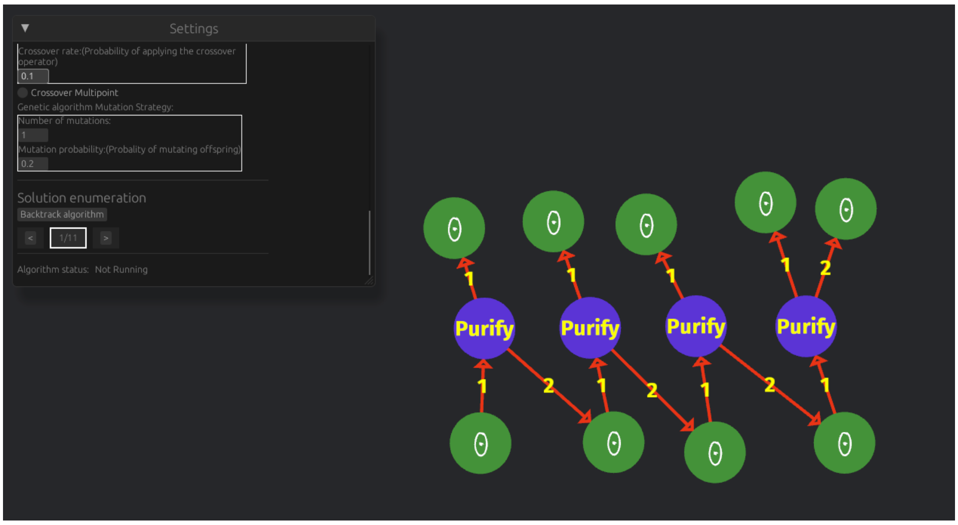
Solution 2



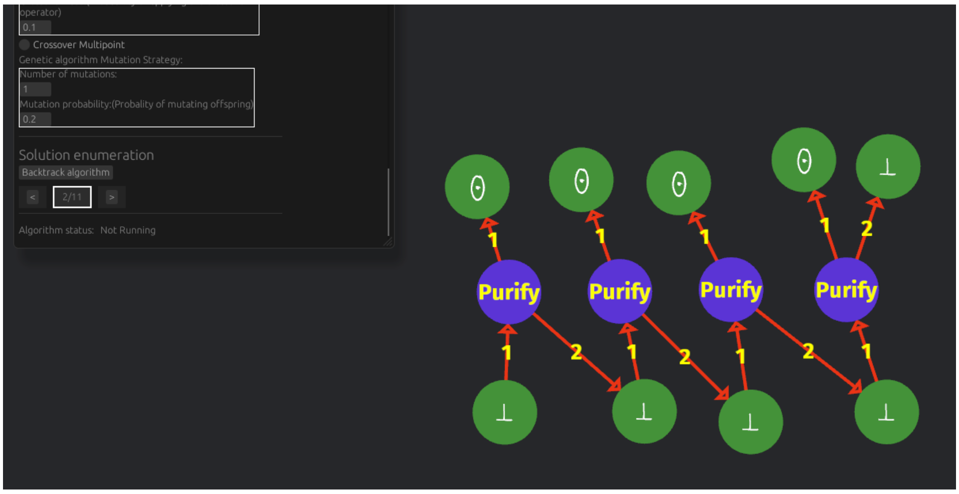
Solution 3

Bit Generator Example

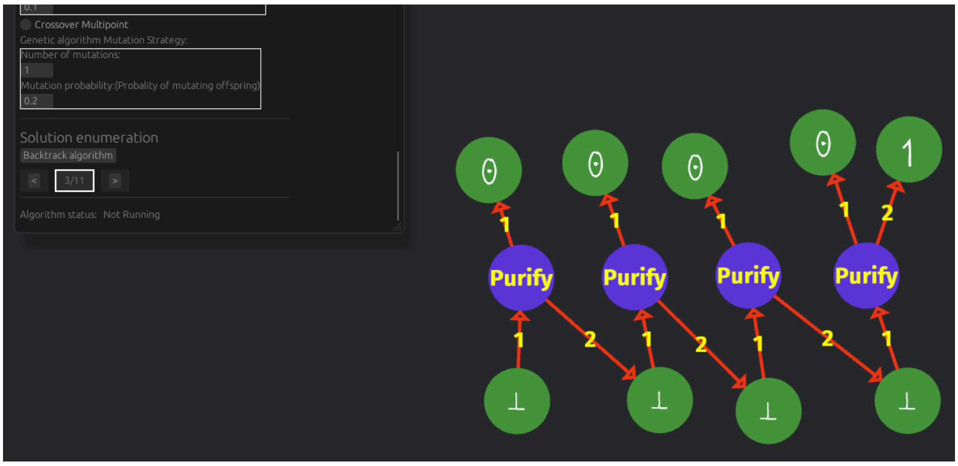
We provide the solution set for the bit generator circuit from section 3.2.2.



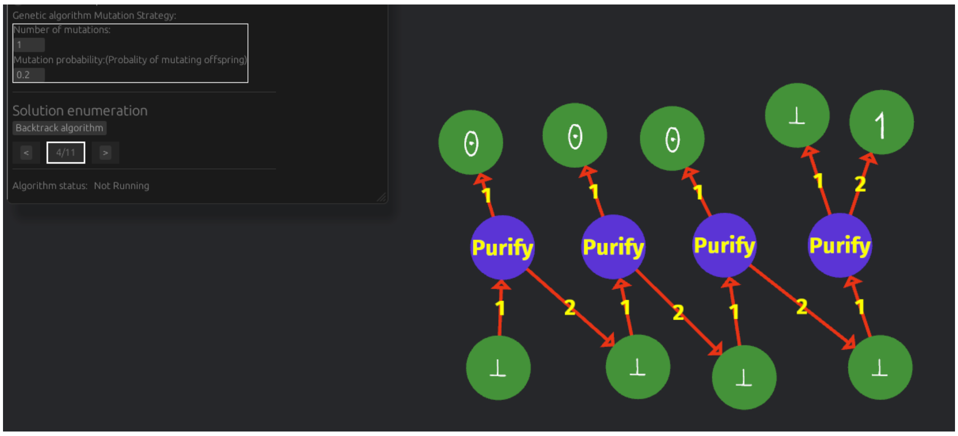
Solution 1



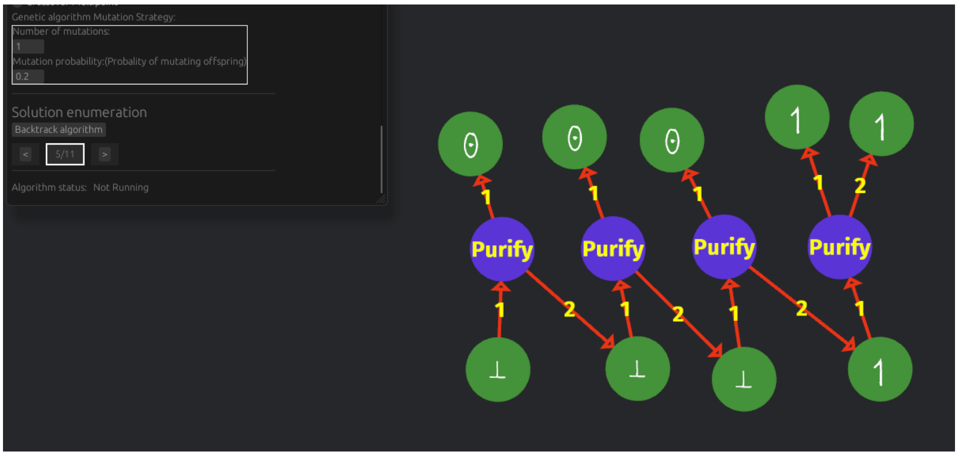
Solution 2



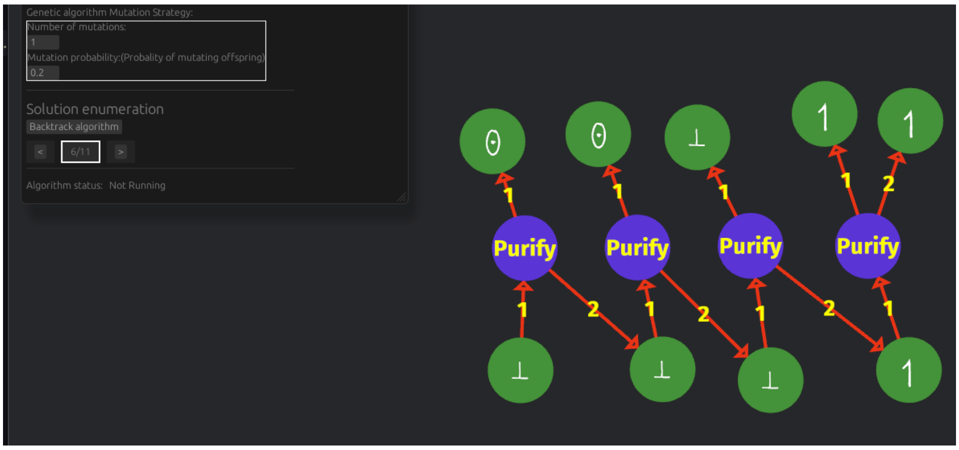
Solution 3



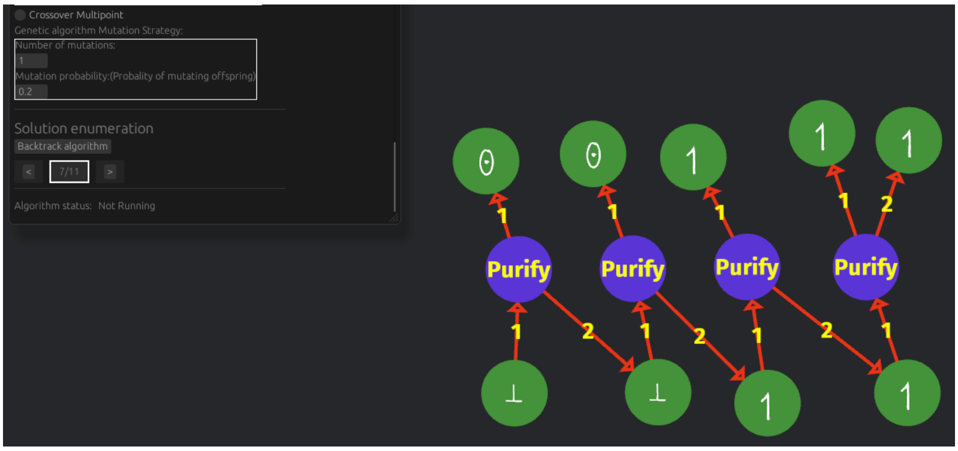
Solution 4



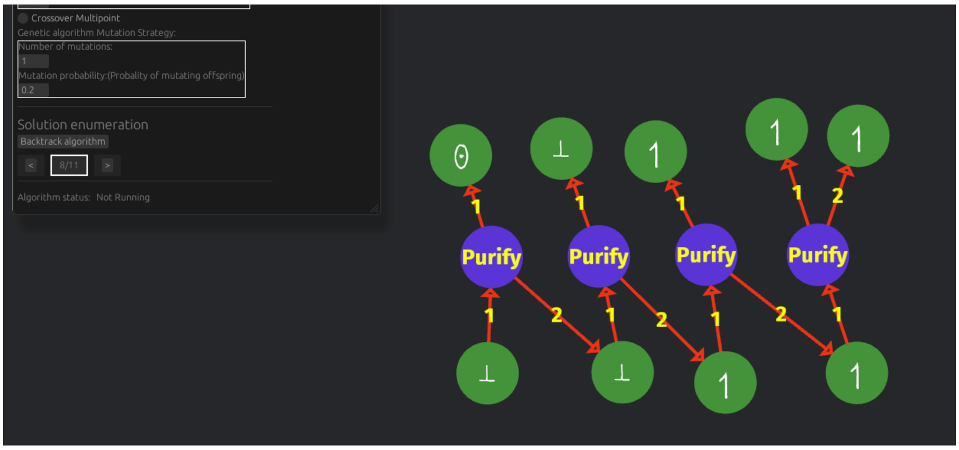
Solution 5



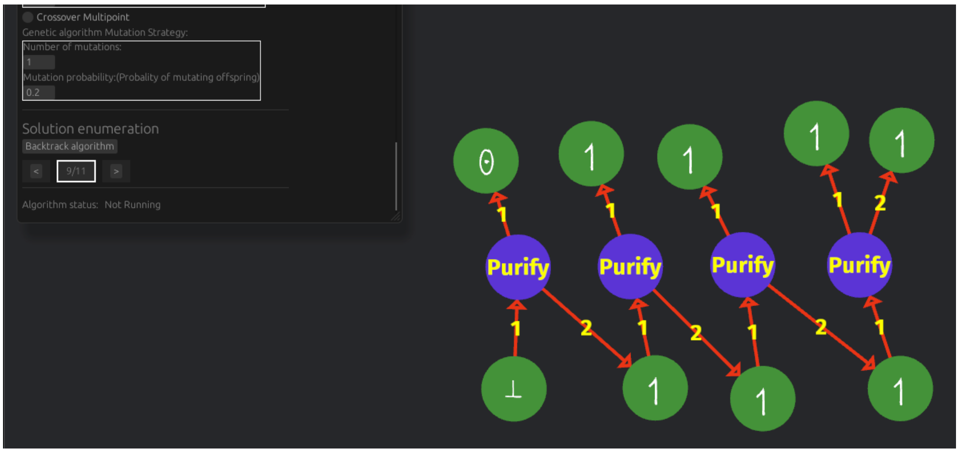
Solution 6



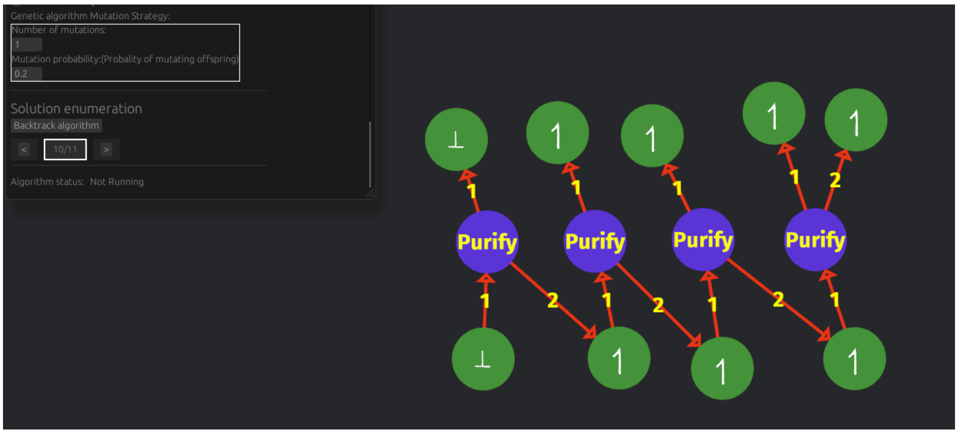
Solution 7



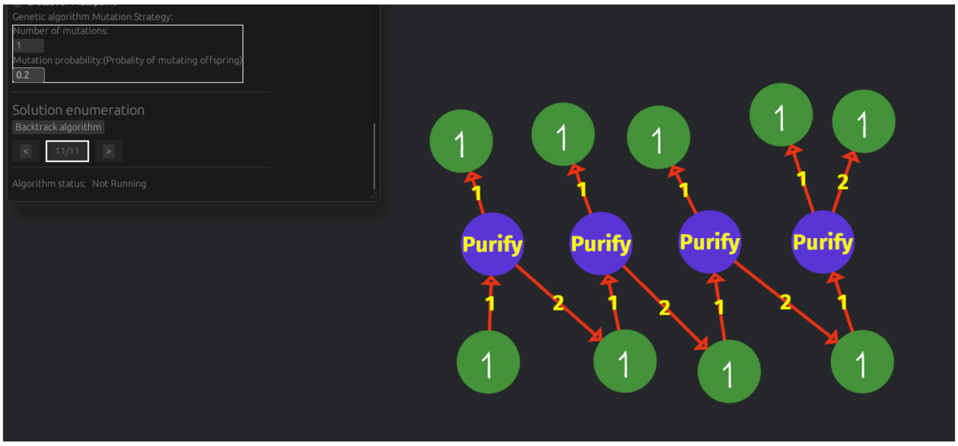
Solution 8



Solution 9



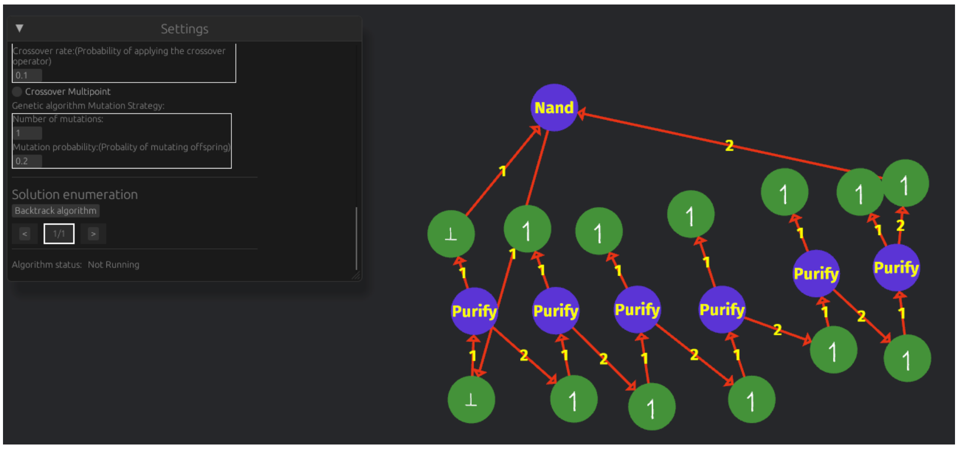
Solution 10



Solution 11

Complex circuit Example

We provide the solution set for the complex circuit from section 3.2.2.


Settings
Crossover rate:(Probability of applying the crossover operator)
0.1
Crossover Multipoint
Genetic algorithm Mutation Strategy:
Number of mutations:
1
Mutation probability:(Probality of mutating offspring)
0.2
Solution enumeration
Backtrack algorithm
1/1
Algorithm status: Not Running
Nand
Purify
⊥
1
2

Solution 1